

Foundations of Generative AI

▼ Teacher resources

▼ Printing Options

This unit aims to build a foundational understanding of text-based generative AI models, focusing on core concepts over technical skills. Students will demystify generative AI models by exploring their internal structures through the familiar lens of input, storage, process, and output. They will gain insights into how these models represent language, the impact of training data on model performance, and the potential for bias. Using this knowledge they will be prepared to have conversations with AI, where they can help educate individuals who feel powerless or lack agency in how AI is impacting their lives, or respond to individuals who have only read the hype headlines and offer feedback or criticism based on their knowledge of how these AI systems work.

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OK

Just Starting this Unit?

We recommend completing the free [Exploring Generative AI Self-Paced Professional Learning Module](#). The module is designed to take 2-3 hours to complete. No previous computer science or AI experience is required.

We also recommend watching the [5-Part Exploring Generative AI Video Series](#) which introduces key concepts for generative AI. The entire video series takes ~15 minutes to watch.

Finished Teaching This Unit?

Answer this [short survey](#) to let the Code.org curriculum team know how the unit went.

Week 1	Lesson 1: Introduction to Generative AI	Lesson 2: Input & Training Data	Lesson 3: Bias in the Machine	Lesson 4: Understanding Embeddings	Lesson 5: Embeddings: How They're Created
Week 2	Lesson 6: Understanding Neural Networks	Lesson 7: Neural Networks: How They're Trained	Lesson 8: Attention Is All You Need	Lesson 9: Outputs & Probabilities	Lesson 10: Hallucinations and Fabrications
Week 3	Lesson 11: Project: Demystifying Generative AI				

Key	☐ Instructional Lesson	☒ Assessment	☒ Unplugged Lesson
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▼ Lesson 1: Introduction to Generative AI

This lesson introduces the learning goals focused on artificial intelligence (AI), transitioning into generative AI. It begins with a discussion on the impact of past technological innovations on society. Students then explore the concept of generative AI through the lens of someone familiar only with its hype, aiming to foster understanding and responsible engagement with AI technology. A key part of the lesson is the introduction of the Input, Storage, Processing, Output (ISPO) framework, essential for understanding generative AI's workings. The lesson concludes with students receiving a graphic organizer for the unit and sharing their anticipations about learning more on generative AI.

☰ 1

Student Pre-Survey

🖥️ 2

Warm Up

☰ 3

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▼ Lesson 2: Input & Training Data

or

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In this lesson, students explore the process of recognizing and categorizing words, comparing it to how AI systems achieve similar tasks. Through discussion and activities, they consider the data sources an AI system requires to build a task-specific vocabulary, leading to the understanding that language models necessitate vast data amounts, like the entire internet. The lesson concludes with students updating a graphic organizer to reflect the Input step of generative AI.

🖥️ 1

Warm Up

☰ 2

Wrap Up

▼ Lesson 3: Bias in the Machine

This lesson formally introduces the concept of bias, where certain information is favored over others. Students experience how bias can implicitly and unintentionally creep into large-language models by participating in an activity where they are asked to source articles and information for a RecipeBot. Once they've used search engine results to generate their training data, they analyze patterns in the data that the class found as a collection, noticing unintended biases such as certain cuisines being unintentionally excluded or all the articles falling within a certain narrow date range. After acknowledging how bias can appear, students analyze several scenarios and discuss how bias could appear in the training data of each one.

🖥️ 1

Warm Up

☰ 2

Wrap Up

▼ Lesson 4: Understanding Embeddings

In this lesson, students explore how language models understand and represent word meanings. Beginning with a game, they investigate the concept of words being "close" in meaning. This leads to an activity where students assume the roles of fictional characters, using a sequence of numbers to represent their "character traits" and comparing these numerical representations to others. The lesson demonstrates that language models use a similar method, known as embeddings, to represent words as sequences of numbers, with similar words being numerically close to each other. The lesson concludes with students updating the storage section of their graphic organizer to encapsulate the idea of word embeddings.



Warm Up



Wrap Up

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▼ Lesson 5: Embeddings: How They're Created

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In this lesson, students delve into the creation of word embeddings. They start by interpreting a passage in a made-up language, using context clues to understand word meanings based on their relationships with other words. Following this, they engage in an activity to construct embeddings from sentences by noting word proximity. The climax reveals that students worked with two distinct passages, illustrating how the same word can have different representations based on its usage context. This insight sets the stage for upcoming lessons on how language models use sentence structure and context to determine word meanings accurately.

[Privacy Policy](#)



Warm Up



Wrap Up

▼ Lesson 6: Understanding Neural Networks

This lesson introduces neural networks through a hands-on activity where students create a lunch recommender for a school cafeteria. It starts with an exercise involving two mad-libs that use the same words in different orders, highlighting how word relations and grammar influence word choice. Students then work together to build a neural network model for the cafeteria's lunch choices, learning about key concepts like hidden layers and weights. The lesson concludes by tying these ideas back to language, prompting students to think about how words can change meaning in different contexts or phrases.



Warm Up



Wrap Up

▼ Lesson 7: Neural Networks: How They're Trained

This lesson explores supervised learning and its role in teaching neural networks language structure. Starting with a game, students learn through trial and error and use their past outcomes to inform future decisions. They then extend this approach to engaging in conversations in a fictional language, initially facing more failures but gradually improving as they identify patterns. This process illustrates how neural networks adjust their behavior based on new data. The lesson concludes with a reflection on whether this process signifies true "intelligence" or merely a sophisticated rule-following system.



Warm Up

▼ Lesson 8: Attention Is All You Need

This lesson delves into the ambiguity of language and the necessity of context for meaning. Students first discuss examples of ambiguous language, followed by an activity where they brainstorm words with multiple meanings and craft sentences to clarify their intended meanings through context. After trading papers with a partner, students guess the intended meanings, identifying which words provided the most context.

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Sign in on this page to save your progress. This activity to how language models use embeddings for word meanings and neural networks for relational meaning, introducing the concept of Attention as the process of updating word meanings based on context.

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1

Warm Up

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2

Wrap Up

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▼ Lesson 9: Outputs & Probabilities

In this lesson, students investigate how language models generate output based on probabilities. The lesson kicks off with the class collaboratively crafting a story, with a twist to alternate between realistic and fantastical elements. This leads into a group activity where students fill in a mad lib with predetermined words, aiming for either coherent or absurd outcomes. They are then introduced to an app that aggregates responses from numerous users to the same mad lib, showcasing a variety of popular choices for each blank. Through selecting the most popular choices based on this data, students experience firsthand how language models select words with the highest probability from training data.



1

Wrap Up



2

[For Teachers] Mad Lib Widget

▼ Lesson 10: Hallucinations and Fabrications

In this lesson, students investigate why language models may produce inaccurate outputs, known as hallucinations. They start by understanding the probabilistic nature of generative AI and how bias in training data can lead to such errors. Following discussions on examples of AI systems recalled for bias and hallucinations, students assume the role of AI Auditors to explore proactive measures for mitigating harm caused by deployed language models. After their exploration, they watch a video of a professional AI Auditor, reflecting on the parallels between the professional's work and their class activity.



1

Warm Up



2

Wrap Up

▼ Lesson 11: Project: Demystifying Generative AI

In this lesson, students create their project for the unit. In the project, students read several generative AI opinion articles that contain misrepresentations or exaggerations of what generative AI is capable of and how it works. Using their newfound knowledge from this unit, students draft a rebuttal and create an artifact to help visualize or conceptualize an aspect of their argument. This project acts as a summative assessment for the content of this unit.

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Lesson 1: Introduction to Generative AI

45 minutes

Overview

This lesson introduces the learning goals focused on artificial intelligence (AI), transitioning into generative AI. It begins with a discussion on the impact of past technological innovations on society. Students then explore the concept of generative AI through the lens of someone familiar only with its hype, aiming to foster understanding and responsible engagement with AI technology. A key part of the lesson is the introduction of the Input, Storage, Processing, Output (ISPO) framework, essential for understanding generative AI's workings. The lesson concludes with students receiving a graphic organizer for the unit and sharing their anticipations about learning more on generative AI.

Standards

Full Course Alignment

AI4K12 National Guidelines 2023

- ▶ **BI-2** - Agents maintain representations of the world and use them for reasoning

CSTA K-12 Computer Science Standards (2017)

- ▶ **AP** - Algorithms & Programming
- ▶ **CS** - Computing Systems
- ▶ **IC** - Impacts of Computing

Agenda

Before the Lesson

Preparing for the Unit

Student Pre-survey

Teacher Pre-Survey

Preparing to Lead Discussions

Student Levels

Warm Up (5 minutes)

Activity (35 minutes)

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Distinguish between different types of AI, including machine learning, deep learning, and generative AI.
- Evaluate the potential impact of generative AI on society and discuss its capabilities and limitations.
- Identify and explain the four key components that make up a computer system.

Preparation

- Print copies of the graphic organizer for each student
- Run through the slide deck and videos ahead of time
- (Optional) Prepare vocabulary posters for class word wall
- (Optional) Print out the slide deck with speaker notes as a resource during class. [Click here for steps on how to do this.](#)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- [Introduction to Generative AI](#) - Slides
▼ Make a Copy
- [LEADERS Framework](#) - Resource
▼ Make a Copy

For the students

- [Foundations of Generative AI](#)
[Graphic Organizer](#) - Resource

▼ Make a Copy

- **Introduction to Large Language Models** - Video ([Download](#))
- **What Is Machine Learning?** - Video ([Download](#))
- **What Makes a Computer a Computer?** - Video ([Download](#))

Vocabulary

- **Artificial Intelligence** - A technology that mimics human intelligence, performing tasks such as understanding language, recognizing patterns, and making decisions
- **Large Language Model** - A computer program that analyzes large amounts of raw human-created data to the point where it can represent and recreate new data to perform a variety of tasks
- **Model** - A computer program designed to make a decision

Teaching Guide

Before the Lesson

Preparing for the Unit

Getting Started with Code.org: Consider watching our [Getting Started with Code.org](#) video series for an overview of how to navigate lesson plans, setup a classroom section, and other important features of the Code.org platform. Each video also has a support article if you'd prefer to read or print instructions - [click here to learn more](#).

Setup a Classroom Section: You can use a class section in Code.org to manage your students, view their progress, and assign specific curriculum - [click here to learn more](#).

If you are using a learning management system, there may be additional steps to sync your classes with Code.org:

- [Click here](#) for steps to setup your classes with Google Classroom
- [Click Here](#) for steps to setup your classes with Clever

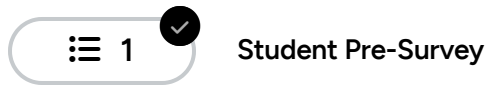
Become a Verified Teacher: Lesson plans and levels have additional resources and answer keys for Verified Teachers, which is quick process that verifies your position at an educational institution. [Click here to complete a form](#) and you should have access to verified teacher resources in ~1 business day.

Professional Development: We recommend completing the free [Exploring Generative AI Self-Paced Professional Learning Module](#) before starting the unit. The module is designed to take 2-3 hours to complete. No previous computer science or AI experience is required.

Student Pre-survey

Before beginning the unit, please have students complete a short 5-question presurvey. When students finish, they can start the warm-up in the next level.

Although their answers are anonymous, the aggregated data will be available to you once at least five students have completed the survey.



Teacher Pre-Survey

Before starting the unit or while students take their survey, please complete [this teacher pre-survey](#). Your feedback is valued and appreciated!

Preparing to Lead Discussions

LEADERS Framework: Throughout this unit, there are opportunities for students to discuss the impacts of generative AI on society, both the intended and unintended consequences. Navigating these conversations so they elevate and honor the experiences of your students, while also keeping the conversation grounded and connected to the lesson content, requires careful consideration and intentionality as the discussions are happening.

To help prepare for these moments, we recommend reviewing the **LEADERS Framework** for facilitating lessons and discussions involving the societal effects of AI, including bias and ethical considerations. Not all of the tenets are applicable to every lesson, but many of the teaching tips and discussion goals in this lesson follow the principles outlined in this framework.

Student Levels

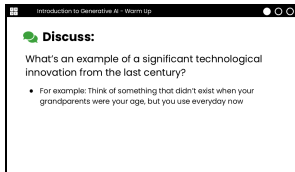
These levels contain the same questions as the **Warm Up** and **Wrap Up** from the lesson slide deck, allowing students to answer the questions within the Code.org platform. These can be used as a digital alternative to physical journals, and lets you take advantage of our [Student Summaries features](#) for checking and displaying student work.

By answering these questions in these levels, you can take advantage of discussion and differentiation strategies that elevate student voice to help you guide the discussion. See our [Teaching Guide for Incorporating Student Responses in Lessons](#) for more information on these strategies.



Warm Up (5 minutes)

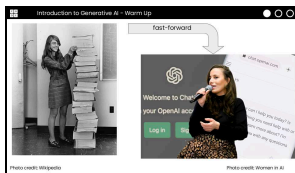
Slide	Lesson Guide
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Discuss: What's an example of a significant technological innovation from the last century? For example: think of something that didn't exist when your grandparents were your age, but you use everyday now.

Have students discuss in small groups before regrouping as a full group. Ask several students from around the class to share what they said, and let students raise their hands or quietly snap their fingers if they said the same thing or something similar.

Discussion Goal: This prompt is intentionally open-ended to invite all students into the conversation and offer some kind of technological innovation. Some potential answers might be the internet, social media, driverless cars, smartphones, and (of course) artificial intelligence.



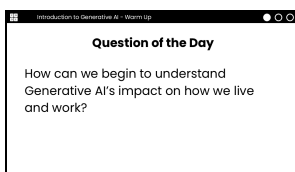
Say: Many devices we use today were once marvels of modern technology. For example, this picture shows Margaret Hamilton in front of the source code used in the computers for the Apollo 11 space mission. At the time, computers were an exciting innovation set to fundamentally change how we live and work

Click the animation

Say: Fast forward to today, and the same conversation is happening around artificial intelligence. For example, this picture shows Mira Murati, the chief technology officer of OpenAI, talking about their chatbot ChatGPT, which is a special kind of "generative" artificial intelligence that lets you chat with an artificial intelligence agent. Just like computers many years go, we're having these same conversations about the impact of artificial intelligence and its potential impact on society and how we live and work.

Teaching Tip

Teaching Tip: Intentional Foreshadowing - The comparison on this slide between the computer and artificial intelligence is very intentional, as we will ultimately use the same framing for understanding how a computer works to also understand how generative AI works. Making this kind of association early and often is important to help avoid anthropomorphizing artificial intelligence or making it seem impossible to understand. If we are capable of understanding a computer and seeing it as a machine (and not a human), then we are capable of doing the same with artificial intelligence.



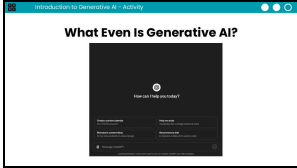
Say: Today we'll start to explore the impact generative AI can have on how we live and work, and uncover some of the key themes we'll see throughout this unit.

Question of the Day: How can we begin to understand Generative AI's impact on how we live and work?

Teaching Tip

Teaching Tip: Questions of the Day - The Question of the Day provides a question that, by the end of the lesson, students should (hopefully) be able to answer. It's presented in the warm-up as a way to frame the content of the lesson in a student-friendly way. It's also provided in the wrap-up if you would like to use the Question of the Day as an additional exit ticket or student journal response. For schools or districts that have requirements around student-friendly objectives or daily formative assessments, these questions can help meet these requirements or meet other external needs.

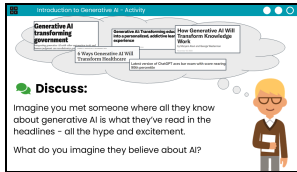
Activity (35 minutes)

Slide	Lesson Guide
	☒ Do This: Show students an example of a text-based generative AI. There are a few options for how to do this: - Option 1: If you are able to access a large-language model at school, such as ChatGPT or Google Gemini or Claude.ai, then you can demonstrate the model for the class. You should come up with 3-4 prompts you plan to demonstrate and test them ahead of time, then run them in front of the class. You can also take questions from the audience, but consider finding a way to preview the responses before students see them to make sure they are appropriate. - Option 2: Play the video embedded in the slides, which shows 3 prompts being given to ChatGPT. Paused the video if needed to look closely at the types of responses being given. - Option 3: Find another video of a person using ChatGPT in an interesting and fascinating way that you think will resonate with your students. Paused the video if needed to look closely at the types of responses being given. Avoid examples with image-based generative AI (like deepfakes or DALL-E) since these technologies are not covered in this unit.  Say: In many ways, generative AI seems like magic - they're able to create results that seem to mimic human intelligence and perform incredible tasks.

Teaching Tip

Teaching Tip: Don't Skip the Demo - It can be easy to assume everyone's heard the hype around generative AI and seen them in action, but this risks isolating the students who may not have seen this technology or are less familiar with it than their peers. Showing one of these demos, even if it's just the shorter video, gives everyone the same shared experience that you can reference throughout the unit.

	 Click the animation  Say: As a result of these amazing advancements, there's been a lot of AI hype. There are new articles every week about how generative AI will revolutionize this thing or that thing, or how it will change a certain industry forever. There's a lot of hype and excitement and buildup about how AI will forever change the world, just like there was about computers when they were first invented.
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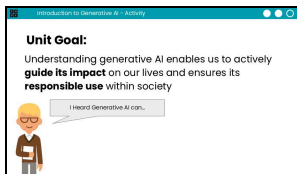


Discuss: Imagine you met someone where all they know about generative AI is what they've read in the headlines - all the hype and excitement. What do you imagine they believe about AI?

Have students discuss in a small group, then have a few students share with the class. If students struggle to come up with ideas, ask them to think of their younger cousins or the elders in their family whose beliefs about artificial intelligence may only be coming from headlines like this.

Discussion Goal: Try to draw out responses that hypothesize that these people believe AI can do just about anything, or potentially don't have any flaws. Encourage students to consider friends or family members and what they believe, and how much of it is built up by all of the hype around AI. Emphasize that this is a thought experiment where we're purposefully imagining a person whose only read these kinds of hyped-up headlines, even if we might already know the facts or reality are much different than what is presented in these types of stories. This kind of dissonance - what a person might think from headlines vs what the realities are - will be a key distinction for the rest of the unit.

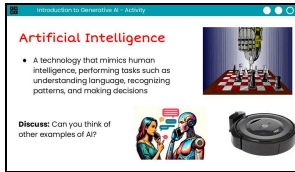
Say: It can be exciting to imagine all the ways a new technology can improve our lives, but it's important to still be critical and cautious to make sure we're not swept up in the hype. This is part of being a responsible and informed citizen, to make sure we're not taken advantage and know the true benefits and limits of technology.



Say: This will be a key theme in this unit - we should understand how generative AI works in order to actively guide its impact on our lives and ensure its responsible use within society.

 Click the animation

Say: Throughout this unit, we're going to return to this fictional "AI Hype" person who might have certain beliefs that may not be fully formed and our job will be to respond and help educate them. And the best way to do that is to make sure we understand this technology ourselves.



Say: Let's start by discussing Artificial Intelligence, or AI, in general. AI is a technology that mimics human intelligence, performing tasks such as understanding language, recognizing patterns, and making decisions. Some examples include

- When you call a help line or a store and you talk to a robot for a brief moment before speaking to a human - this kind of AI understands human language
- A robot that's learned to play chess and can beat a human opponent - this robot recognizes patterns from previous games to learn the optimal strategy
- A robot that can navigate around your house and avoid your furniture - this robot uses sensors to make decisions

Discuss: What are other examples of AI you can think of?

Have students discuss with a neighbor before sharing with the class. Consider asking students to share something interesting they heard their partner say, rather than sharing their own idea.

Discussion Goal: This is a purposefully open-ended discussion - students will share a variety of answers. For example, students may recall examples from movies or television, or they will reference AI technologies on their devices (like image filters), or they may reference recommendation systems. Try to involve as many student voices in the conversation as you can before continuing to the next slide.

Teaching Tip

Teaching Tip: Vocabulary Management - Artificial Intelligence may seem ubiquitous, but we don't want to assume everyone in the room has heard this term before so we introduce it here so everyone starts with the same foundation. However, this sometimes introduces tension with vocabulary management - how should students keep track of vocabulary in this unit so they can reference it when needed?

This unit provides a partial solution - students will get a graphic organizer at the end of this lesson that includes key vocabulary and concepts for the entire unit. However, not all words (like this one) are represented in that organizer. If you have your own system for tracking vocabulary that you've used in previous units, consider applying it here as well so students have a consistent experience.

If you don't have a system for managing vocabulary, you might consider the following strategies:

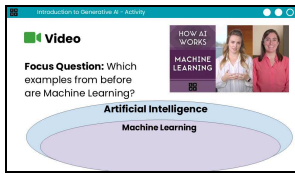
- Create a Word Wall or posters for these vocabulary terms and hang them up in your classroom as they are introduced throughout the unit.
- Provide a vocabulary handout that is updated after lessons and lives near desks or computers so students can reference as needed.

Say: There are different types of artificial intelligences that focus on different things. One important type is called Machine Learning - let's watch a video to learn a bit more. After the video, we'll reflect on which of the examples we just named are also machine learning examples.

Video: Watch the [How AI Works: Machine Learning video](#)

Discuss: Which examples from before are machine learning?

Discussion Goal: Students won't have enough context about machine learning to give perfect answers here - instead, help students make connections between things they may have mentioned and examples they saw in the video, such as recommendation algorithms or image recognition / filters. If students ask "is this an example of machine learning?", it's okay to say we're not sure and suggest students do some research if they're interested. If needed, you can skip to 1:06 in the video where it shows a list of various machine learning tasks. Once a few students have reflected on different machine learning tasks, continue to the next slide.



Say: This hierarchy of AI types goes even deeper!

Click the animation

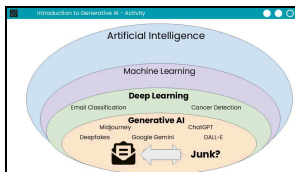
Say: Deep Learning uses complex tools called Neural Networks, which we'll learn more about in this unit. A few examples of Deep Learning are cancer detection or email classification, where we give the AI system an email and asked "is this junk?".

Click the animation

Say: But what if we flipped this around? What if we started by saying "write me an email that sounds like junk mail" and asked it to generate the text instead?

Click the animation

Say: This is the latest evolution in AI technology, called Generative AI. It works similar to these other Deep Learning systems, just in reverse - generating new text or image or video rather than trying to classify it.



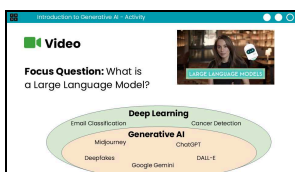
Say: Let's hear a little bit about some of the advances in generative AI from the people who are actively working on it!

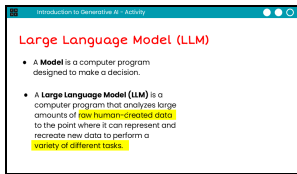
Video: Watch the [Introduction to Large Language Models \(LLMs\) video](#)

Discuss: What is a Large Language Model?

Discussion Goal: The answer to this flies by in the video and is easy to miss, so students aren't expected to answer it perfectly at this point. Instead, this question sets up for the next slide where we introduce a formal definition. For the discussion, have students share with a neighbor, but don't have students share as a full group. Instead, continue to the next slide where you can ask how closely their definitions here align with the definition we'll use as a class.

Say: Let's see how well your definitions match up with the one we'll use in this class





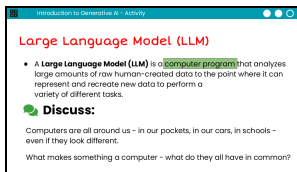
Say: First let's define one of the words in this phrase: model. An AI model is a computer program designed to make a decision, like deciding whether or not to unlock your phone based on your facial ID. Regular machine learning models are designed to solve only one type of problem, and the data they need has to be very specialized for the task it's designed to complete.

Click the animation

Say: Then we have large-language models, which is a computer program that analyzes large amounts of raw human-created data to the point where it can represent and recreate new data to perform a variety of tasks.

Click the animation

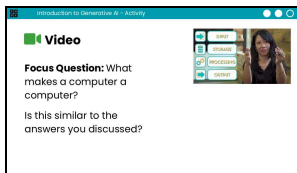
Say: The two reasons large-language models are such a big technological breakthrough is the capacity to look at the raw human-created data and perform a variety of different tasks with the same model - like passing lawyer exams and science exams with the same model



Say: We also have our first clue to help us understand how all the magic of a large-language model works: the fact that it's a computer program!

Discuss: Computers are all around us - in our pockets, in our cars, in schools - even if they look different. What makes something a computer - what do they all have in common?

Have students share with their neighbors, but don't regroup for a full-group share-out. Instead, watch the video on the next slide before debriefing.

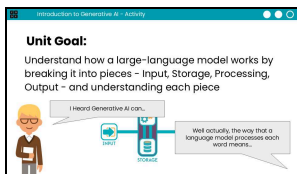


Video: Watch the [What Makes a Computer a Computer](#) video?

Discuss: What makes a computer a computer? Is this similar to the answers you discussed?

Have students share out as a full class

Discussion Goal: Students should have picked up that a computer is made up of Input <> Storage <> Processing <> Output components. When comparing to the earlier discussion, they may make comparisons to these structures in their answers - for example, all computers have some kind of keyboard or touch screen interface (ie: input), or all computers can perform calculation of some kind (processing).



Say: This is another key theme in this unit - we can understand large-language models by breaking it into pieces - input, storage, processing, output - and understanding each piece.

Click the animation

Say: And using this information, we'll have a better understanding of the opportunities and limitations of generative AI and be better equipped to talk to our imaginary friends who only believe the hype.

	Discuss: Based on the generative AI demo we saw earlier, what might you predict is the input, storage, processing, and output for a chatbot language model? Have students discuss in groups before sharing as a class. Discussion Goal: This is an open-ended foreshadowing question just to help students start making predictions and thinking about what may come in later lessons. Some students might already think the input are words, like the thing you type into the chatbot, and the output is the new paragraph that the chatbot generates. Students aren't expected to have correct answers here especially around storage and processing - it's just a prediction that can be returned to in later lessons.
	Distribute: Pass out the Generative AI Graphic Organizer Say: We are going to use this graphic organizer throughout the entire unit to understand how language models work. By the end of the unit, these will be filled in and we'll have definitive answers for each of these components!

Wrap Up (5 minutes)

Slide	Lesson Guide
	Do This: Have students answer the following questions on their own - What is something new you learned about AI today? - What is a question you still have about AI? Consider collecting these as an exit ticket to review what knowledge and questions students have coming into this unit.

Teaching Tip

Teaching Tip: Getting Organized - This exercise also gives time for students to organize what they will do with their graphic organizer - students may want to put it in a folder or notebook they keep themselves, or you may decide to offer a dedicated place for students to store this document.

	Question of the Day: How can we begin to understand Generative AI's impact on how we live and work?
	Vocabulary: The key vocabulary from this lesson is provided on this slide and in the lesson plan.

Teaching Tip

Teaching Tip: Key Vocabulary - The vocabulary from each lesson is provided at the end of the slide deck. This can be useful to print and keep available for students throughout the unit, or this slide can help facilitate your own vocabulary strategies. One possible strategy is to build a word wall throughout the unit, providing both a written definition and a visual that students can reference as-needed during the rest of the unit.



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Lesson 2: Input & Training Data

45 minutes

Overview

In this lesson, students explore the process of recognizing and categorizing words, comparing it to how AI systems achieve similar tasks. Through discussion and activities, they consider the data sources an AI system requires to build a task-specific vocabulary, leading to the understanding that language models necessitate vast data amounts, like the entire internet. The lesson concludes with students updating a graphic organizer to reflect the Input step of generative AI.

Standards

Full Course Alignment

AI4K12 National Guidelines 2023

- **BI-2** - Agents maintain representations of the world and use them for reasoning

CSTA K-12 Computer Science Standards (2017)

- **CS** - Computing Systems

Agenda

Before the Lesson

Student Levels

Warm Up (5 minutes)

Activity (35 minutes)

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Explain how a language model uses data to recognize words as input
- Identify potential data sources relevant to a task being performed by a language model

Preparation

- Print copies of the activity guide for each student
- Run through the slide deck and videos ahead of time
- (Optional) Print out the slide deck with speaker notes as a resource during class. [Click here for steps on how to do this.](#)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- [Input & Training Data](#) - Slides

▼ Make a Copy

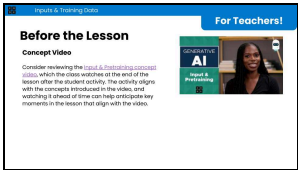
For the students

- [Generative AI: Input & Pre-training](#) - Video ([Download](#))
- [Training Data](#) - Activity Guide

▼ Make a Copy

Teaching Guide

Before the Lesson

Slide	Lesson Guide
 The thumbnail shows a slide titled 'Before the Lesson' with a 'Concept Video' section. It includes a small video player and a 'For Teachers!' box.	Before the Lesson: This activity uses a pedagogy rooted in discovery learning, where students first complete an activity and reflect on their process, which illuminates the underlying AI concepts. A video is provided at the end of the lesson that explains and summarizes the key learning from the lesson. Concept Video: Consider reviewing the Input & Pretraining concept video, which the class watches at the end of the lesson after the student activity. The activity aligns with the concepts introduced in the video, and watching it ahead of time can help anticipate key moments in the lesson that align with the video.

Student Levels

These levels contain the same questions as the **Warm Up** and **Wrap Up** from the lesson slide deck, allowing students to answer the questions within the Code.org platform. These can be used as a digital alternative to physical journals, and lets you take advantage of our [Student Summaries features](#) for checking and displaying student work.

By answering these questions in these levels, you can take advantage of discussion and differentiation strategies that elevate student voice to help you guide the discussion. See our [Teaching Guide for Incorporating Student Responses in Lessons](#) for more information on these strategies.

1

Warm Up

2

Wrap Up

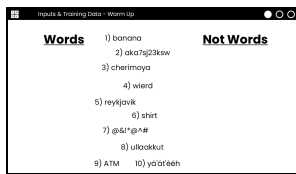
Warm Up (5 minutes)

Slide	Lesson Guide
 The thumbnail shows a comic strip of Calvin and Hobbes playing Scrabble. The text on the slide reads: 'In this comic, Calvin and Hobbes are playing Scrabble - a game where you score points based on the words you can create with random letters. On the next screen, you will see a list of "words". Decide which ones are truly words like you can play in Scrabble and which are not.'	Say: In this comic, Calvin and Hobbes are playing Scrabble - a game where you score points based on the words you can create with random letters. On the next screen, you will see a list of "words". Decide which ones are truly words like you can play in Scrabble and which are not.

Teaching Tip

Teaching Tip: Classroom Prep - Before beginning this warmup, consider how you will manage the discussion on the next slide where students are sorting terms into "words" or "not words". Consider having two sheets of poster paper that you can write on as the class sorts, or consider making a table on a whiteboard and writing

the results there. You could also consider presenting these slides in “edit” mode so you can move the text boxes around on the screen, but make sure you are able to reset the choices between class periods if you teach this lesson to multiple classes in the same day.



Words	Not Words
1) banana	
2) aka7sj23ksw	
3) cherimoya	
4) wierd	
5) reyjavik	
6) shirt	
7) @&!*@^#	
8) ullaakut	
9) ATM	
10) yá'át'ééh	

Discuss: Have students discuss in groups before sharing out as a full class. Instead of going line-by-line through these words, ask students which decisions they are “most confident” about and sharing their classification for those words first. Using this strategy, students will likely sort “1) banana” and “6) shirt” as words, and “2) aka7sj23ksw” and “7) @&!*@^#” as not words. The remaining words are deliberately chosen to motivate certain discussions and create moments of cognitive dissonance and curiosity, hopefully allowing students to lead the discussion by asking questions or respectfully challenging ideas.

Let students know they will likely change their minds as the discussion continues and that’s okay, and allowing students to state their confidence level for their initial thoughts can make them more comfortable speaking up since it feels less “high-stakes”. Be prepared to address the following situations in this discussion:

3) **cherimoya** is a fruit common to South America, but may not be a common fruit your students have heard of. It’s possible many students may say “this isn’t a word”, but one or two students may speak up because they’re familiar with this fruit. Be sure to elevate those voices and highlight how our own knowledge is sometimes limited by lack of exposure to different words or phrases. Or, if no one raises this point, be sure to point this out to students explicitly such as showing them the Wikipedia page. In either case, make sure cherimoya is classified as a “word” before leaving the discussion

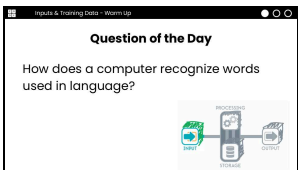
8) ullaakut and 10) yá’át’ééh are both words in other languages - **ullaakut means hello in Inuktitut**, a language spoken by Inuit peoples in Canada and Greenland; and **yá’át’ééh means “hello” in the Navajo / Diné language**. Similar to above, students may initially classify these as not words because they are unfamiliar with them, but it is important to surface the meaning and use of these words to students either with a quick google search or elevating the voice of a student in class who is already familiar with these words. As a result, make sure these terms don’t end up in the “Not Words” category by the end of the discussion.

4) wierd is a misspelling of either the word “weird” or the word “wired”. Students may initially say that this is “not a word”, but another student or yourself should challenge them by pointing out that if this is typed into a device with autocorrect, it will automatically correct it to one of these other words. Even in this slide deck, the word has a red squiggly underline representing that the computer recognizes it as something, even if it’s not a correctly spelled word. This may inspire the class to create a new category to represent this term - it’s not a correctly spelled word, but it’s also not a made-up word, so maybe there’s a new category for “misspelled words” or “important terms that aren’t words”. Let students guide how they want to handle this situation, even if they decide not to make a new category and it remains in “not word”.

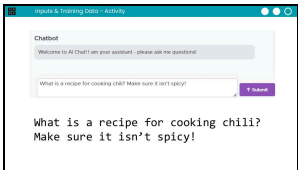
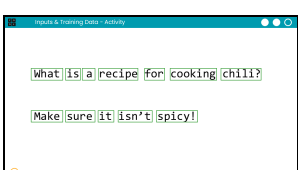
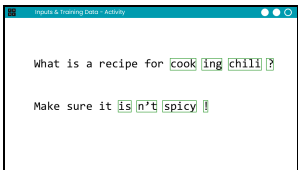
At some point, students who are more familiar with the Scrabble game may try to invoke the “rules of Scrabble” which restrict words in other languages, proper nouns, or abbreviations. From this perspective, words (8) and (10) above would be restricted, as would 5) Reykjavik (the capital of Iceland) and 9) ATM (since it is an abbreviation for Automatic Teller Machine). If this comes up, use this as an opportunity to question the effect that arbitrary rules have on what we consider valid language, since these are all definitely words that are used in society even if not officially allowed in Scrabble. You could consider creating new categories to represent these phrases, such as “Still words but not allowed in Scrabble” or “Part of language, but not allowed in Scrabble”. Or, the class could decide these belong in the Words category, regardless of what Scrabble thinks, and that’s the end of the story.

Discussion Goal: Ultimately, this is an exercise to surface that the way we identify words versus nonsense - much like Calvin and Hobbes in the comic panel - can be contextual to our lived experiences and influences what we consider important information versus random noise that can be ignored. It’s okay if the discussion ends without closure, so long as students have wrestled with how what they initially considered “words” may have been

expanded as they were exposed to new contexts in the discussion. However, be very intentional that the discussion ends with very few items in the “Not Words” category, especially making sure ullaakkut and yá’át’ééh are not in this category and are instead classified as either a Word or a new category the class has created.

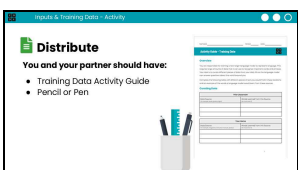
 Question of the Day How does a computer recognize words used in language?	Say: Just from this warm-up, we’ve already seen that it can be tricky trying to recognize “words” from non-words. For a large-language model, those words are the inputs it uses to start generating it’s response. Today, we’re going to explore how a computer is able to recognize words used in language. This is the first step in understanding how a large-language model words - by looking at the inputs it uses! Question of the Day: How does a computer recognize words used in language?
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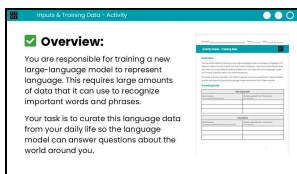
Activity (35 minutes)

Slide	Lesson Guide
 Chatbot Welcome to AI Chat! I am your assistant - please ask me questions! What is a recipe for cooking chili? Make sure it isn't spicy! What is a recipe for cooking chili? Make sure it isn't spicy!	Say: When you use a large-language model, you start by giving it a prompt. In this example, we’re asking it for a recipe for making chili.
 What is a recipe for cooking chili? Make sure it isn't spicy!	Say: As a first step, the model breaks apart this input into individual pieces. Click the animation Say: Usually this is word by word
 What is a recipe for cook ing chili ? Make sure it is n't spicy !	Say: But sometimes it’s even smaller, like when there is a root word or for punctuation. Technically, these are called tokens - they’re the smallest unit of data that a large-language model can represent. Say: So then how does a large-language model learn how to recognize these words and tokens? And what words can it use when generating a response? Well - that’s exactly what we’re going to learn about today!

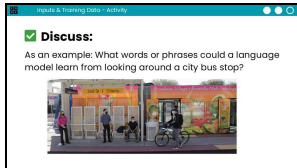
Teaching Tip

Teaching Tip: Viewing Tokens - If your school district allows access, OpenAI has a website that lets you preview how it interprets words and converts them into tokens: <https://platform.openai.com/tokenizer>. This can be a helpful visualization of how words are broken apart and interpreted by large language models. Consider demonstrating this for the class and typing in example sentences. Most words will likely be interpreted as single words, but compound words like “decapod”, “anthology”, “pretense”, and “arachnophobia” are interpreted as two tokens instead of one.

 Distribute You and your partner should have: - Training Data Activity Guide - Pencil or Pen	Distribute: Pass out the Training Data Activity Guide
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☒ **Do This:** Read the overview of this document as a class. This text is also provided on the slide.



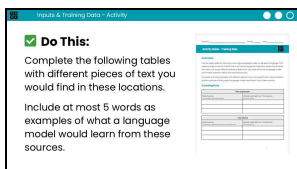
 **Discuss:** As an example: What words or phrases could a language model learn from looking around a city bus stop?

Have students study the image, then respond as a class with different words or phrases they see.

 **Discussion Goal:** Students will likely share words they can see in the image, such as the street names or the advertisement for sushi on the bus itself. As students respond, broaden the conversation to the categories of language they might see at a bus stop - for example: street signs, advertisements, map legends, or even signs for businesses near bus stops. This task mirrors how students will complete the activity guide - noting the source types of data they would find (ie: street signs), and then the words they might learn from that source (ie: "stop" or "yield")

Teaching Tip

Teaching Tip: Make it Local - Rather than using the provided image of a bus stop, consider finding an image of a bus stop from your own community to use instead. Or, go out and take a picture of something similar near your school and use that image instead. This is an opportunity to make the lesson more personalized and relatable to your students by using a local example.



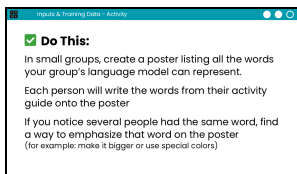
☒ **Do This:** Have students complete each of the tables individually, listing examples of data sources in each location that could be used to help a large-language model represent different words. This requires identifying objects in these locations with text - literally any object - and imagining the words a language model could learn if it scanned the object.

Since the first example is "This Classroom", you could do an example together as a class and point out any posters or books or anything with text. An easy example is the slide itself being projected, which would help a language model represent words like "Do", "Complete", "Include", etc.

It's not vital that students finish every aspect of this guide, since they will work in a group in the next section of this lesson. With that in mind, consider setting a timer to ensure students finish with enough time to move to the next part of the activity.

 **Circulate:** Monitor students as they complete this guide, noticing especially unique words or examples that can be useful to highlight in the next activity. If students are struggling, encourage them to visualize what it would look like if a computer sat in the middle of a space and scanned anything it saw - what words or symbols would it encounter?

The next part of this activity requires making a poster - during this time, also be thinking about how you will distribute supplies to groups so they are ready to start right away with the next task.



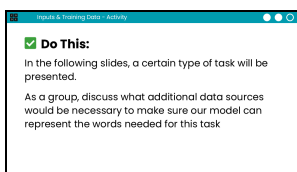
☑ **Do This:** Read the text on the slide to the class. Have students form groups of 3-4 people with a poster paper and markers. Each person should write the words from their activity guide onto the posters. They do not have to write the source they got it from. Groups should also find a way to emphasize words that are common across group members, such as writing it large or with special emphasis.

Set a short timer for this activity. When a group is finished, their poster paper essentially represents a language model trained from a variety of sources curated by each student. These can be referred to in later parts of the unit as our understanding of language models evolves.

🔄 **Circulate:** Ensure all students are able to participate and add their words to the poster. Emphasize and celebrate situations where a unique word appears on a poster that other models in the classroom likely won't have.

💡 Teaching Tip

Teaching Tip: What Do Models “Know” At This Point? - The language being used in this lesson is very intentional, saying that models can “represent” words as opposed to “know” or “understand”. Students may pick up on this and ask what the models really “know” or “understand” at this point, or you may hear students saying the model “knows what this word means” which isn’t quite correct. At this stage, all we’re doing is building a “word bank” that a model can recognize and eventually reproduce. But the model doesn’t have any inherent understanding of the meaning of these words - this happens in later steps and later lessons. If we asked a model to respond to a prompt just with the training done today, it could generate any of the words students are putting on their posters, but the results would be random and nonsensical since it doesn’t have any tools to help it make decisions about which words are best used in which situations. Nevertheless, this is a necessary step in understanding how a language model works - a model can’t even begin to “understand” a word if it’s not even included in it’s training data.

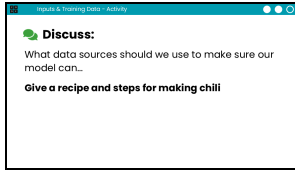


🗣️ **Say:** These are some great models we’re building - such a wide variety of words they’re able to represent just from the sources we have close to us. But let’s imagine we want to make sure we’re providing enough data so it can eventually be used for certain tasks.

☑ **Do This:** In the following slides, a certain type of task will be presented. Students will discuss in groups what additional data sources would be necessary to make sure our model can represent the words needed for this task.

✅ Assessment Opportunity

Assessment Opportunity: Formative Assessment - The following prompts can be used as a formative assessment to see how members of the class understand the relationship between data sources used as input and the types of words a language model can represent from that source. You can get a pulse-check on this understanding by listening to how students respond as a class, or having them discuss in groups first and listen in on the conversations, or have students use the blank space on the back of their activity guide to write down their answers before responding. If you notice students are struggling to answer these questions, consider doing one or two as a class to model how students should think about this relationship between training data and what a model can represent.

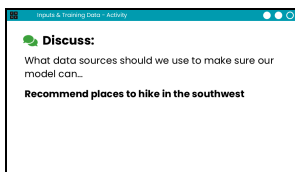


Discuss: What data sources should we use to make sure our model can... Give a recipe and steps for making chili

Discussion Goal: Possible answers can include:

- Recipe books
- Instruction manuals for kitchen appliances
- History of chili cookoffs

Optional: If any student posters seem to have any words relevant to this task, consider highlighting that poster as already having some relevant training data to this task.

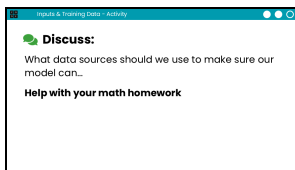


Discuss: What data sources should we use to make sure our model can... Recommend places to hike in the southwest

Discussion Goal: Possible answers can include:

- Travel books
- Maps of Arizona
- Hike Reviews

Optional: If any student posters seem to have any words relevant to this task, consider highlighting that poster as already having some relevant training data to this task.

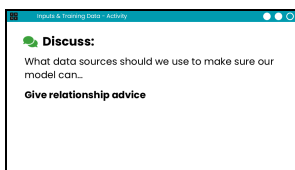


Discuss: What data sources should we use to make sure our model can... Help with math homework

Discussion Goal: Possible answers can include:

- Math textbooks
- Multiplication tables
- How to be a good tutor books

Optional: If any student posters seem to have any words relevant to this task, consider highlighting that poster as already having some relevant training data to this task.

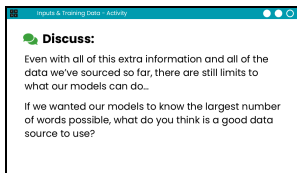


Discuss: What data sources should we use to make sure our model can... Give relationship advice

Discussion Goal: Possible answers can include:

- Teen magazines (teen vogue / cosmo)
- Self-help books
- Fashion magazines

Optional: If any student posters seem to have any words relevant to this task, consider highlighting that poster as already having some relevant training data to this task.



Discuss: Even with all of this extra information and all of the data we've sourced so far, there are still limits to what our models can do... If we wanted our models to know the largest number of words possible, what do you think is a good data source to use?

Discussion Goal: This is intended to be a quick question with a single right answer: The internet. If students give answers other than this, continue to prompt them with "Yah, but what's even bigger than that..." and work towards getting students to consider the internet as a large data source. If the conversation reaches a lull, just tell the class "The Internet".

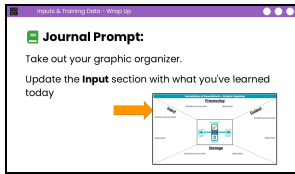
Say: The models we've started to make today are a great start and represent a good starting collection of words, but can't possibly encompass all of the data we'd need to represent the entire world and all of its complexity. To solve this, most modern large language models are trained on huge collections of information usually from the internet, such as the entirety of wikipedia. By using such large quantities of data, large-language models are able to build up huge libraries of words they recognize and represent natural language.

Teaching Tip

Teaching Tip: Looking Inside the Box - The article "[Inside the secret list of websites that make AI like ChatGPT sound smart](#)" from the Washington Post has a detailed list of how the internet and various websites have been used to train current large-language models. We don't recommend using this article for classroom use as aspects may not be appropriate, but the interactive web version of the article ([link here](#)) includes an interactive component where you can type in the name of a website and see if it's been used to train large language models (if the website is behind a paywall, consider investigating if your local library allows limited free access). This can be a useful illustration of just how much data was used to train large-language models.

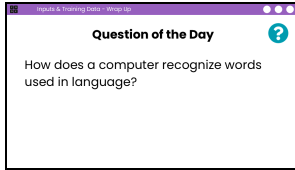
Wrap Up (5 minutes)

Slide	Lesson Guide
	Say: We've been exploring the first aspect of how a language model works - how it processes words as input. Let's review what we've seen today and see how it applies to large-language models. Video: Watch the <u>Generative AI: Input + Pretraining Video</u> Discuss: What is the pre-training stage of a language model? Discussion Goal: Students should recall what they heard in the video, but it's okay if their answers aren't 100% accurate since this will be reinforced in the next slide. Once a few students have shared, transition to the next slide to present the definition.



☑ **Do This:** Take out your graphic organizer. Update the Input section with an explanation of this stage.

🔄 **Circulate:** Check in with students as they complete their graphic organizer and summarize their thoughts. There is no "one right answer" that should go in these boxes - instead, help students synthesize what they've learned in the lesson and seen in the video into their own words. Similarly, cue students to draw an illustration or a diagram representing this process. It's okay for it to be abstract, as long as the student understand what it means to them.



Question of the Day: How does a computer recognize words used in language?



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Lesson 3: Bias in the Machine

45 minutes

Overview

This lesson formally introduces the concept of bias, where certain information is favored over others. Students experience how bias can implicitly and unintentionally creep into large-language models by participating in an activity where they are asked to source articles and information for a RecipeBot. Once they've used search engine results to generate their training data, they analyze patterns in the data that the class found as a collection, noticing unintended biases such as certain cuisines being unintentionally excluded or all the articles falling within a certain narrow date range. After acknowledging how bias can appear, students analyze several scenarios and discuss how bias could appear in the training data of each one.

Standards

Full Course Alignment

AI4K12 National Guidelines 2023

- **BI-2** - Agents maintain representations of the world and use them for reasoning

CSTA K-12 Computer Science Standards (2017)

- **IC** - Impacts of Computing

Agenda

Before the Lesson

Student Levels

Warm Up (5 minutes)

Activity (35 minutes)

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Describe patterns in collections of data sources that contribute to bias

Preparation

- Print copies of the activity guide for each student
- Run through the slide deck and videos ahead of time
- (Optional) Prepare vocabulary posters for class word wall
- (Optional) Print out the slide deck with speaker notes as a resource during class. [Click here for steps on how to do this.](#)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- [**Bias in the Machine**](#) - Slides

▼ Make a Copy

For the students

- [**Algorithmic Bias with Dr Deborah Raji**](#) - Video ([Download](#))
- [**RecipeBot Data Curation**](#) - Activity Guide ▼ Make a Copy

Vocabulary

- **Bias** - When a model favors certain information over others

Teaching Guide

Before the Lesson

Slide	Lesson Guide
	Before the Lesson: This activity uses a pedagogy rooted in discovery learning, where students first complete an activity and reflect on their process, which illuminates the underlying AI concepts. A video is provided partway through the lesson that explains and summarizes the key learning from the lesson. Concept Video: Consider reviewing the <u>Algorithmic Bias with Deborah Raji video</u>, which the class watches at the end of the lesson after the student activity. The activity aligns with the concepts introduced in the video, and watching it ahead of time can help anticipate key moments in the lesson that align with the video.

Student Levels

These levels contain the same questions as the **Warm Up** and **Wrap Up** from the lesson slide deck, allowing students to answer the questions within the Code.org platform. These can be used as a digital alternative to physical journals, and lets you take advantage of our [Student Summaries features](#) for checking and displaying student work.

By answering these questions in these levels, you can take advantage of discussion and differentiation strategies that elevate student voice to help you guide the discussion. See our [Teaching Guide for Incorporating Student Responses in Lessons](#) for more information on these strategies.



Warm Up



Wrap Up

Warm Up (5 minutes)

Slide	Lesson Guide
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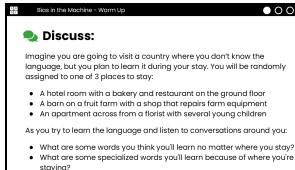
🗣 Say: Imagine you are going to visit a country where you don't know the language, but you plan to learn it during your stay. You will be randomly assigned to one of 3 places to stay:

- A hotel room with a bakery and restaurant on the ground floor
- A barn on a fruit farm with a shop that repairs farm equipment
- An apartment across from a florist with several young children

🗣 Discuss: As you try to learn the language and listen to conversations around you:

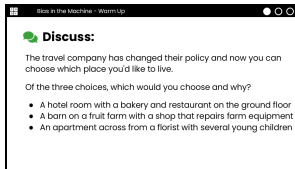
- What are some words you think you'll learn no matter where you stay?
- What are some specialized words you'll learn because of where you're staying?

💡 Discussion Goal: Students should call out how they may learn the same conversational words and phrases in all three places - things like "Hello", "goodbye", "where is the bathroom?" - but they should call out how the secondary details in these scenarios may influence the types of words and phrases they learn. For example, in the hotel room, they are more likely to learn words associated with cooking and baking, whereas in the barn they are more likely to learn mechanical and technical terms. Emphasize how these differences are primarily due to circumstance and exposure rather than intent - whether you or not you wanted to learn certain phrases, the proximity to these situations makes it somewhat inevitable.



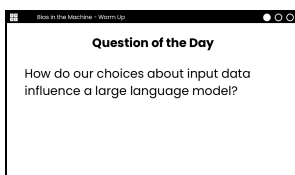
🗣 Discuss: The travel company has changed their policy and now you can choose which place you'd like to live. Of the three choices, which would you choose and why?

💡 Discussion Goal: This can be a quick discussion and there is no single right answer - let students share their thoughts and justify their responses in whatever ways make sense to them. The important thing is that there is a choice involved - we can decide which type of experience we want to have when learning a language, and that choice will influence the types of words and phrases we learn. This will be an important point to make in the following slide.



🗣 Say: This situation explores how the choices we make can affect the way we learn language purely based on the types of experiences we are exposed to. The same thing can happen with language models - during pre-training when it's looking at different sources to learn words and phrases, the types of data it is exposed to makes a difference in how it represents language. And, like in this example, we have choices about the data we use in pre-training, and those choices are important in how they influence language models.

Question of the Day: How do our choices about input data influence a large language model?

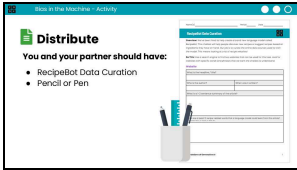
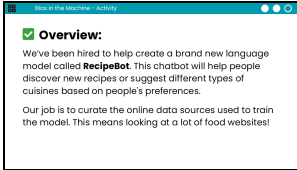
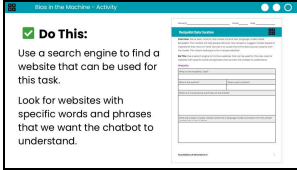


🧠 Ethics Opportunity

Ethics Opportunity: Real-World Connection - To emphasize this point, consider sharing with students the work of Michael Running Wolf, an indigenous computer scientist who is working to add Native languages like Salish, Cheyenne, or Blackfoot to voice assistants. Part of this work includes creating the data that the assistants can

use when generating their response, since that data was missing when these systems were first created. Consider [sharing this news article](#) with students or completing one of the activities from the [CS Education Week CS Heroes series](#)

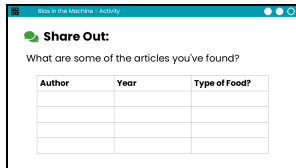
Activity (35 minutes)

Slide	Lesson Guide
	 Distribute: Pass out the RecipeBot Data Curation activity guide.
	☒ Do This: Read the overview of this document as a class. This text is also provided on the slide.
	☒ Do This: Use a search engine to find a website that can be used for this task. Look for websites with specific words and phrases that we want the chatbot to understand.  Circulate: Give students time to review the articles and respond to the prompts on the activity guide. Encourage students to avoid articles that aggregate information or with clickbait headlines - for example, "top 10 recipes" or "will these foods ruin your teeth?" - instead, encourage students to find primary sources such as actual recipes or articles on different food cultures. Students do not need to read each article in depth - only enough to get the gist at a high-level. The larger learning outcome is in the next part of the activity where students will compare similarities in the articles they've found.  Say: At this point, everyone has found an article that could be used to train RecipeBot to help provide recipes or food information. And we've done this in a similar process that actual language models do: looking at data readily available on the internet.
 Assessment Opportunity 	
Assessment Opportunity: Formative Assessment - The last question on this worksheet can be used as a formative assessment for the learnings from yesterday's lesson, where students should understand that the training data used by a language model helps it recognize words. As students scan these articles, they should be able to identify different words that a chatbot will start to recognize because it is introduced in these articles. If the class struggles with how to answer this, consider regrouping and revisiting this concept from the previous lesson.**	

Share Out: What are some of the articles you've found?

Have students volunteer and share out the article they found and the information on their sheet. As they do, quickly record some of the key information in a space where the entire class can see. Specifically, try to list the following for each article:

- The first name of the author
- The year it was written
- The type of food it focuses on (ie: American, Indian, Mexican, etc. Or, focus on: fast-food, home-cooking, fine-dining, etc)



Author	Year	Type of Food?

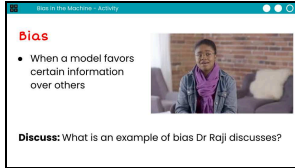
 **Discussion Goal:** At first, students are only sharing out their articles. However, as you listen to student responses, be looking for patterns and potential biases in the data that's been collected. A few patterns that are likely to emerge are:

- Most of the articles are by authors of the same gender (mostly male, or mostly female)
- Most of the articles are written within the last decade
- Most of the articles are about a specific cuisine (ie: mostly American).
- If none of these patterns emerge, consider noticing something that's missing. For example, none of the articles seem to address a specific cuisine (ie: there are no articles yet with Korean food).

Once you've identified at least one of these patterns, pause and ask the whole class to raise their hand if their articles also followed this pattern. For example: "I'm noticing most of these articles are written within the last 10 years. Can you raise your hand if your article was also written in the last 10 years?", or "I'm noticing none of these articles are about food from the Middle East. Can you raise your hand if your article was about food from this region?". Once you've called out at least one of these patterns, continue to the next slide.

Teaching Tip

Teaching Tip: Natural "Aha" Moments - A key "aha" moment in this lesson is: when looking at data sources in aggregate, it's likely certain biases will emerge unintentionally. The most likely example to occur in this lesson is a recency bias (where most of the articles are only a few years old), but this activity also lends itself to noticing gender bias (where all the authors are the same gender) and a cultural bias (where certain cultural foods are emphasized while others are excluded). In an ideal world, one of these patterns emerges quickly as students share their responses and you can move on to the next slide in a timely manner. However, it is possible that no clear biased pattern emerges even after calling on 6-8 students. In this case, it may be helpful to continue to the next slide anyway and discuss bias, but then ask the class to imagine ways bias could have manifested in this activity. This is less impactful than when students notice these patterns themselves, but cueing students this way still lets you continue with the lesson in a timely manner.



Say: These patterns in our data collection are an example of bias, which is when a model favors certain information over others. For example, the front page of most search engines tend to have a recency bias where they favor new or recent articles over older ones. Biased data means that our language model will give biased answers. Let's hear a little more about Bias from Dr Deborah Raji

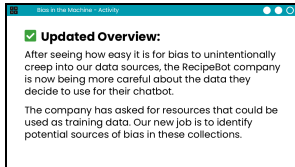
Video: Watch the [Algorithmic Bias with Dr Deborah Raji](#) video

Discuss: What is an example of Bias Dr Raji discusses?

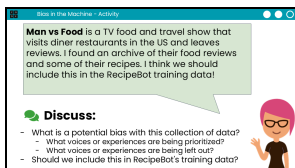
Discussion Goal: Use this as an opportunity to check-in on how well students understood the video. A key connection is how the curation and selection of data can contribute to bias, which is what has happened so far in this activity. Consider letting students respond more freely and open-endedly to the video as they reflect on the role of bias in machine learning, as well as Dr Raji's personal story about experiencing bias in a project she worked on.

Ethics Opportunity

Ethics Opportunity: Pause and Reflect - Consider showing the video [3 Types of Bias in AI by Google](#). This video was originally created for machine learning and not generative AI, but the types of bias highlighted here are also relevant in machine learning systems and may appear in this lesson. The video also aligns to the (E) Errors Occur for Many Reasons and (S) Show Solutions strategies from the [LEADERS framework](#), highlighting how humans are in the loop both in the creation of biased data and in being part of the solution. After watching the video, ask the class to reflect on whether they've noticed these kinds of patterns in other uses of technology and discuss different ways bias may be occurring in their everyday use of technology.



Do This: Read the updated overview as a class: The RecipeBot company has asked for collections of resources that could be used as training data. Our new job is to identify potential sources of bias in these collections. It's impossible to avoid bias completely. The best we can do is ensure we have a diverse set of data and identify potential biases ahead of time.

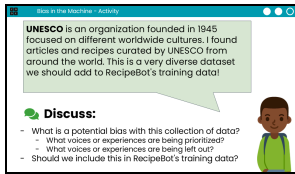


Do This: Read the quote from the student: Man vs Food is a TV food and travel show that visits diner restaurants in the US and leaves reviews. I found an archive of their food reviews and some of their recipes. I think we should include this in the RecipeBot training data!

Discuss: What is a potential bias with this collection of data? Should we include this in RecipeBot's training data?

Have students discuss in small groups before sharing as a full-class. Consider having students discuss the first question and then regroup to share out before discussing the second question.

Discussion Goal: Students should identify that US-centric restaurants are being prioritized, which could lead to a potential bias. Students may also identify that since the show is also posting reviews, the content of the articles may be overly positive and create a false impression of how "good" these recipes really are. Students may decide to include this in RecipeBot's training data, but might suggest balancing this with additional travel TV shows that visited more diverse places or including more customer-provided reviews to get balanced opinions.

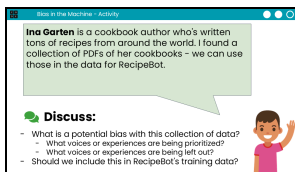


☑ **Do This:** Read the quote from the student: UNESCO is an organization founded in 1945 focused on different worldwide cultures. I found articles and recipes curated by UNESCO from around the world. This is a very diverse dataset we should add to RecipeBot's pretraining data!

💬 **Discuss:** What is a potential bias with this collection of data? Should we include this in RecipeBot's training data?

Have students discuss in small groups before sharing as a full-class. Consider having students discuss the first question and then regroup to share out before discussing the second question.

💡 **Discussion Goal:** This example is pretty ideal and it may be hard to find potential biases with this data collection without being able to look at it specifically. Students will likely agree that this is a good data source to include for RecipeBot.



☑ **Do This:** Read the quote from the student: Ina Garten is a cookbook author who's written tons of recipes from around the world. I found a collection of PDFs of her cookbooks - we can use those in the data for RecipeBot.

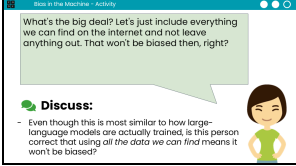
💬 **Discuss:** What is a potential bias with this collection of data? Should we include this in RecipeBot's training data?

Have students discuss in small groups before sharing as a full-class. Consider having students discuss the first question and then regroup to share out before discussing the second question.

💡 **Discussion Goal:** Students may identify the author as a potential source of bias, since all the recipes are coming from one person, and may suggest including multiple sources in the training data. However, when discussing "should we include this in the training data", push students to consider the broader ethical question of including an author's direct work in the training data for a language model, especially since we haven't asked their permission to do so. This is an opportunity to discuss issues related to Copyright and language models, since including this information in training data means the chatbot could generate these recipes verbatim without the author's consent. Once this perspective is surfaced, let students consider the implications for the author and include this in the debate on whether or not to include this data while training.

Ethics Opportunity

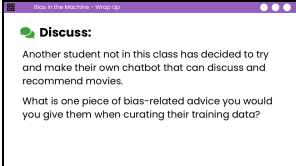
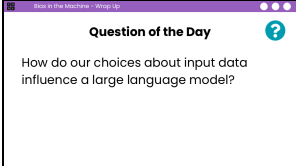
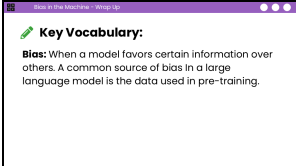
Ethics Opportunity: Real-World Connection - Empathizing with this author and worrying about their work being regurgitated by a chatbot is not a hypothetical situation - for example, an investigation showed **chatbots could generate works of poetry verbatim** without the knowledge or consent of their authors. These concerns have prompted **several lawsuits alleging that language models violated copyright laws** by ingesting this raw data without an author's consent. As you teach this unit, consider keeping an eye on the news and see if there are any new developments related to how generative AI must respect copyright laws and consider having students review and respond to these news stories, especially if they are from the perspective of the authors whose work is being used to train these systems. For example, consider having students **listen to this podcast episode that includes author interviews** and have them respond with how they think the situation should be addressed.

	☑ Do This: Read the quote from the student: What's the big deal? Let's just include everything we can find on the internet and not leave anything out. That won't be biased then, right? 💬 Discuss: Even though this is most similar to how large-language models are actually trained, is this person correct that using all the data we can find means it won't be biased? Have students discuss in small groups before sharing as a full-class. Consider having students discuss the first question and then regroup to share out before discussing the second question. 💡 Discussion Goal: Students should identify two potential issues with this: (1) the majority voices from the data may outnumber and overwhelm smaller voices, and (2) there's the potential for incorrect or misinformation to be included if we're not careful to curate this data. For example, if it was possible to include all recipe-related data, the majority of that data would likely be English-speaking and focusing on Western culinary traditions, and foods and recipes from minority groups - like Indigenous recipes - might be harder to identify amongst all the other recipes. For (2), it's possible that some data sources will include outdated information that could be harmful - for example, recipes that include ingredients that were later discovered to be unsafe or unhealthy.
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💡 Teaching Tip

Teaching Tip: Real-World Connection - Being concerned about AI systems making nonsensical or biased recommendations is not a hypothetical situation - for example, when Google released their Gemini-assisted search summary feature, it would **sometimes include nonsensical or incorrect recommendations such as eating rocks or putting glue on pizza**. This was caused by the language model relying on satirical articles or irreverent posts on user forums to generate its response, which were not reliable sources. Consider surfacing this example to students as an example of how ingesting all data could lead to unintended consequences.

Wrap Up (5 minutes)

Slide	Lesson Guide
	💬 Discuss: Another student not in this class has decided to try and make their own chatbot that can discuss and recommend movies. What is one piece of bias-related advice you would give them when curating their training data?
	Question of the Day: How do our choices about input data influence a large language model?
	📖 Vocabulary: The key vocabulary from this lesson is provided on this slide and in the lesson plan.



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Lesson 4: Understanding Embeddings

45 minutes

Overview

In this lesson, students explore how language models understand and represent word meanings. Beginning with a game, they investigate the concept of words being "close" in meaning. This leads to an activity where students assume the roles of fictional characters, using a sequence of numbers to represent their "character traits" and comparing these numerical representations to others. The lesson demonstrates that language models use a similar method, known as embeddings, to represent words as sequences of numbers, with similar words being numerically close to each other. The lesson concludes with students updating the storage section of their graphic organizer to encapsulate the idea of word embeddings.

Standards

Full Course Alignment

AI4K12 National Guidelines 2023

- ▶ **BI-2** - Agents maintain representations of the world and use them for reasoning
- ▶ **BI-4** - Intelligent agents require many kinds of knowledge to interact naturally with humans

CSTA K-12 Computer Science Standards (2017)

- ▶ **CS** - Computing Systems

Agenda

Before the Lesson

Warm Up (5 minutes)

Warm Up Option 1

Warm Up Option 2

Activity (35 minutes)

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Describe how embeddings show the similarity or difference between two items
- Explain how language models use embeddings to store the meaning of words

Preparation

- Have markers, crayons, or colored pencils prepared for students
- Print copies of the activity guide for each student
- Print copies of the activity instructions and cut them into quarter-sheets
- Run through the slide deck and videos ahead of time
- (Optional) Prepare vocabulary posters for class word wall
- (Optional) Print out the slide deck with speaker notes as a resource during class. [Click here for steps on how to do this.](#)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- [Lesson Instructions](#) - Resource
▼ Make a Copy
- [Understanding Embeddings](#) - Slides
▼ Make a Copy

For the students

- [Character Quiz](#) - Activity Guide

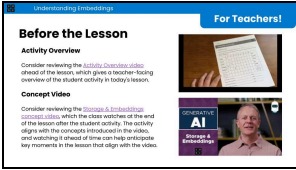
▼ Make a Copy
- [Generative AI: Storage & Embeddings](#) - Video ([Download](#))

Vocabulary

- **Embedding** - A set of numbers used to represent a word in a language model. They are used to represent the discrete meaning of words.

Teaching Guide

Before the Lesson

Slide	Lesson Guide
	Before the Lesson: This activity uses a pedagogy rooted in discovery learning, where students first complete an activity and reflect on their process, which illuminates the underlying AI concepts. A video is provided at the end of the lesson that explains and summarizes the key learning from the lesson. Activity Overview: Consider reviewing the Activity Overview video ahead of the lesson, which gives a teacher-facing overview of the student activity in today's lesson. Concept Video: Consider reviewing the Storage & Embeddings concept video, which the class watches at the end of the lesson after the student activity. The activity aligns with the concepts introduced in the video, and watching it ahead of time can help anticipate key moments in the lesson that align with the video.

Warm Up (5 minutes)

Warm Up Option 1

There are two options for warm-ups in this lesson. This first option is unplugged and involves students playing a game, requiring movement and lots of talking. The second activity involves playing a game from a website, with the option to have students play the game as well. Consider factors such as your classroom layout, access to devices, and whether or not you can access the website in the second warm-up when deciding which warm-up to choose.

If you choose this warm-up, skip or delete the slides associated with Warm Up 2

Slide	Lesson Guide
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☒ **Do This:** Have the class read the rules of this game. The next slide includes these same instructions as a visual - consider asking a student volunteer to help demonstrate the game before allowing students to play.

Teaching Tip: Printing Handouts - This Instructions Student Resource includes the same instructions on this slide so students can reference them during the activity and to help with keeping track of their steps in the activity. The resource is intended to be printed double-sided and cut into fourths, since it contains instructions for both this warm-up activity and the in-class activity later.

☒ **Do This:** These are the same instructions from the last slide, but in visual form. Consider modeling this process with students. Be sure to say explicitly that it's very unlikely students will say the same word the first time, but that's part of the point - from this starting position, they can start to plan their next words until they "converge" on a single word. After demonstrating, have students pair up and play the game and change partners a few times.

🔄 Circulate: Encourage to play this game a few times. The classroom will likely look slightly chaotic, with a lot of talking and (hopefully) a lot of laughing. After a few minutes, regroup the class.

(Optional) Share Out: Consider asking students to share out some of the word-sequences they went through with their partner (ie: which words they said first, then next and next and next until saying the same word). However, only do this if you have an extended class period - otherwise, continue with the lesson so students have time to complete the activity.

🎙️ **Say:** This exercise introduces this idea of “closeness” or “similarity” between words. At its core, it’s based on the idea that when you have an input of a single word...

 Click the animation

Say: you're able to access additional information about that word in your head...

 Click the animation

🗣️ **Say:** which then lets you think of similar words related to the one you started with...

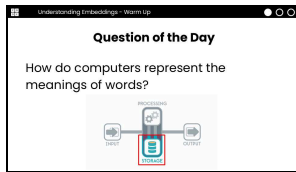
🗣️ Say: It turns out this is our next step in understanding how language models work. We learned in the last few lessons how language models use training data to recognize individual words.

 Click the animation

🎙 Say: And the next step is to learn how models can group these words in a logical way.

 Teaching Tip

Teaching Tip: NYT Connections Game - The screenshot on this slide is from the **New York Times Connections Game**, which is another fun game that is related to the topic of today's lesson. To play the game, students are given 16 words and must group them into 4 groups with a similar theme. You may not have time to introduce this game in this lesson, but since the game resets daily, you could consider making it an additional part of the warm-up for the remainder of the unit or something for students to do before the bell rings when they enter class.



Say: Today, we'll learn about the special representation it uses for these words. This is what allows models to approximate the meaning of words and determine how "close" or similar words are to each other, just like we did in this warm-up.

Question of the Day: How do computers represent the meanings of words?

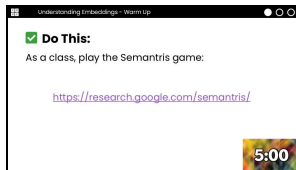
Warm Up Option 2

There are two options for warm-ups in this lesson. This first option is unplugged and involves students playing a game, requiring movement and lots of talking. The second activity involves playing a game from a website, with the option to have students play the game as well. Consider factors such as your classroom layout, access to devices, and whether or not you can access the website in the second warm-up when deciding which warm-up to choose.

If you choose this warm-up, skip or delete the previous slides associated with Warm Up 1

1

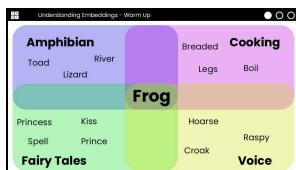
Warm Up



Before Class: Have the **Semantris game** loaded on your computer. Consider having it displayed as students walk in instead of this slide.

Do This: Play the Semantris Arcade game as a class, with you driving the game while students shout out words to type in. After a few rounds, continue on with the lesson. A timer is supplied on the slide to help with time management.

(Optional): If your class has easy access to devices, have students play this game on their own before continuing.



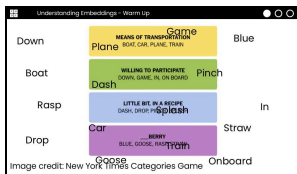
Say: This exercise introduces this idea of "closeness" or "similarity" between words. At its core, it's based on the idea that when you have an input of a single word...

Click the animation

Say: you're able to access additional information about that word in your head...

Click the animation

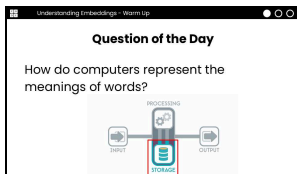
Say: which then lets you think of similar words related to the one you started with...



Say: It turns out this is our next step in understanding how language models work. We learned in the last few lessons how language models use training data to recognize individual words.

Click the animation

Say: Today, we'll learn about the special representation it uses for these words. This is what allows models to approximate the meaning of words and determine how "close" or similar words are to each other, just like we did in this warm-up.



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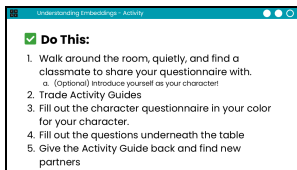
Question of the Day: How do computers represent the meanings of words?

Activity (35 minutes)

Slide	Lesson Guide
	Say: To help understand how words can be similar, we're going to investigate a similar idea but with characters from stories or movies. We'll be completing a "personality quiz" as different characters, and then comparing them to see which characters are the most similar or most different. Distribute: Pass out a copy of the Character Quiz Activity Guide to each student. Also give each student a marker or crayon - ideally, each student has a different color. They will use these tools to draw dots on the appropriate part of the scale from 0-10 on their activity guides. If you use colored pencils, make sure students make large, dark marks - this will be important for later in the activity.
	Do This: Provide directions for students: - Choose a character you are familiar with to fill out this form. Write this person's name at the top of the activity guide. - Use the colored pencil to place a mark on the scale to represent how your character aligns with each trait. - Come up with two additional traits and add them to the bottom of the table. Circulate: Students can also choose characters from television cartoons, or historical figures. If you know a book they are reading in ELA, you may suggest those characters. If students are having a hard time picking a character, encourage them to just fill out the character questionnaire for themselves.

Teaching Tip

Teaching Tip: Personality Traits - These traits are not all necessarily meant to align with "positive" / "negative" behaviors - just traits on a spectrum. For instance, one can stubbornly stick to something, or one can quit easily - these are different ways of reacting to problems but they're not necessarily inherently good or bad. Similarly, being stubborn can sometimes be a negative trait and make people hard headed, or it can be a positive trait that makes people persistent and not quit. These considerations can help students who are stuck on deciding their additional traits for their sheet.



🗣 Say: So far, we've created a representation of our character based on their different traits. Now it's time to compare our characters to others in the class. We'll talk to at least 3 people to see how their characters compare with ours, numerically.

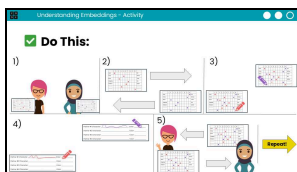
☑ Do This: Read through the instructions. Students will:

- Walk around the room to find a classmate to share with
- Trade activity guides
- Fill out the other person's questionnaire as your character
- Fill in the additional questions under the table to help with tracking
- Trade activity guides back

The next slide has these same instructions as a visual.

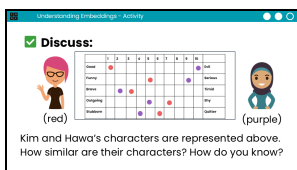
💡 Teaching Tip

Teaching Tip: Printing Handouts - This **Instructions Student Resource** includes the same instructions on this slide so students can reference them during the activity and to help with keeping track of their steps in the activity. If students aren't participating in the first warm-up activity, consider just printing the instructions for this activity and cutting the sheets into fourths.



☑ Do This: If possible, demonstrate this process with a volunteer from the class. Otherwise, reiterate this process for students. Once students understand, they can begin finding partners.

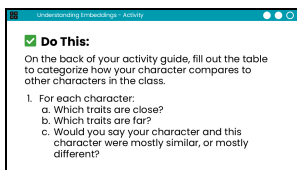
🔄 Circulate: Make sure students are finding people to match up with. Students should try to find at least 3 different partners to compare with before regrouping. This stage of the activity may feel very straightforward, as students are essentially gathering data for the next part of the activity. Students may naturally start commenting on which characters are similar or different based on the data - if you hear this, ask students who they know and listen for explanations that use the "closeness" or "farness" of dots for each trait to justify their answers.



💬 Discuss: Kim and Hawa's characters are represented on the slide. How similar are their characters? How do you know?

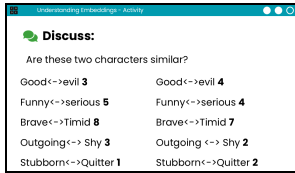
Have students discuss in small groups before having a full-class discussion

💡 Discussion Goal: Students should be comparing the two characters row-by-row, noticing when dots are close together or far apart. Even though this isn't explicit in the question, be sure to emphasize the specific traits that are close or far - for example, these two characters are on opposite ends of the good/bad spectrum, but are close in several other areas. Emphasize responses that are explicitly counting spaces between dots and comparing distances.



☑ Do This: Have students flip to the back of the activity guide. They will do a similar activity they was just completed as a class: decide how similar their character was to their classmates' character based on the marks on the activity guide, and write their answers down.

🔄 Circulate: Make sure students are on-task. If they need help deciding how similar characters are, you can encourage them to just count the distance between their marks and their classmates marks



Say: I'm changing up the format of the quiz - instead of using dots on a spectrum, now we just have raw numbers.

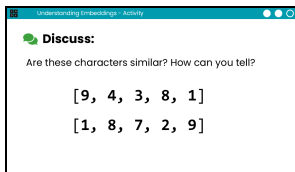
Discuss: Are these two characters similar?

Have students respond with thumbs up/ thumbs down. It may take a while for students t

Discussion Goal: Scan the class for student answers. Ask one student with their thumb up why they think these characters are similar. Students will probably think these numbers are all similar, even though none of them are the same. This means the characters are similar.

✓ Assessment Opportunity ▲

Assessment Opportunity: This slide is meant to be a quick check for understanding for the entire class – hopefully after the activity students will recognize these characters as similar because they have similar numbers associated with individual traits, which would lead to most of the class having their thumbs up. If you see a mix of thumbs up/down, check that students are comparing across the same set of traits, and ensure students that “similar” doesn’t mean “identical” - it’s okay for numbers to be close but not exact.**

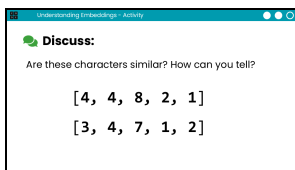


Say: That last slide looked a little complicated trying to represent each trait. On this slide, we've condensed that to just a list of numbers, where each number represents a different trait from your character questionnaire.

Discuss: Are these characters similar? How can you tell?

Have students chat with a neighbor first, then repeat the thumbs up/down. Call on someone with their thumbs down to share their response.

Discussion Goal: Students should leave this discussion understanding that they can compare numbers in each position to determine how similar or different characters are. This can be done by comparing each number from left-to-right. Students may also get caught up on the lack of labels, asking “what does each position represent?”. This can lead to a powerful insight: it doesn’t matter what each position represents, only how close or far the numbers are in that same position. This is a key concept that helps us generalize the idea of “character traits” to other uses, which we’ll see in a few slides.



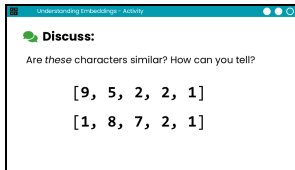
Discuss: Are these characters similar? How can you tell?

Scan the room to see that students still have their thumbs up.

Discussion Goal: Students should still put their thumbs up for these characters being similar – we’ve gotten rid of the human-readable words, but left the computer-readable numbers. Humans can still look at these numbers and realize that they’re similar.

💡 Teaching Tip ▲

Teaching Tip: Differentiating Responses - If your students are feeling antsy, you may wish to have them get out of their seat and do a spectrum activity with these next several thumbs up/down slides, or possibly stand up if they think the words are similar rather than the thumbs up to at least get them out of their chairs.

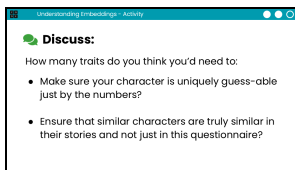


Discuss: Are these characters similar? How can you tell?

Scan the room for who has thumbs up or down; some may have their thumb halfway between; allow that.

Discussion Goal: Unlike the previous several questions, this question does not necessarily have one easy answer – while the first 3 numbers are far from each other the last two are exact. Call on a student with their thumb down, who will probably note that the first three numbers are very far apart, and validate that answer. Then call on someone with their thumb up, who will probably note that the last two are exact. Validate that answer as well.

Say: Characters can be similar in different ways, that may be more or less important in a given situation – it's context dependent. For example, if two characters have the same sense of humor, they may behave similarly in a comedy scene. But if those same characters happen to be the hero and the villain, those differences will appear in other ways in different types of situations.



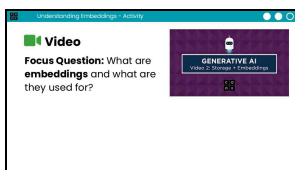
Say: There were only 5 traits we looked at for these characters. This means it's possible that even though two personality quizzes have very similar traits, they're actually different quizzes. They could even be very different from each other, but just happen to be similar with these specific traits.

Discuss: How many traits do you think you'd need to:

- Make sure your character is uniquely guess-able just by the numbers?
- Ensure that similar characters are truly similar in their stories and not just in this questionnaire?

Discussion Goal: Give some thought time, then take hands. There isn't a specific number of traits these questions looking for, merely that the more traits you are keeping track of the more accurately you can describe the character – this should answer both questions.

Say: This is why it's important when describing concepts as numbers, to use a lot of numbers. Fewer numbers are less accurate.

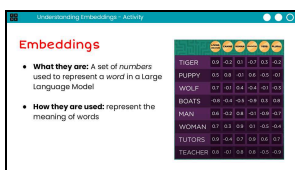


Say: We've been talking about personality traits for different characters and using a list of numbers to represent those characters. It turns out we can do the same thing with words - using a list of numbers to represent the meaning of words! We're going to learn more about that in this video

Video: Watch the [Generative AI: Storage + Embeddings](#)

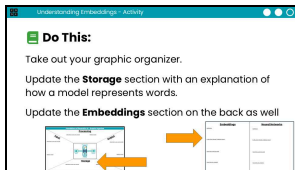
Discuss: What are embeddings and what are they used for?

Discussion Goal: Students should recall what they heard in the video, but it's okay if their answers aren't 100% accurate since this will be reinforced in the next slide. Once a few students have shared, transition to the next slide to present the definition.



Say: Embeddings are the set of numbers used to represent a word in a large-language model. These numbers, when taken altogether and compared with other words, can be used to represent the meaning of words.

Teaching Tip: Embeddings Demo - This lesson has emphasized embeddings as a list of numbers and comparing these lists to figure out similarity between words. Another common way to make this comparison is visually, by graphing certain dimensions of words in a 2D or 3D space. If you'd like to explore this further, we recommend the **Word Embedding Demo from Carnegie Mellon University**. When the page loads, it provides a visualization of several words mapped onto a 3D space. They have a **tutorial available** that explains more about how this works. This can be an option to show students if they are interested, but graphing words on 2D or 3D spaces is not something students are expected to do in this course so we have avoided it in the lesson and instead present it as an option.



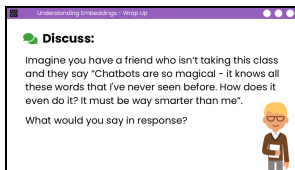
Do This: Take out your graphic organizer. Update the Storage section with an explanation of how a model represents words. Update the Embeddings section on the back as well

Circulate: Check in with students as they complete their graphic organizer and summarize their thoughts. It may be helpful to return to the previous slide so students can see the definition of Embeddings. There is no "one right answer" that should go in these boxes - instead, help students synthesize what they've learned in the lesson and seen in the video into their own words. Similarly, cue students to draw an illustration or a diagram representing this process. It's okay for it to be abstract, as long as the student understand what it means to them.

Wrap Up (5 minutes)

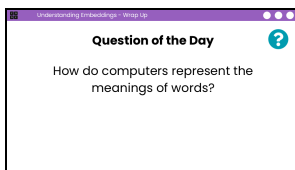
2

Wrap Up

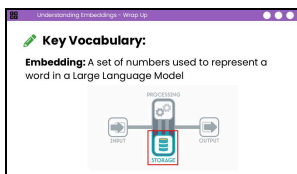


Discuss: Imagine you have a friend who isn't taking this class and they say "Chatbots are so magical - they know all these words that I've never seen before. How do they even do it? They must be way smarter than me". What would you say in response?

Discussion Goal: This discussion is an opportunity to de-mystify how generative AI works. Students might try to clarify that the reason a chatbot knows so many words is because of all the pre-training it's been through. And the way it "learns" these words isn't really magical at all - it's just a bunch of numbers. Also encourage students to lead with empathy and reassure this person that even if they think the chatbot "must be way smarter than me", that's not necessarily true and once you understand how it works, it's just another piece of technology.



Question of the Day: How do computers represent the meanings of words?



 **Vocabulary:** The key vocabulary from this lesson is provided on this slide and in the lesson plan.



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Lesson 5: Embeddings: How They're Created

45 minutes

Overview

In this lesson, students delve into the creation of word embeddings. They start by interpreting a passage in a made-up language, using context clues to understand word meanings based on their relationships with other words. Following this, they engage in an activity to construct embeddings from sentences by noting word proximity. The climax reveals that students worked with two distinct passages, illustrating how the same word can have different representations based on its usage context. This insight sets the stage for upcoming lessons on how language models use sentence structure and context to determine word meanings accurately.

Standards

Full Course Alignment

AI4K12 National Guidelines 2021

- **BI-3** - Computers can learn from data

AI4K12 National Guidelines 2023

- **BI-2** - Agents maintain representations of the world and use them for reasoning
- **BI-4** - Intelligent agents require many kinds of knowledge to interact naturally with humans

CSTA K-12 Computer Science Standards (2017)

- **AP** - Algorithms & Programming

Agenda

Before the Lesson

Student Levels

Warm Up (5 minutes)

Activity (35 minutes)

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Construct an embedding for a text passage
- Explain how different passages can create different embeddings

Preparation

- Print copies of **both versions** of the activity guide for each student. Ensure some students get Version A while others get Version B.
- Run through the slide deck and videos ahead of time
- (Optional) Print out the slide deck with speaker notes as a resource during class. [Click here for steps on how to do this.](#)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

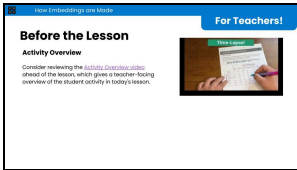
- **How Embeddings are Made** - Slides
▼ Make a Copy

For the students

- **How Embeddings are Made (Version A)** - Activity Guide ▼ Make a Copy
- **How Embeddings are Made (Version B)** - Activity Guide ▼ Make a Copy
- **Warm Up Cutouts** - Resource
▼ Make a Copy

Teaching Guide

Before the Lesson

Slide	Lesson Guide
	Before the Lesson: This activity uses a pedagogy rooted in discovery learning, where students first complete an activity and reflect on their process, which illuminates the underlying AI concepts. Activity Overview: Consider reviewing the Activity Overview video ahead of the lesson, which gives a teacher-facing overview of the student activity in today's lesson.

Student Levels

These levels contain the same questions as the **Warm Up** and **Wrap Up** from the lesson slide deck, allowing students to answer the questions within the Code.org platform. These can be used as a digital alternative to physical journals, and lets you take advantage of our [Student Summaries features](#) for checking and displaying student work.

By answering these questions in these levels, you can take advantage of discussion and differentiation strategies that elevate student voice to help you guide the discussion. See our [Teaching Guide for Incorporating Student Responses in Lessons](#) for more information on these strategies.



Warm Up



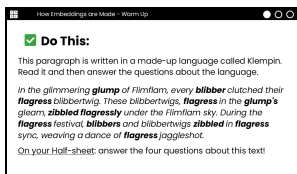
Wrap Up

Warm Up (5 minutes)

Prepare: There are two versions of the How Embeddings are Made activity guide. The first page of both versions are identical, but the second page contains different passages. Try to distribute different versions of the guide without students noticing they are different. You can accomplish this by intermixing the two versions, or purposefully giving one group Version A and another group Version B. Don't call out that the guides are different - this will be revealed later as part of the lesson.

- [How Embeddings are Made \(Version A\)](#)
- [How Embeddings are Made \(Version B\)](#)

Slide	Lesson Guide
	 Distribute: Make sure each student has a copy of the Warm Up half-sheet. Students need a writing utensil as well.



This paragraph is written in a made-up language called Klempin. Read it and then answer the questions about the language.

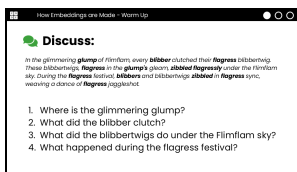
In the glimmering glump of Flimflam, every blibber clutched their flagress blibbertwig. These blibbertwigs, flagress in the glump's gleam, zibbled flagressly under the Flimflam sky. During the flagress festival, blibbers and blibbertwigs zibbled in flagress sync, weaving a dance of flagress jaggleshot.

☒ **Do This:** Read the instructions and paragraph from the half-sheet and available on the slides. Then direct students to their half-sheet to answer four questions about the text:

- 1) Where is the glimmering glump?
- 2) What did the blibber clutch?
- 3) What did the blibbertwigs do under the Flimflam sky?
- 4) What happened during the flagress festival?

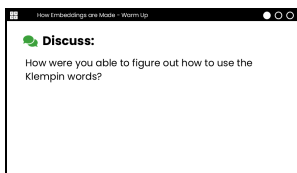
Teaching Tip

Teaching Tip: Lean Into the Absurdity - Consider ways you can engage students in engaging with this text in a playful way, such as having them read the text aloud or a version of a call-and-response where you read the paragraph and students shout-out the bolded silly words. It can also be fun to pretend to take the text and questions very seriously, for example by pretending that “this passage will be part of a quiz later”. In general, leaning into the absurdity in a playful way can help keep the task engaging and make it easier to shift to the debrief later that leads into the rest of the lesson.



 **Discuss:** Ask students to answer the questions. Don't ask any follow-up questions yet about “how do you know?” - wait for the next discussion to expand on how students were able to answer these questions

- 1) Where is the glimmering glump? Flimflam
- 2) What did the blibber clutch? Flagress blibbertwig
- 3) What did the blibbertwigs do under the Flimflam sky? Zibbled flagressly
- 4) What happened during the flagress festival? Blibbers and blibbertwigs zibbled in flagress sync



 **Discuss:** How were you able to figure out the Klempin words?

 **Discussion Goal:** Guide students to notice how the other words in the sentence helped them respond to the questions. This is sometimes called using “context clues”, where the context of the sentence helps you learn new words. Some students may try to say they “understand” or “know” these new words - if this comes up, consider asking some leading questions about whether or not this is really true. Just because we could answer some very direct questions about the text, does that mean we truly “understand” it? Digging into this may reveal that regurgitating responses from the text doesn't always mean we know what we're talking about. You can also relate this to student experiences where they may have earned a high-score on a test, but never really feel like they “learned” the material - a similar phenomenon has happened in the warm up.

 **Say:** Even without really understanding any of the words in this paragraph, we were able to figure out how to use the words from context and answer some questions. This is what a Large Language Model does as well - they figure out how to use words from the context around them rather than somebody telling the model explicitly what the word means.

Question of the Day
How do language models learn the embeddings they use to represent words?

Vocabulary
Embedding - A set of numbers used to represent a word in a language model.

Say: Last class we explored how computers represent words as a group of numbers, called embeddings, and can use those numbers to represent the “meaning” of words. In this class, we will explore how computers create those embeddings in the first place.

Question of the Day: How do language models learn the embeddings they use to represent words?

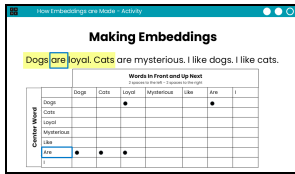
Activity (35 minutes)

Slide	Lesson Guide
	Say: Today we’ll explore how language models create their own embeddings represent the meaning of words. We’re going to start with a simple sentence: “Dogs are loyal. Cats are mysterious. I like dogs. I like cats.”. Our goal is to create embeddings in the same way language models would, starting from nothing and building a simple representation of each word’s meaning.

Teaching Tip

Teaching Tip: Simplicity - This sentence may seem very simple, which is partially due to the lesson needing a simple sentence as a first example. However, sentences like this are also similar in structure and complexity to popular children’s books, which serve a similar purpose to what we’re doing here: helping children learn new words and define their meaning. This is a useful comparison to imagine the type of text language models are able to represent at this point - understanding simple children’s books before advancing to more complex sentences and structures.

	Say: To make our embeddings, we want to track of how often a word appears within a few words of other words. We’ll keep track of this in a table that contains all the words in these sentences, and we’ll add dots whenever words are close together. Eventually, each row of the table will become our word embedding representing the meaning of each word.
	Say: In the warm-up, we used “context clues” to answer the questions in the Klepin language. Thinking about this in a more step-by-step fashion, that means we looked at words that were “close” to other words in the sentence. Let’s try that here too with the first word in the sentence, Dogs. This word doesn’t have 2 words before, so we’ll just count the 2 words after. Click the animation Say: Next I’ll go to the row in my table with that word Click the animation Say: And then I’ll mark a dot for each word close to it within two words.



Say: Then we start over, but not we look at the next word in the sentence. This is our "Center Word".

Click the animation

Say: Then look at the two words in front, and the two words up next. We're going to ignore the punctuation and just focus on the words.

Click the animation

Say: Then for each word, we put a dot in the Center Word row to indicate they're close together.

Click the animation



Say: Let's skip ahead and try an example in the middle of the sentence. Let's look at the word Mysterious.

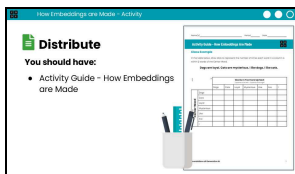
Do This: Ask the class to assist you with the next step in the process. Consider this set of questions to help assist with this part of the process:

- Which row should we look at? Mysterious
- Which columns are we looking at? "Cats" and "are" from the left of the word, and "I" and "like" to the right of the word.
- Once we have these words, what do we do in the table? Go to the "Mysterious" row and put dots for each of these words.

Once students have answered, click the animation to show the answer.

Click the animation

Say: This example shows how embeddings start to work - this row shows that "Mysterious" is close to "Cats" but not "Dogs", which is also what the sentence tells us.

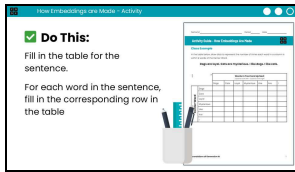


Distribute: Pass out a copy of the How Embeddings are Made activity guide. Make sure students secretly have both A and B versions of the guide.

Say: We're going to finish this example, then practice with one more. After these two examples, we'll reflect on how well this process approximates the "meaning" of words for a language model.

Teaching Tip

Teaching Tip: Transparent Sheets & Dry Erase Markers - Consider putting each activity guide in a transparent sheet protector (or laminating each activity guide). This allows students to use dry erase markers on the worksheet. This can help with circling the "center word" and highlighting the words in front or up next, then quickly erasing that scratch work and moving on to the next center word.



Do This: Fill in the table for the sentence. For each word in the sentence, fill in the corresponding row in the table

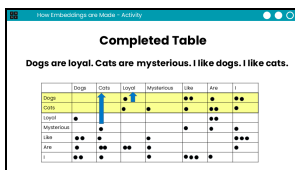
Circulate: Monitor students as they work. Consider letting students pair for this first example to help cement their understanding of this process. Students may struggle keeping track of which word they're currently looking at - in these cases, you can suggest students underline words or use a highlighter to show how far in the sentence they've moved.

As students progress and get about halfway through, show the next slide as a way for students to check their progress.



Say: We've been working for a while - let's pause and check in. Here's what your table should look like when you've processed the first half of the phrase "Dogs are loyal. Cats are". Your table may not look exactly like this, but it should be similar. If yours looks completely different, let's chat!

Circulate: Continue to check-in with students as they complete their table



Say: Here's our completed table. Let's check and see how we did.

Give students a chance to check their work before continuing to the next slide.

Say: Let's remind ourselves what we've done: each row in this table is an example of an embedding that language models create to represent the meaning of words and how they are related.

Click the animation

Say: For example, if we wanted to know "Which pets are mysterious?", the table already gives us some signal that cats are mysterious.

Click the animation

Say: Or if we wanted to know "how would you describe a dog?", we already have some signal that dogs are loyal

Click the animation

Say: We can also already start to see which words are "similar". For example, since "cats" and "dogs" are both animals and are used in similar ways in sentences, their embeddings will start to look similar too. This means these words are "close" to each other in how a language model represents language.

Do This:

- Flip to the back page of the activity guide and try a new example.
- Use 3 words in front and up next instead of 2!
- Only focus on the **bolded words** - ignore the others.

	bat	team	the	base	count	the
bat						
team						
the						
base						
count						
the						

☑ **Do This:** Have students flip to the back of their activity guide.

🗣 **Say:** Let's try another example. In this case, our "closeness" window is expanding and we'll look at words that are within 3 of our center word. We're also not looking at every word of the paragraph, just a few specific words which have been bolded.

🔄 **Circulate:** Make sure students are working on the correct paragraph. Help students count words if they are struggling to do so; it may help to encourage them to draw in the bounds of which words should be counted. It may help to have a few different colored pencils to draw boxes with.

💡 Teaching Tip

Teaching Tip: Different Passages - Be sure to have distributed different versions of the activity guide, but don't tell students. If this starts to come up or students discover this naturally, feel free to start regrouping the class early to have a discussion about the discrepancy.

Discuss:

- What word do we know the most about?
- What word do we know the least about?
- In the embedding table, what words are "Bat" most related to? Could you construct a simple definition of "bat" using just the embedding table?

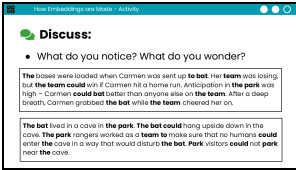
🗣 **Discuss:** These questions are available on the activity guide as well:

- What word do we know the most about? Explain how we know using the embedding table.
- What word do we know the least about? Explain how we know using the embedding table.
- In the embedding table, what words are "Bat" most related to? Could you construct a simple definition of "bat" using just the embedding table?

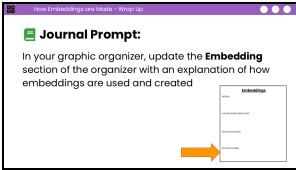
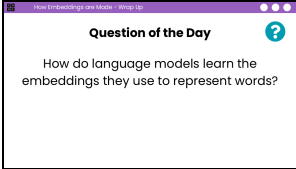
💡 **Discussion Goal:** These answers change depending on the passage students were given, which will come up in this group exercise. In Passage 1, students will know the most about "team" and the least amount "park", where Passage 2 will be the opposite. In Passage 1, "bat" is associated with the sports equipment baseball bat; in Passage 2, "bat" is associated with the flying animal. In the discussion, students may initially be surprised by these answers and feel like they "did something wrong" - if this happens, move quickly to the next slide to reveal that there are two different passages.

✅ Assessment Opportunity

Assessment Opportunity - Formative Assessment: - Evaluating how students respond to these questions and apply their knowledge of embeddings can act as a formative assessment.

	Discuss: What do you notice? What do you wonder? Discussion Goal: Students should notice that the words contain the same bolded words, but the subject of the passages are different. Specifically, the word “bat” is being used differently in each passage. As students notice this, point out that this also manifests in the embeddings that were created - one set was capturing bat as an animal while the other was capturing bat as the sporting equipment. Say: This exercise highlights why it’s important to have lots and lots of data when making language models - in order to capture the full meaning of words, especially words with multiple meanings, a language model needs to see data that uses those words in the appropriate context. However - this also highlights a new question: how does a language model know which meaning of the word to use? This is something we’ll learn about in the next few lessons!
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Wrap Up (5 minutes)

Slide	Lesson Guide
	Discuss: In your graphic organizer, update the Embedding section of the organizer with an explanation of how embeddings are used and created
	Question of the Day: How do language models learn the embeddings they use to represent words?



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Lesson 6: Understanding Neural Networks

45 minutes

Overview

This lesson introduces neural networks through a hands-on activity where students create a lunch recommender for a school cafeteria. It starts with an exercise involving two mad-libs that use the same words in different orders, highlighting how word relations and grammar influence word choice. Students then work together to build a neural network model for the cafeteria's lunch choices, learning about key concepts like hidden layers and weights. The lesson concludes by tying these ideas back to language, prompting students to think about how words can change meaning in different contexts or phrases.

Standards

Full Course Alignment

AI4K12 National Guidelines 2021

- **BI-3** - Computers can learn from data

AI4K12 National Guidelines 2023

- **BI-2** - Agents maintain representations of the world and use them for reasoning

CSTA K-12 Computer Science Standards (2017)

- **CS** - Computing Systems
- **IC** - Impacts of Computing

Agenda

Before the Lesson

Student Levels

Warm Up (5 minutes)

Activity (35 minutes)

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Construct a neural network to represent a situation and make decisions based on inputs

Preparation

- Print copies of the activity guide for each student
- Run through the slide deck and videos ahead of time
- (Optional) Prepare vocabulary posters for class word wall
- (Optional) Print out the slide deck with speaker notes as a resource during class. [Click here for steps on how to do this.](#)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- [Understanding Neural Networks](#) - Slides [▼ Make a Copy](#)

For the students

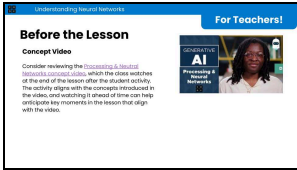
- [Generative AI: Processing & Neural Networks](#) - Video ([Download](#))
- [Neural Networks](#) - Activity Guide [▼ Make a Copy](#)

Vocabulary

- **Neural Network** - An interconnected network that makes decisions using weights and hidden layers. They are

Teaching Guide

Before the Lesson

Slide	Lesson Guide
	Before the Lesson: This activity uses a pedagogy rooted in discovery learning, where students first complete an activity and reflect on their process, which illuminates the underlying AI concepts. A video is provided at the end of the lesson that explains and summarizes the key learning from the lesson. Concept Video: Consider reviewing the Processing & Neural Networks concept video, which the class watches at the end of the lesson after the student activity. The activity aligns with the concepts introduced in the video, and watching it ahead of time can help anticipate key moments in the lesson that align with the video.

Student Levels

These levels contain the same questions as the **Warm Up** and **Wrap Up** from the lesson slide deck, allowing students to answer the questions within the Code.org platform. These can be used as a digital alternative to physical journals, and lets you take advantage of our **Student Summaries features** for checking and displaying student work.

By answering these questions in these levels, you can take advantage of discussion and differentiation strategies that elevate student voice to help you guide the discussion. See our **Teaching Guide for Incorporating Student Responses in Lessons** for more information on these strategies.

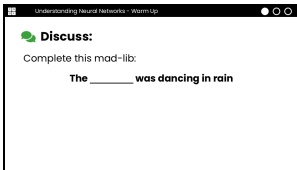


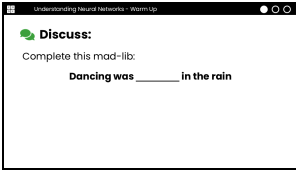
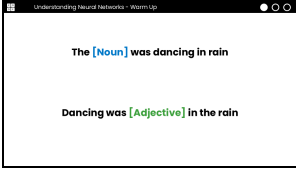
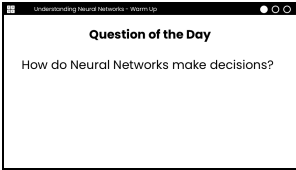
Warm Up



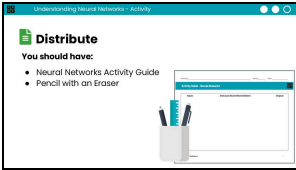
Wrap Up

Warm Up (5 minutes)

Slide	Lesson Guide
	Discuss: Complete this mad-lib: The _____ was dancing in rain Discussion Goal: Notice that most answers are likely nouns, like “dog” or “firefly” or “leaf”

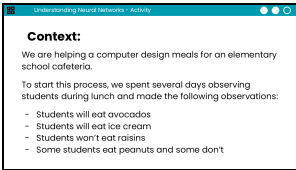
	Discuss: Complete this mad-lib: Dancing was _____ in the rain Discussion Goal: Notice that most of the answers are likely adjectives, like “fun” or “disastrous” or “difficult”
	Say: All languages have grammatical rules that help construct valid sentences. You can’t just put random words together - you need things like nouns and verbs and adjectives, and they need to be in a certain order so sentences make sense. These rules create certain relationships within sentences themselves, like how a noun usually follows the word “The”.
	Say: Language models represent these relationships with a special kind of network called a Neural Network. Today we’re going to learn what neural networks are and how they function, and then tomorrow we’ll learn how they’re used with language models Question of the Day: How do neural networks make decisions?

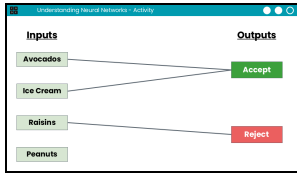
Activity (35 minutes)

Slide	Lesson Guide
	Say: Today we're going to participate in an activity where we develop our own neural network to solve a problem. We're going to complete this as a class and following my lead for parts of the activity, sometimes erasing and re-drawing as we go along. Our goal is not to become an expert on neural networks, but just to understand what they look like and how they're used. Distribute: <u>Pass out the Neural Network Activity Guide</u>

Teaching Tip

Teaching Tip: Erasable Materials - Students will be drawing their own neural network and iterating on the design. This means students will be doing some trial and error, which will involve lots of erasing. If your classroom has them available, consider using whiteboards for this activity instead of the activity guide so students can quickly and easily erase their work as the lesson progresses. Or if students have 1:1 devices, consider using a collaborative drafting software such as Figma or Miro or Freeform. If students are using the printed activity guide, encourage students to use pencil so they can easily erase.

	Do This: Read the slide to the class, or ask the class to read it aloud. We are helping a computer design meals for an elementary school cafeteria. To start this process, we spent several days observing students during lunch and made the following observations: - Students will eat avocados - Students will eat ice cream - Students won't eat raisins - Some students eat peanuts and some don't
---	--



Say: We can represent this with a network. On the left side, we have our inputs - in this case, avocados, ice cream, raisins, and peanuts.

Click the animation

Say: On the right side, we have our output states: accept or reject.

Click the animation

Say: We can draw lines that connect the inputs to the output. These connections represent our decisions - when the model sees avocados, it will accept the meal and eat it. And when it sees raisins, it will reject the meal and won't eat it.

Teaching Tip

Teaching Tip: Structure vs Creativity - The remainder of this activity uses these inputs, especially Avocados and Ice Cream, to teach new topics - we will frequently refer back to these inputs and iterate. Students will apply what they learn by adding new inputs and connections based on their personal preferences. This means student activity guides should look very similar around these inputs and what is presented on the slides, but will diverge quickly when students make these networks their own. Keep this in mind when looking over student shoulders and checking that students are on-task and following along.

Do This:

- 1) Draw the network so far on your paper
- 2) Since the data was inconclusive, decide how you want your network to handle peanuts - will your model accept or reject?
- 3) Add two other foods to your network and decide if you want to accept or reject them

Do This:

- 1) Draw the network so far on your paper
- 2) Since the data was inconclusive, decide how you want your network to handle peanuts - will your model accept or reject?
- 3) Add two other foods to your network and decide if you want to accept or reject them

Circulate: Make sure students are representing the network correctly based on the last slide. Also make sure students decide how they want to handle peanuts and have added 2 more foods to their network. By the end of this section, students should have 6 inputs on their network.

1 Week Later:

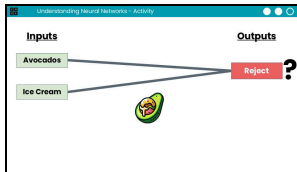
While testing our network at the school, something unexpected happened: **the network suggested an avocado covered in ice cream!**

And no one would eat it!!

Do This: Read the slide to the class, or ask the class to read it aloud.

While testing our network at the school, something unexpected happened: the network suggested an avocado covered in ice cream! And no one would eat it!!

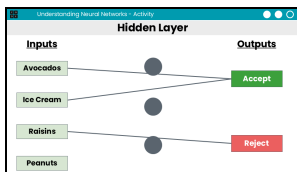
[Image credit: Created with DALL-E with the prompt "generate an image of an avocado covered in ice cream in a clip art style 100px by 100px"]



Say: Even though we might think this is unusual, our network thought this was very reasonable since our data showed everyone likes Ice Cream and everyone likes Avocados, so it thought everyone would like Ice Cream on Avocados.

Click the animation

Say: But it turns out, when these foods were combined, everyone had a different reaction and rejected them! With this new data, it's time to update our network



Click the animation

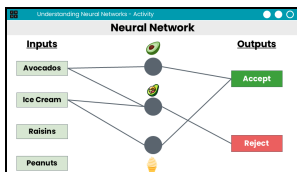
Say: And instead we're going to add a column of circles in the middle of our network...

Click the animation

Say: These circles are called nodes, and we can draw connections from our inputs to represent the relationships between our input foods.

Click the animation

Say: This is called a hidden layer in our network because it's not part of the inputs or outputs we usually see when making a decision. It's used to add extra information that helps make a decision and represent relationships between inputs.



Say: Let's use this hidden layer to represent the relationship between ice cream and avocados

Click the animation

Say: Here's how we can connect our network to represent what we knew from our initial data. When we just have avocados, or when we just have ice cream: we will eat those foods

Click the animation

Say: Here's how we can connect our network to represent what we learned about avocados and ice cream combined. When we have both of these foods, we want to reject that meal since no one will eat it

Click the animation

Say: Since the hidden layers are "activated" by previous layers, it can feel similar to how our brains work and how some neurons can activate other neurons to create chain-reactions and complex thoughts. These hidden layers and how they activate is what makes this a Neural Network.

Understanding Neural Networks - Activity

Do This:

- 1) Draw additional **nodes** in your **hidden layer**
- 2) Update your network to represent different relationships and decisions
 - What happens if every single food was on the same plate?
 - Are some combinations still good? Are some combinations bad?

Do This:

- 1) Draw additional nodes in your hidden layer
- 2) Update your network to represent different relationships and decisions
 - What happens if every single food was on the same plate?
 - Are some combinations still good? Are some combinations bad?

Circulate: Make sure students are adding more nodes to their hidden layer and representing different combinations. Ask students what kind of relationships they are trying to represent with the network connections. Networks will start to get complex, which is okay since it can lead to students realizing how simple rules like “making a network” can lead to very complex behaviors. By the end of this section, students should have a hidden layer with multiple inputs and connections to the outputs.

Understanding Neural Networks - Activity

1 Week Later:

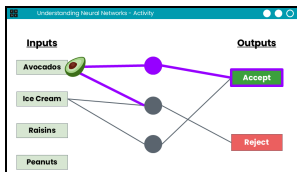
While testing our network at the school, we found a weird bug: sometimes the network will freeze and won't make any decision at all!

This tends to happen when both avocado and ice cream are suggested, even if they're given individually instead of combined.



Do This: Read the slide to the class, or ask the class to read it aloud.

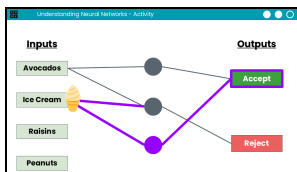
While testing our network at the school, we found a weird bug: sometimes the network will freeze and won't make any decision at all! This tends to happen when both avocado and ice cream are suggested, even if they're given individually instead of combined.



Say: We can try testing the network with just avocado...

 Click the animation

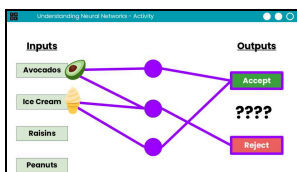
Say: And we see that one pathway is activated, so we know this is a meal students are likely to accept



Say: We can also try testing the network with just ice cream...

 Click the animation

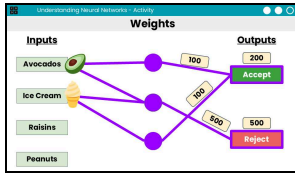
Say: And we see that one pathway is activated, so we know this is a meal students are likely to accept



Say: But when we try with both an avocado and ice cream...

 Click the animation

Say: Multiple paths are lit up! And it looks like two different decisions are being made! This must be the bug we saw where the network would freeze - it doesn't know which decision to make since both are activated!



Say: One thing neural networks use to help solve this problem is adding numbers to connections, called Weights.

Click the animation

Say: For example, we can add lower weights to the “just avocados” and “just ice-cream” connections, and a higher weight to the “both” connection.

Click the animation

Say: When we combine the weights at the end, the network can look at the total weight and use that to make a final decision

Click the animation

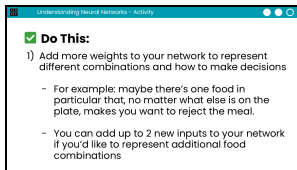
Say: In this case, the Reject decision has a higher weight, so that’s what the model would decide to do.

Teaching Tip

Teaching Tip: Explaining Weights - The key takeaway from this example should be: neural networks use weights to help them make decisions. Students may have additional questions - like how the weights are created, or what numbers to use, or if every connection has weights - which aren't fully answered in this slide. Some of these questions are answered by the video at the end of this lesson, and some are answered in tomorrow's lesson and the Neural Networks video at the end of that lesson. If you would like to be more prepared to talk about weights in neural networks, you may want to watch these videos first to be better prepared to discuss weights with students. Or, you can tell students to wait until the end of the lesson and for tomorrow's lesson to learn more.

Content Corner

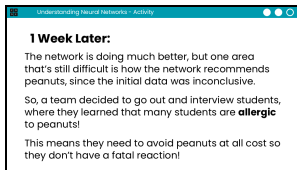
Content Corner: Simplifying Neural Networks - Neural Networks are complex structures with many moving parts - to fully explain them at an advanced level requires several hours and advanced mathematics like calculus and linear algebra! This unit makes a decision to simplify what is presented to students, focusing on weights and hidden layers as the key things to understand in this introduction to neural networks. As students take additional courses or progress their understanding of artificial intelligence, they will likely learn more advanced components of these networks such as bias parameters and activation functions and the mathematics required to activate the network - these details are abstracted in this curriculum, but could be explored in other lessons. If you are interested in a deeper technical dive into neural networks for your own curiosity, we recommend the video **But what is a neural network?** from the threeblueonebrown Youtube channel. We do not recommend showing this to students, but it does highlight the additional technical components of a neural network that are not included in these lessons.



☑ Do This:

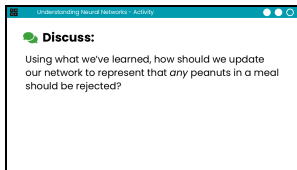
- 1) Add more weights to your network to represent different combinations and how to make decisions
- For example: maybe there's one food in particular that, no matter what else is on the plate, makes you want to reject the meal.
 - You can add up to 2 new inputs to your network if you'd like to represent additional food combinations

🔄 Circulate: Check in with students as they are adding weights. Ask students what they are trying to represent with their decisions, listening for intentional decisions behind certain weights and combinations. Students may be unsure what numbers to use in specific situations - reassure them that the specific numbers aren't important right now, but it's their relationship that matters: important connections should have larger numbers compared to the rest of the network. If students are stuck, cue them to consider situations where several items may "add up" to a positive meal (like just ice-cream isn't enough, but ice-cream with peanuts is delicious). Or consider how things they dislike might outweigh everything else. By the end of this section, students should have weights on most of their connections to their outputs.



☑ Do This: Read the slide to the class, or ask the class to read it aloud.

The network is doing much better, but one area that's still difficult is how the network recommends peanuts, since the initial data was inconclusive. So, a team decided to go out and interview students, where they learned that many students are allergic to peanuts! This means they need to avoid peanuts at all cost so they don't have a fatal reaction!



💬 Discuss: Using what we've learned, how should we update our network to represent that any peanuts in a meal should be rejected?

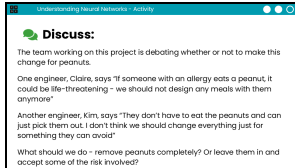
💡 Discussion Goal: Students should apply what they've learned about hidden layers and weights to represent this situation, aiming for a representation where peanuts is connected to "Reject" with a very very high weight. One possible way to do this is to create a new node in the hidden layer with a single connection from peanuts to this node and then to "Reject", and give this connection a high weight. This makes it so anytime peanuts appear, this node is activated, which causes a very high "Reject" signal to be sent through the network.



Assessment Opportunity



Formative Assessment: Neural Networks - This prompt can be used as a formative assessment of how well students understand the lesson so far, specifically around how networks can represent decisions. You could decide to have students discuss in a group and then draw their decision on their network, allowing you to circulate and check how students are answering this question. Or you can have students discuss in small-groups and circulate to overhear conversations, then facilitate a class-wide discussion about different solutions to see how the class interprets and responds to each idea. Both of these are strategies that give signal on how well students are understanding the lesson so far and, if they struggle, can be an opportunity to clarify and reinforce concepts.



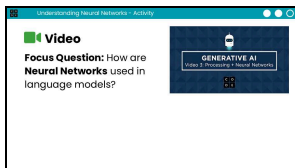
Discuss: The team working on this project is debating whether or not to make this change for peanuts. One engineer, Claire, says “If someone with an allergy eats a peanut, it could be life-threatening - we should not design any meals with them anymore”. Another engineer, Kim, says “They don’t have to eat the peanuts and can just pick them out. I don’t think we should change everything just for something they can avoid”. What should we do - remove peanuts completely? Or leave them in and accept some of the risk involved?

Discussion Goal: This is an open-ended discussion hitting on some of the ethical considerations with machine learning. This particular discussion focuses on safety concerns and what an acceptable level of risk is when making decisions with machines like this. Some students may think the consequences - a life-threatening reaction - may necessitate removing this risk from the system altogether. Others may think that, since this is something where personal judgement can be applied by just a subset of users, it may be fine to let this network keep designing meals with peanuts. Focus the conversation on wrestling with both sides of the debate, elevating student voices to think critically and carefully about the scenario. It’s likely - and acceptable - for the discussion to end in a place of non-closure - instead, aim for students having come to their own decision that they can justify from the scenario and their lived experiences.

Ethics Opportunity

Pause and Reflect: From a food perspective, this scenario mirrors real considerations that kitchens take into account with food preparation. Some kitchens avoid nuts altogether so consumers can be sure they won’t have an allergic reaction. Others apply warnings saying “These items were baked in a facility that includes nuts” as a warning for people to use their own judgement and discretion.

From a machine-learning perspective, this scenario is relatable to stories where an AI model or chatbot was released and then needed to be adjusted after the release to account for a bias or harm they had not considered or intended. Consider pausing the lesson here and have students reflect on their own experiences or knowledge with AI or other technologies where they witnessed unintended impacts. One example could be the **Twitter Cropping Algorithm**, which was discovered to make biased decisions on what it decided to crop out of photos. This blog post is a great example of an organization taking action once the unintended harm was discovered, although it is likely too technical for students to read and process on their own.

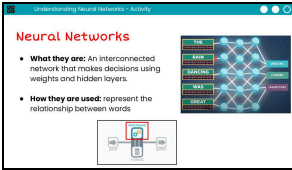
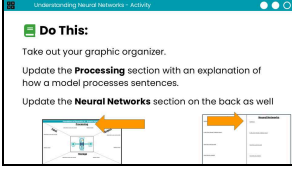


Say: We’ve now constructed a basic neural network that helps us make a decision. Along the way, we’ve learned some of the key components like hidden layers and weights. Let’s review what we’ve learned in class, and see how they can be used by large language models.

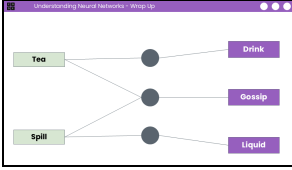
Video: Watch the **Generative AI: Processing + Neural Networks**

Discuss: How are Neural Networks used in language models?

Discussion Goal: Students should recall what they heard in the video, but it’s okay if their answers aren’t 100% accurate since this will be reinforced in the next slide. Once a few students have shared, transition to the next slide to present the definition.

	🎤 Say: Neural Networks are how language models represent relationships between words in a sentence. These networks are created in such a way that they approximate common sentence structures like subject/noun or verb/object grammatical relationships, based on how the training data is structured.
	✅ Do This: Take out your graphic organizer. Update the Processing section with an explanation of how a model processes sentences. Update the Neural Networks section on the back as well. 🔄 Circulate: Check in with students as they complete their graphic organizer and summarize their thoughts. It may be helpful to return to the previous slide so students can see the definition of Neural Networks. There is no "one right answer" that should go in these boxes - instead, help students synthesize what they've learned in the lesson and seen in the video into their own words. Similarly, cue students to draw an illustration or a diagram representing this process. It's okay for it to be abstract, as long as the student understand what it means to them.

Wrap Up (5 minutes)

Slide	Lesson Guide
	🎤 Say: Today we saw how neural networks can represent relationships between inputs (like avocados and ice-cream) and use those relationships to make decisions. Language models do something similar, using relationships between words to help them understand the context of a sentence. For example: 🕒 Click the animation 🎤 Say: The word “tea” on its own is a noun associated with drinks or fancy cups or British culture 🕒 Click the animation 🎤 Say: The word “spill” on its own is a verb associated with liquids 🕒 Click the animation 🎤 Say: But the phrase “spill the tea” is a slang term for gossiping and sharing secrets, which isn’t directly related to any of the individual words in the sentence

Discuss:

Think of a phrase where each word means something individually that is different from when the phrase is combined together.

Using what we've learned today, describe how a neural network could be used to represent these different meanings.

Discuss: Think of a phrase where each word means something individually that is different from when the phrase is combined together. Using what we've learned today, describe how a neural network could be used to represent these different meanings.

Discussion Goal: It may take some time for students to think of responses here - if so, encourage them to consider "slang" or "jargon" they use casually and then reflect on the meaning of each word compared to how they use it. If students struggle, a few other examples might be:

- The Big Apple - the phrase means New York City, but each individual word is not related to the city at all
- Throwing Shade - the phrase means to subtly or passive-aggressively disparage someone
- Break a Leg - the phrase means "good luck!" when in a theatre, even though each individual word doesn't involve luck or the theatre at all

Ethics Opportunity

Ethics Opportunity: Real-World Connections - Consider highlighting the origins of some of these phrases - for example, the article **"Let's Talk Tea" from Merriam-Webster Dictionary** traces the origin of the phrase "Spill the Tea" as it was popularized in Black drag culture and shows like Rupaul's Drag Race, along with phrases like "Throwing Shade". This can lead to discussions about language appropriation and erasure of minoritized groups that contribute to mainstream culture. Tying the discussion back to generative AI: unclear guidelines around data privacy and ownership mean that personal data may be used to train large-language models leading to these phrases being re-generated without this original history or context, and without credit or context from the groups that originated them. Consider highlighting how issues of data privacy can lead to these scenarios, and consider highlighting how some groups are working towards solutions by advocating for data sovereignty agreements, **described in this article from the Native Land Information System**.

Question of the Day

How do Neural Networks make decisions?

Question of the Day: How do Neural Networks make decisions?

Key Vocabulary:

Neural Network: An interconnected network that makes decisions using weights and hidden layers. They are used to represent the relationship between words



Vocabulary: The key vocabulary from this lesson is provided on this slide and in the lesson plan.



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Lesson 7: Neural Networks: How They're Trained

45 minutes

Overview

This lesson explores supervised learning and its role in teaching neural networks language structure. Starting with a game, students learn through trial and error and use their past outcomes to inform future decisions. They then extend this approach to engaging in conversations in a fictional language, initially facing more failures but gradually improving as they identify patterns. This process illustrates how neural networks adjust their behavior based on new data. The lesson concludes with a reflection on whether this process signifies true "intelligence" or merely a sophisticated rule-following system.

Standards

Full Course Alignment

AI4K12 National Guidelines 2021

- **BI-3** - Computers can learn from data

AI4K12 National Guidelines 2023

- **BI-2** - Agents maintain representations of the world and use them for reasoning

CSTA K-12 Computer Science Standards (2017)

- **AP** - Algorithms & Programming

Agenda

Before the Lesson

Student Levels

Warm Up (5 minutes)

Green Glass Door

Activity (35 minutes)

Wrap Up (5 minutes)

Wrap Up - Option 1

Wrap Up Option 2

Objectives

Students will be able to:

- Use trial-and-error and past data to increase their accuracy in a supervised learning process

Preparation

- Print copies of the activity guide for each student
- Print and cut out copies of the language cards for each group of students
- Run through the slide deck and videos ahead of time
 - Prepare to play the Green Glass Door game with the class
- (Optional) Prepare vocabulary posters for class word wall
- (Optional) Print out the slide deck with speaker notes as a resource during class. **[Click here for steps on how to do this.](#)**

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **Language Cards** - Resource
▼ Make a Copy
- **Neural Networks: How They're Made**
- Slides ▼ Make a Copy

For the students

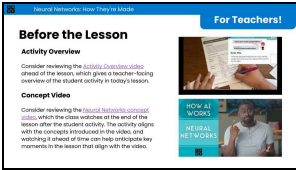
- [How AI Works: Neural Networks](#) - Video
 - [Learning a Language](#) - Activity Guide
- ▼ Make a Copy

Vocabulary

- **Supervised Learning** - A process that uses examples that are already labeled to find patterns and make decisions

Teaching Guide

Before the Lesson

Slide	Lesson Guide
	Before the Lesson: This activity uses a pedagogy rooted in discovery learning, where students first complete an activity and reflect on their process, which illuminates the underlying AI concepts. A video is provided at the end of the lesson that explains and summarizes the key learning from the lesson. Activity Overview: Consider reviewing the Activity Overview video ahead of the lesson, which gives a teacher-facing overview of the student activity in today's lesson. Concept Video: Consider reviewing the Neural Networks concept video, which the class watches at the end of the lesson after the student activity. The activity aligns with the concepts introduced in the video, and watching it ahead of time can help anticipate key moments in the lesson that align with the video.

Student Levels

These levels contain the same questions as the **Warm Up** and **Wrap Up** from the lesson slide deck, allowing students to answer the questions within the Code.org platform. These can be used as a digital alternative to physical journals, and lets you take advantage of our [Student Summaries features](#) for checking and displaying student work.

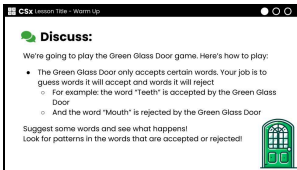
By answering these questions in these levels, you can take advantage of discussion and differentiation strategies that elevate student voice to help you guide the discussion. See our [Teaching Guide for Incorporating Student Responses in Lessons](#) for more information on these strategies.


1

Warm Up

Warm Up (5 minutes)

Green Glass Door

Slide	Lesson Guide
	 Discuss: We're going to play the Green Glass Door game. Here's how to play: The Green Glass Door only accepts certain words. Your job is to guess words it will accept and words it will reject - For example: the word "Teeth" is accepted by the Green Glass Door - And the word "Mouth" is rejected by the Green Glass Door Suggest some words and see what happens! Look for patterns in the words that are accepted or rejected!

For the Teacher: In this game, you know the secret rules (explained below) and are reacting to the suggestions that the class makes. As the class suggests words, you tell them whether or not words are accepted or rejected. This game requires students to notice patterns in the words that are accepted or rejected, and use those patterns to inform the guesses they make. The goal of the activity is for students to eventually discern the rules of the game from these patterns.

The Rules: The secret of the game is: words with double letters are accepted and all others are rejected. This is why Teeth is accepted (because of the double e's) while Mouth is not (because no letter is repeated). This is also where the name of the game comes from - Green Glass Door all have double letters.

Playing the Game: Ask students to suggest words and write them down somewhere visible, then tell the class whether the Green Glass Door accepts or rejects each word. Instead of doing this one-at-a-time, it can be helpful to take words in batches - like 3 to 5 at a time - and then mark each one as accept or reject in a batch. It's important to leave all guesses in a visible spot so students can look at the data to inform their next guesses. Over time, students may start to notice the pattern - that words with double-letters are accepted - and start guessing those more frequently. If students are struggling to catch on, it can be helpful to rewrite the list so several accepted words are together and several rejected words are together. You can also provide your own examples, or reference the name of game - Green Glass Door - as a hint.

Managing Expectations: This activity has a specific type of engagement that some students may struggle with: the activity deliberately starts with 0 knowledge, so most answers are likely to be incorrect rather than correct. Some students may struggle with randomly guessing and getting it incorrect more often in the beginning. However, this is exactly the mindset that is necessary to carry forward into the rest of the lesson and is an important part of how machine learning really works: machines make an enormous amount of incorrect guesses that are gradually refined over time before eventually getting more and more accurate.

Ending the Game: As the game continues, some students will discovered the rule while others may not. Encourage the students who think they know to keep it a secret until others have caught on. Using your judgement, when enough students feel like they know the rule, continue to the next slide to reveal the rule of the game

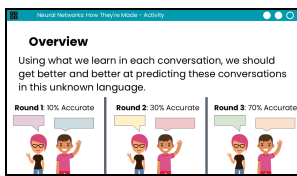
Teaching Tip

Teaching Tip: Play the Game Yourself - A digital version of this game is available in our **AI & Machine Learning unit at this level**. You can play this game yourself to get a sense of how it works, but we don't recommend having students play it since no other part of this lesson involves computers, and the game may go quicker if done as a class rather than a more involved on-device experience.

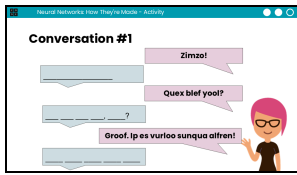
	Discuss: Can you guess the rule for the Green Glass Door game? How did you figure it out? Discussion Goal: Ask a student to share the rule with the class and share how they figured it out. Emphasize answers that rely on noticing patterns within the responses, and responses that used an educated-guess to test their hypothesis (for example: "I suspected it was double letters, so I guessed a word like that to see if it was correct, and that confirmed my suspicions"). To help manage emotions for students who may not have figured out the pattern, remind students that it takes both correct and incorrect answers to determine the rule, which makes all responses valuable.
	Say: Yesterday we saw how neural networks can be used to represent relationships between inputs. Today we'll learn how these networks learn these relationships in the first place. It turns out it's very similar to what we did in this warm-up: a lot of trial and error to learn from our mistakes and identify patterns. Question of the Day: How are neural networks trained to represent relationships between words?

Activity (35 minutes)

Slide	Lesson Guide
	Say: The neural networks in large language models use a form of machine learning called Supervised learning. Supervised learning is when you use examples that are already labeled to find patterns and make decisions. This is similar to the activity we just did - as we got more data labeled "accept" and "reject", we could notice patterns which helped us decide which words to guess. Today we're going to apply this concept in a similar way to how a language model tries to represent language for the first time.
	Group: Organize students into groups of 2-4 Distribute: Pass out the Learning Languages Activity Guide and the Language Cards to each group.
	Do This: Either read through these slides yourself, or have students read aloud. This text is also on the activity guide. Say: This activity simulates the machine learning process that language models use when trying to represent and respond to language. We will look at three example conversations in a gibberish language as if we were learning from a textbook.
	Say: We'll see one side of the conversation, then guess what the response should be. After we write our guess, we'll compare with the expected response see how well we did.



Say: Using what we learn in each conversation, we should get better and better at predicting these conversations in this unknown language.



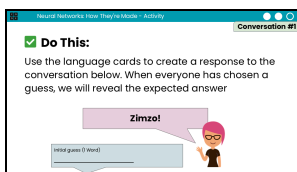
Say: We're going to start the same way a language model does - without any information at all! All we know are a bunch of words, but not necessarily how to use them in a sentence. This means in this first round, we don't really have any idea what to do so we're probably going to get it wrong just like in the warm-up. But remember - this data can help us with the next conversation we have.

Teaching Tip

Teaching Tip: Strategies for Conversation #1 - In this first conversation, students have no information at all yet to base their responses off of and will tend to get most responses incorrect. However, this is a necessary step because it helps gather the data they will need for future conversations where they can look for patterns. To help facilitate this initial conversation and gather our initial data, don't take it too seriously and consider leaning into the playful aspect of just making guesses and seeing what happens. It also helps to move quickly through this round, since any correct answers will likely be due to luck rather than strategy.

Teaching Tip

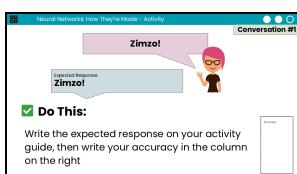
Teaching Tip: Laughing in a Conversation - Some students may be very skeptical that it's possible to try and have a conversation when you don't understand any of the words. As a potential talking point, consider asking students if they've ever had trouble hearing someone and instead of asking them to clarify, they just laugh or smile or nod. Conversation strategies like this make it possible to continue conversations without having full context. In some ways, we are doing something similar in this first conversation of this activity.



Do This: Use the language cards to create a response to the conversation. When everyone has chosen a guess, we will reveal the expected answer

Circulate: Students should be deciding which word to use in their response. They can use the language card manipulatives to decide how they want to response. They should also write their response on their activity guide. Encourage students to lean-in to the fact that they don't have any information right now, so any word they pick might be a good one. Once students have decided on a word, continue to the next slide.

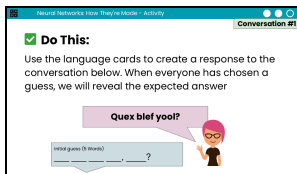
Say: Now it's time to find out what the expected response is...



Say: It turns out the expected response in this conversation was "Zimzo!". Even if we didn't get it correct this time, maybe that will be helpful in future conversations. Let's record the result so we can remember it for next time.

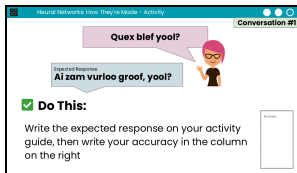
Do This: Write the expected response on your activity guide, then write your accuracy in the column on the right.

Circulate: Make sure students are recording this on their activity guide - it will be important for the next conversation. Also help students record their accuracy - at this stage, it will either be 0% (they guessed incorrectly) or 100% (they guessed correctly)



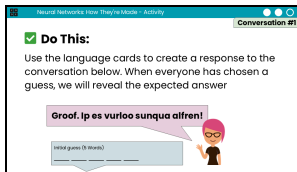
☑ **Do This:** Use the language cards to create a response to the conversation. When everyone has chosen a guess, we will reveal the expected answer

🔄 **Circulate:** Students should be deciding which word to use in their response. They can use the language card manipulatives to decide how they want to response. They should also write their response on their activity guide. Encourage students to lean-in to the fact that they don't have any information right now, so any word they pick might be a good one. Once students have decided on a word, continue to the next slide.



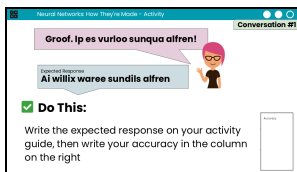
☑ **Do This:** Read the expected answer aloud, then have students record their answers in their accuracy box. Playfully ask students how they're doing and whether they managed to guess any of the answers correctly this time around.

🔄 **Circulate:** Make sure students are recording this on their activity guide - it will be important for the next conversation. Also help students record their accuracy - they can represent it as a fraction if it's easier (for example: 0/5 or 2/5, etc)



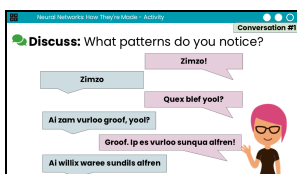
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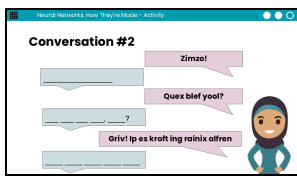


🗣️ **Discuss:** Now we have our first piece of real data, we can start to notice patterns. What patterns do you notice in this conversation?

💡 **Discussion Goal:** Students should be prompted to consider ways that this conversation might inform future conversations, and any response that makes observations or speculates on potential patterns are good to highlight. A few patterns students may notice are:

- "Zimzo" is used by both people at the start of the conversation. They may speculate this could be some kind of greeting, like "Hello"
- "Yool" is used at the end of sentences and with question marks, which may be useful for the next conversation
- Alfren appears at the end of sentences and Ai appears at the start of sentences, which may be useful for the next conversation

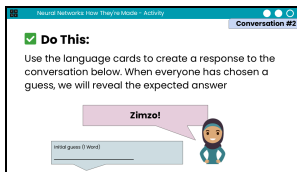
Additionally: it can be fun to read the conversation out-loud as a class to add some levity to this activity.



Say: For this next conversation, you can refer to the data we just gathered with conversation number 1. Flip over your activity guide to get started

Teaching Tip

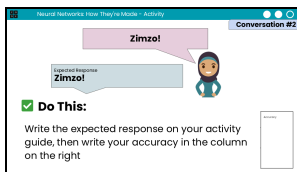
Teaching Tip: Strategies for Conversation #2 - Now that we have some initial data, it can be useful to slow down a bit and encourage students to be strategic when picking their conversation responses. While circulating, listen for students justifying why they chose certain words based on previous data. Students shouldn't be choosing "randomly" anymore, and can use the fact that certain words appeared in the previous conversation to help inform this conversation. There will still be instances where students may be surprised, but by using patterns from the previous conversation, their accuracy should increase.



Do This: Use the language cards to create a response to the conversation. When everyone has chosen a guess, we will reveal the expected answer

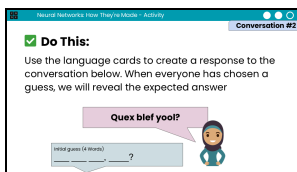
Circulate: Students should be deciding which word to use in their response. Listen to how students justify their responses - ideally, students will see that this conversation starts the same as the last one and will consider choosing "Zimzo" as their response, but it's okay if not everyone realizes this yet.

Say: Now it's time to find out what the expected response is...



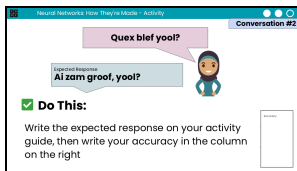
Say: It turns out the expected response in this conversation was "Zimzo!". Even if we didn't get it correct this time, maybe that will be helpful in future conversations. Let's record the result so we can remember it for next time.

Do This: Write the expected response on your activity guide, then write your accuracy in the column on the right.



Do This: Use the language cards to create a response to the conversation. When everyone has chosen a guess, we will reveal the expected answer

Circulate: Students should be deciding which word to use in their response. They can use the previous response to help generate this one, but this response has less words than in conversation #1 so they will have to guess which word to remove from the previous response.



Do This: Read the expected answer aloud, then have students record their answers in their accuracy box. Ask if any students got the answer 100% correct and if so, how they decided on which words to use.

Circulate: Make sure students are recording this on their activity guide - it will be important for the next conversation. Also help students record their accuracy - they can represent it as a fraction if it's easier (for example: 0/5 or 2/5, etc)



☑ **Do This:** Use the language cards to create a response to the conversation. When everyone has chosen a guess, we will reveal the expected answer

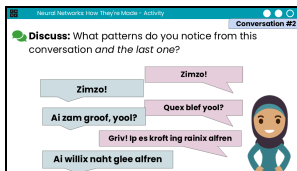
🔄 **Circulate:** Students should be deciding which word to use in their response. They can use the language card manipulatives to decide how they want to response. They should also write their response on their activity guide. Encourage students to lean-in to the fact that they don't have any information right now, so any word they pick might be a good one. Once students have decided on a word, continue to the next slide.



☑ **Do This:** Read the expected answer aloud, then have students record their answers in their accuracy box.

Ask students how well they did with this response. It's unlikely groups will get 100%, but it's reasonable some of the patterns from the previous conversation are helpful here, such as using "Ai" at the start of sentences and "alfren" at the end of sentences. And now this is another piece of data for learning more about this language.

🔄 **Circulate:** Make sure students are recording this on their activity guide - it will be important for the next conversation.



🗣️ **Discuss:** What patterns do you notice from this conversation and the last one?

💡 **Discussion Goal:** Now that students have a second conversation to compare to, they can reflect on whether their predictions were true or not. For example, students who noticed "Zimzo!" was re-used in the opening of the conversation are rewarded for guessing correctly here. A few other patterns students might notice are:

- The second response is similar to the first conversation, but with one word removed. Students might hypothesize that word is an adjective or adverb that modifies other parts of the sentence.
- Students may notice the third response has the greatest variation between the two conversations, so it may be harder to draw patterns for that particular response.

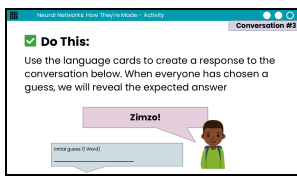
Additionally: it can be fun to read the conversation out-loud as a class to add some levity to this activity.



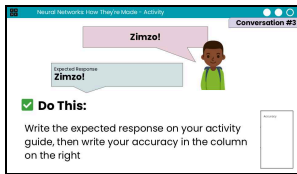
🗣️ **Say:** For this next conversation, you can refer to the data we just gathered with conversations 1 and 2

💡 Teaching Tip

Teaching Tip: Strategies for Conversation #3 - This conversation should see the largest accuracy increase, as many phrases are re-used from previous conversations. This is a good round to explicitly compare to conversation #1 and point out how much improvement has happened just from two conversations and the data involved. Also, if you are running short on class time, this conversation can be skipped and instead just focus on the wrap-up



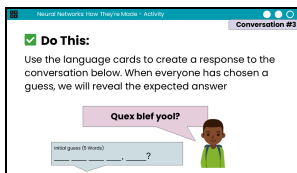
☒ **Do This:** Use the language cards to create a response to the conversation. When everyone has chosen a guess, we will reveal the expected answer



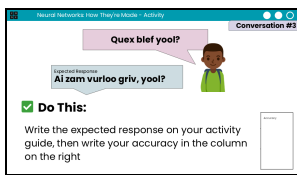
☒ **Do This:** Write the expected response on your activity guide, then write your accuracy in the column on the right.

Teaching Tip

Teaching Tip: Celebration - By this third round, hopefully most of the class was able to answer this correctly. If so, be sure to celebrate the class as having noticed a particular pattern in the structure of conversations - that when one person says "Zimzo", a typical response can also be "Zimzo". You can even connect this directly to the English language, letting students know they may have noticed a pattern based around how people greet each other in this new language, much like how we say "Hello" when greeting people.



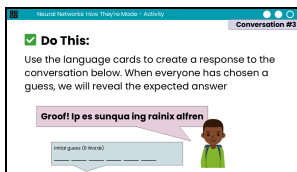
☒ **Do This:** Use the language cards to create a response to the conversation. When everyone has chosen a guess, we will reveal the expected answer



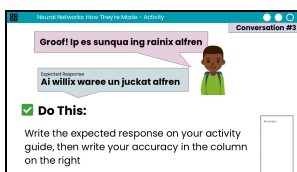
☒ **Do This:** Read the expected answer aloud, then have students record their answers in their accuracy box. Students should notice their accuracy increasing.

Teaching Tip

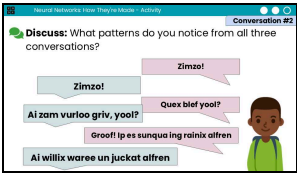
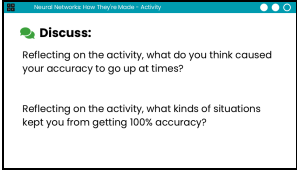
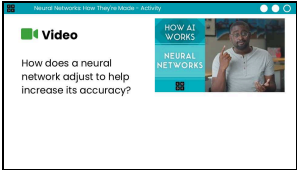
Teaching Tip: Patterns Across Conversations - Even though this question has been the same across all conversations, the answer has been slightly different each time by varying the words "vurloo", "groof", and "griv". You may decide to point this out to students and connect back to the concept of embeddings, which can be used to represent the meaning of words and how "close" words are to each other. From these three conversations, it seems like these words are all "close" to each other since they are used interchangeably in response to this question. This means their embeddings are likely close to each other, which means they might be synonyms or antonyms or modifiers for each other.



☒ **Do This:** Use the language cards to create a response to the conversation. When everyone has chosen a guess, we will reveal the expected answer



☒ **Do This:** Read the expected answer aloud, then have students record their answers in their accuracy box.

	Discuss: What patterns do you notice from all three conversations? Discussion Goal: This is a chance to reinforce and celebrate any patterns students have identified throughout the activity, especially related to responses 1 and 2. However, try to keep this discussion brief to focus on the final discussion and wrap-up in the lesson.
	Discuss: Reflecting on the activity, what do you think caused your accuracy to go up at times? Reflecting on the activity, what kinds of situations kept you from getting 100% accuracy? Discussion Goal: This is an open-ended opportunity to debrief this activity and reflect on the process of “learning” a language without any prior experience. Listen for responses that highlight how previous data informed their decisions, such as the opening “Zimzo” greeting. Also listen to examples of how, even though they might be confident in their guesses based on the data, they were never 100% sure. In fact, one of the things that often stops 100% accuracy is when a new word is used in a new way. All of these reflections mirror how large language models work when generating responses to language, especially using past data to “guess” responses and never being 100% sure what a correct response should be.
	Say: Let’s take a closer look at how neural networks handle this kind of process Video: Play the How AI Works: Neural Networks video Discuss: How does a neural network adjust to help increase its accuracy? Discussion Goal: This can be a quick discussion - students should call out how neural networks adjust their weights based on feedback, similar to how the class adjusted their responses based on feedback to become more accurate.

Teaching Tip

Teaching Tip: Additional Example - Depending on the interest of your students, you may decide to show them an additional example of the [Mar/IO neural network](#) that learned to play Super Mario Bros. This video reinforces a few topics key to this unit, such as neural networks and supervised learning, and reinforces the idea that machine learning spends most of its time being wrong before it starts working as expected. However, it also discusses technical details that are beyond the scope of this unit - such as evolutionary learning and NEAT - which may be a distraction to students. Use your judgement on whether or not to show this video, and if so, be sure to reinforce the ways it connects to the unit and let students know which parts are just additional information that may be interesting to think about but are beyond the scope of this course.

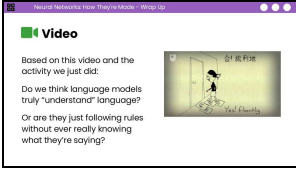
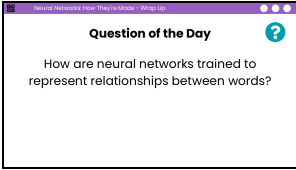
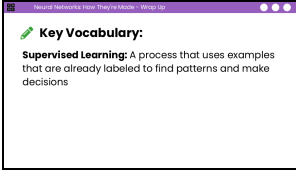
Wrap Up (5 minutes)

Wrap Up - Option 1

There are two options for wrap-ups in this lesson. This first option involves watching a 1-minute video about the nature of intelligence and responding to a prompt, while the second option involves considering a scenario involving data collection and privacy and responding to a prompt. This first option is more abstract addressing the hype around “intelligence machines” and how well language models “learn”, while the second option is more aligned to the real-world ethical considerations of data collection and privacy with opportunities for students to bring their

lived-experiences into the discussion. Both wrap-ups can help students construct arguments for the final project in the unit - but for the sake of time, you may only be able to do one. If possible, consider doing the second activity during another lesson so students can reflect and discuss both.

If you choose this wrap-up, skip or delete the slides associated with Wrap Up 2

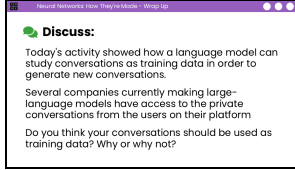
Slide	Lesson Guide
 Video Based on this video and the activity we just did: Do we think language models truly "understand" language? Or are they just following rules without ever really knowing what they're saying?	Video: Show the Chinese Room Experiment video Discuss: Based on this video and the activity we just did: Do we think language models truly "understand" language? Or are they just following rules without ever really knowing what they're saying? Discussion Goal: This is an open-ended discussion where students may fall into different opinions on how relevant this scenario is to how artificial intelligence works. Some students may find themselves agreeing with the thought experiment, especially after the activity - that being able to communicate and respond appropriately isn't a true barometer for understanding. Other students may think that this doesn't actually matter - as long as the computer can solve problems and communicate appropriately as a tool for how we use it, then it doesn't matter what it does or doesn't understand. In the discussion itself: elevate student voices who are able to explain their thinking, no matter where they fall in this argument.
 Question of the Day How are neural networks trained to represent relationships between words?	Question of the Day: How are neural networks trained to represent relationships between words?
 Key Vocabulary: Supervised Learning: A process that uses examples that are already labeled to find patterns and make decisions	Vocabulary: The key vocabulary from this lesson is provided on this slide and in the lesson plan.

Wrap Up Option 2

There are two options for wrap-ups in this lesson. This first option involves watching a 1-minute video about the nature of intelligence and responding to a prompt, while the second option involves considering a scenario involving data collection and privacy and responding to a prompt. This first option is more abstract addressing the hype around "intelligence machines" and how well language models "learn", while the second option is more aligned to the real-world ethical considerations of data collection and privacy with opportunities for students to bring their lived-experiences into the discussion. Both wrap-ups can help students construct arguments for the final project in the unit - but for the sake of time, you may only be able to do one. If possible, consider doing the second activity during another lesson so students can reflect and discuss both.

If you choose this wrap-up, skip or delete the slides associated with Wrap Up 1

Slide	Lesson Guide
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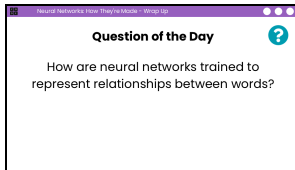


Discuss: Today's activity showed how a language model can study conversations as training data in order to generate new conversations. Several companies currently making large-language models have access to the private conversations from the users on their platform. Do you think your conversations should be used as training data? Why or why not?

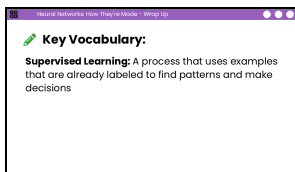
Discussion Goal: This is an open-ended discussion where the act of reflecting on this scenario is more important than the conclusion the class comes to. Encourage students to put themselves in the position of someone on these platforms and consider how they would feel if their private conversations became part of a language model training set. Encourage students to consider issues of privacy and transparency when deciding their opinions, such as whether or not they should be informed if their data is used in this way and if they can opt out. It's okay for the discussion to reach a place of non-closure as students reflect on how they may already be a part of this system with the apps and devices they engage in.

Ethics Opportunity

Ethics Opportunity: Real-World Connections - This is not a hypothetical situation - for example, consider sharing the article [How to opt out of Meta's AI training](#) (June 2024, MIT Technology Review) with students and having them respond to the concerns and strategies raised in the article. Consider having students research the privacy policies of the apps they use most often, or the policies of the devices themselves (such as Google or Apple privacy policies for messages sent on their devices). Most companies include a way to opt-out of using their personal data for AI training data - consider suggesting this to students and have them decide whether or not they want to opt-out and justify their decision.



Question of the Day: How are neural networks trained to represent relationships between words?



Vocabulary: The key vocabulary from this lesson is provided on this slide and in the lesson plan.



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Lesson 8: Attention Is All You Need

45 minutes

Overview

This lesson delves into the ambiguity of language and the necessity of context for meaning. Students first discuss examples of ambiguous language, followed by an activity where they brainstorm words with multiple meanings and craft sentences to clarify their intended meanings through context. After trading papers with a partner, students guess the intended meanings, identifying which words provided the most context. Reflecting on this, the lesson connects the activity to how language models use embeddings for word meanings and neural networks for relational meaning, introducing the concept of Attention as the process of updating word meanings based on context.

Standards

Full Course Alignment

AI4K12 National Guidelines 2021

- ▶ **BI-3** - Computers can learn from data

AI4K12 National Guidelines 2023

- ▶ **BI-2** - Agents maintain representations of the world and use them for reasoning
- ▶ **BI-4** - Intelligent agents require many kinds of knowledge to interact naturally with humans

CSTA K-12 Computer Science Standards (2017)

- ▶ **AP** - Algorithms & Programming
- ▶ **CS** - Computing Systems
- ▶ **IC** - Impacts of Computing

Agenda

Before the Lesson

Student Levels

Warm Up (5 minutes)

Activity (35 minutes)

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Construct examples of words and sentences that require context clues to understand fully
- Explain how embeddings and neural networks are used in the attention process

Preparation

- Print copies of the activity guide for each student
- Run through the slide deck and videos ahead of time
- (Optional) Prepare vocabulary posters for class word wall
- (Optional) Print out the slide deck with speaker notes as a resource during class. [Click here for steps on how to do this.](#)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- [Attention is All You Need](#) - Slides

▼ Make a Copy

For the students

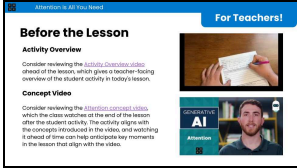
- [Generative AI: Attention](#) - Video ([Download](#))
- [Meaning from Context](#) - Activity Guide ▼ Make a Copy

Vocabulary

- **Attention** - A process that updates the meaning of each word by using the context from the other words in the input

Teaching Guide

Before the Lesson

Slide	Lesson Guide
	Before the Lesson: This activity uses a pedagogy rooted in discovery learning, where students first complete an activity and reflect on their process, which illuminates the underlying AI concepts. A video is provided at the end of the lesson that explains and summarizes the key learning from the lesson. Activity Overview: Consider reviewing the Activity Overview video ahead of the lesson, which gives a teacher-facing overview of the student activity in today's lesson. Concept Video: Consider reviewing the Attention concept video, which the class watches at the end of the lesson after the student activity. The activity aligns with the concepts introduced in the video, and watching it ahead of time can help anticipate key moments in the lesson that align with the video.

Student Levels

These levels contain the same questions as the **Warm Up** and **Wrap Up** from the lesson slide deck, allowing students to answer the questions within the Code.org platform. These can be used as a digital alternative to physical journals, and lets you take advantage of our [Student Summaries features](#) for checking and displaying student work.

By answering these questions in these levels, you can take advantage of discussion and differentiation strategies that elevate student voice to help you guide the discussion. See our [Teaching Guide for Incorporating Student Responses in Lessons](#) for more information on these strategies.



1

Warm Up

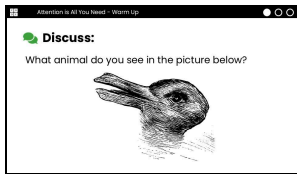


2

Wrap Up

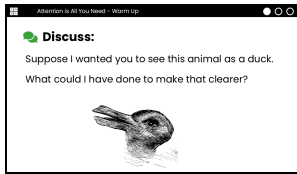
Warm Up (5 minutes)

Slide	Lesson Guide
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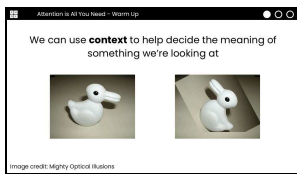
Discuss: What animal do you see in the picture below?

Discussion Goal: This image is an optical illusion - some students may see a duck, while others will see a rabbit. Students who see the duck will think the animal is facing to the left, with the two long appendages acting as the beak. Students who see the rabbit will think the animal is facing to the right, with the two long appendages acting as the ears. Once students know how to view the image, they may find themselves switching back and forth in how they view the image. In the class discussion, both views are correct and let students surface that there are two different ways to view the image before pointing it out explicitly.

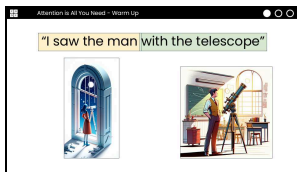


Discuss: Suppose I wanted you to see this animal as a duck. What could I have done to make that clearer?

Discussion Goal: There are many possible answers to this, such as adding color or drawing more items in the scene to indicate a setting (like water for a duck or a hole in the ground for a rabbit) or drawing more of the body. Emphasize answers that focus on how having additional information in the image helps add context that your brain can use to make sense of the image. Avoid answers that suggest re-drawing the image or clarifying details - the focus should be on how, even with this same ambiguous drawing, the surrounding details (like color or setting) can still give clues as to what this image is supposed to be.



Say: A key detail in viewing this illusion is context - the surrounding details that help us decide the meaning of what we're looking at. In this version of the illusion, it's the same object but its orientation is the context we need to know what we're supposed to be looking at - a duck on the left, and a rabbit on the right.



Say: The same thing can happen with language, where words or sentences can sometimes have ambiguous meanings. For example, is this sentence saying "I saw the man..." [pause] "...with the telescope"

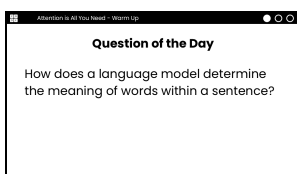
Click the animation

Say: Or is it saying "I saw..." [pause] "... the man with the telescope"

Click the animation

Image credit: ChatGPT Generated image Prompt: png white background, a woman looking through a telescope, in the style of animated illustrations, standing in a window frame, full body, text-based --style raw

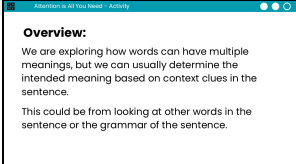
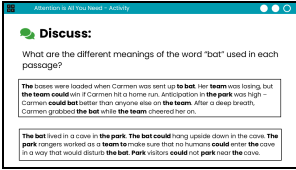
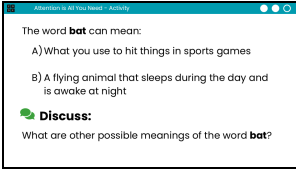
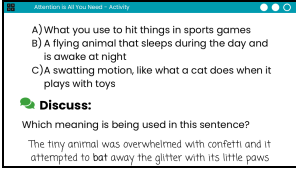
ChaptGPT Generated Image Prompt: png white background, a man standing next to a telescope, in the style of animated illustrations, standing in classroom, full body, text-based --style raw



Say: This can be a problem for language models when they try to generate responses - how do they know the intended meaning of sentences as they get more complex? This is what we're going to investigate today.

Question of the Day: How does a language model determine the meaning of words within a sentence?

Activity (35 minutes)

Slide	Lesson Guide
 Overview: We are exploring how words can have multiple meanings, but we can usually determine the intended meaning based on context clues in the sentence. This could be from looking at other words in the sentence or the grammar of the sentence.	Say: We are exploring how words can have multiple meanings, but we can usually determine the intended meaning based on context clues in the sentence. This could be from looking at other words in the sentence or the grammar of the sentence.
 Discuss: What are the different meanings of the word “bat” used in each passage? The buses were loaded when Carmen was sent up to bat. Her team was losing, but the team could win if Carmen hit a home run. Anticipation in the park was high – Carmen could bat better than anyone else on the team. After a deep breath, Carmen grabbed the bat while the team cheered her on. The bat lived in a cave in the park. The bat could hang upside down in the cave. The park rangers worked as a team to make sure that no humans could enter the cave in a way that would disturb the bat. Park visitors could not park near the cave.	Discuss: What are the different meanings of the word “bat” used in each passage? Have students share with a partner before regrouping as a full class. Discussion Goal: These passages were used in an earlier lesson, so students should quickly recall these differences: one passage uses “bat” in the context of a team sport, while another uses “bat” in the context of the flying animal.
 The word bat can mean: A) What you use to hit things in sports games B) A flying animal that sleeps during the day and is awake at night Discuss: What are other possible meanings of the word bat?	Discuss: What are other possible meanings of the word bat? Discussion Goal: Students may come up with a variety of meanings, or link the word as parts of phrases. For example, “didn’t bat an eyelash” is a phrase meaning “not surprised”. Or combining, “bat-man” describes a certain superhero. There may be other new slang meanings that didn’t exist at the time this lesson was written that you can rely on. However, if no one suggests it, make sure to explicitly call out “bat” as a verb meaning a swatting motion like when a cat plays with toys. This will be important for the next activity.
 A) What you use to hit things in sports games B) A flying animal that sleeps during the day and is awake at night C) A swatting motion, like what a cat does when it plays with toys Discuss: Which meaning is being used in this sentence? The tiny animal was overwhelmed with confetti and it attempted to bat away the glitter with its little paws.	Discuss: Which meaning is being used in this sentence: the tiny animal was overwhelmed with confetti and it attempted to bat away the glitter with its little paws. Have students discuss in small groups before quickly voting as a class. Discussion Goal: Students should vote for choice (C). Wait to ask students to justify their choice until the next slide.

Teaching Tip

Teaching Tip: Whiteboards or Posters - Consider having this sentence written on a whiteboard or poster paper, since the next slide involves drawing arrows to represent how other words influenced how we understood the word ‘bat’ in this sentence.

 **Discuss:** How did the other words in the sentence help you determine the meaning? Draw arrows from the other words in the sentence that helped you figure out the intended meaning

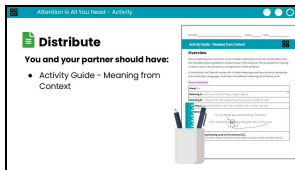
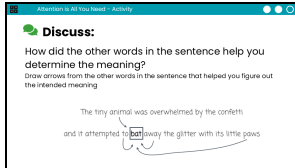
If you are able to draw on the screen or write the sentence on a whiteboard or another surface, it's best to ask students to respond and draw arrows in real-time based on their answers. Prompt students to explain why these particular words helped with an understanding of the word "bat".

 **Discussion Goal:** There are many possible answers as long as students can justify their reasoning. However, make sure to emphasize the following words as contributing to this word's meaning:

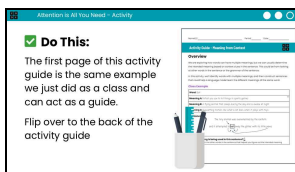
- "Animal" - because bats could be animals or they could be a motion used by animals, this word provides a strong signal
- "To" and "Away" - these together give clues that the word may be a verb rather than a noun
- "Paws" - because "batting" is something that happens with animal paws

 Click the animation

 **Say:** Here's one way a language model could have used other words in the sentence to help understand which meaning of "bat" should be used. Notice that some words help in different ways - for example, "to" and "away" give clues to the grammar of the sentence and indicate "bat" is likely a verb, but "animal" and "paws" give clues to the meaning of the word itself. Now that we've done one example, it's your turn to come up with words like this!



 **Distribute:** Pass out the **Meaning from Context Activity Guide**



 **Do This:** Point out to students that the first page of the activity guide is the same example just completed as a class, then have them flip to the back side

 **Say:** 1) Come up with a word that has at least 3 different meanings

 Click the animation

 **Say:** 2) Write a sentence that could help a language model determine one of these intended meanings

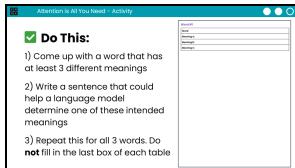
 Click the animation

 **Say:** Repeat this for all 3 words. Do not fill in the last box of each table

 **Circulate:** Give students time work through each example and generate their own words. If students are struggling, encourage them to think of phrases that mean something different at home or with friends than they do at school. If a student is really stuck, consider suggesting any of the following words to get them started:

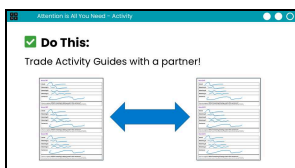
- Date
- Spring
- Bark
- Ring
- Bow
- Match
- Light
- Bank
- Rock
- Fly

Notice any words and sentences that stand out and could be shared as a full class. Once students have progressed far enough, continue to have them trade papers with a partner.

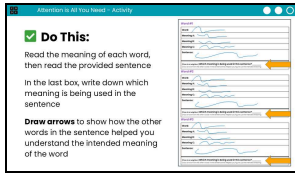


Teaching Tip

Teaching Tip: Complex Sentences - As an extension, some students may find it enriching or challenging to design a single sentence that uses all three meanings of the word. For example, this sentence uses three different definitions of the word “fly”: “The angler, wanting to match his fly to the insects that were hatching that afternoon, quickly snatched a small fly from a flower to examine it closer, before it could escape his grasp and fly away”. If a student decides to do this, their partner would draw arrows to all three uses of the word in the sentence and justify what each one means.



 **Do This:** Trade activity guides with a partner! If students worked together or "overheard" what a neighbor was putting on their activity guide: make sure they trade with someone far away to avoid spoiling the activity.

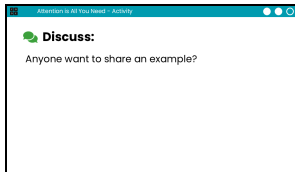


☑ **Do This:** Have students read the meaning of each word, then read the provided sentence. In the last box, write down which meaning is being used in the sentence. Draw arrows to show how the other words in the sentence helped you understand the intended meaning of the word

🔄 **Circulate:** Monitor students as they guess the meaning the other person included. Make sure they are also drawing arrows from the words that helped inform the meaning. The number of arrows will vary from sentence to sentence, but look for at least 2 arrows with each sentence as a minimum. Consider also asking students if they'd be willing to share with the class in the next part of the lesson so they have a heads-up before being called on and can choose to opt-out if they don't feel comfortable sharing.

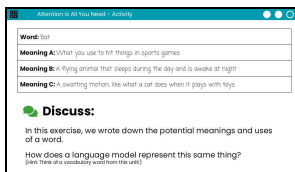
✓ Assessment Opportunity ▲

Formative Assessment: Contextual Meanings - A key learning goal in this lesson is modeling how words in a sentence add context to help determine the correct meaning of words, which is also a process language models use. Monitoring students as they complete this section of the activity guide can help inform how well students are grasping this concept. If students are struggling, consider regrouping and doing a few examples as a class.



🗣️ **Discuss:** Give students an opportunity to share out examples, or call out any examples you noticed while circulating. Use this as an opportunity to highlight student creativity in choosing words and designing sentences that help us understand an intended meaning.

🗣️ **Say:** We've now examined how we, as humans, determine the meanings of words from context clues. Now let's examine how a language model does something similar



🗣️ **Discuss:** In this exercise, we wrote down the potential meanings and uses of a word. How does a language model represent this same thing?

Have students discuss with a neighbor before regrouping. You can also encourage students to use their graphic organizer to help with this question.

💡 **Discussion Goal:** This question has a single intended answer: Embeddings. Try to get to this answer quickly rather than having students continually guess and try to "read your mind". If students don't answer correctly after two or three guesses, it's okay to tell them explicitly.

✓ Assessment Opportunity ▲

Formative Assessment: Key Unit Concepts - A key learning goal throughout the unit is how embeddings represent the semantic "definition" meaning of words, and neural networks represent the relational meanings of words. This discussion question and the next one are an opportunity to see how well students have internalized these concepts. If students aren't able to think up these vocabulary words in response to these questions, consider finding time for students to review the big ideas and vocabulary from this unit in a later class period, but avoid the instinct to dwell on it in this moment.

Attention is All You Need - Activity

Embeddings

- **What they are:** A set of numbers used to represent a word in a large Language Model
- **How they are used:** represent the meaning of words

Say: Embeddings are how language models represent words using numbers. These embeddings are created in such a way that they approximate the meaning of words, based on how they've been used in the training data.

Attention is All You Need - Activity

Sentence: The tiny animal was overwhelmed by the context and it attempted to ~~bury~~ bury the gift with its little paws

Discuss:

In this exercise, we drew arrows to represent the relationships between words within a sentence. How does a language model represent this same thing? (Hint: Think of a vocabulary word from this unit)

Discuss: In this exercise, we drew arrows to represent relationships between words within a sentence. How does a language model represent this same thing?

Have students discuss with a neighbor before regrouping. You can also encourage students to use their graphic organizer to help with this question.

Discussion Goal: This question has a single intended answer: Neural Networks. Try to get to this answer quickly rather than having students continually guess and try to "read your mind". If students don't answer correctly after two or three guesses, it's okay to tell them explicitly.

Attention is All You Need - Activity

Neural Networks

- **What they are:** An interconnected network that makes decisions using weights and hidden layers.
- **How they are used:** represent the relationship between words

Say: Neural Networks are how language models represent relationships between words in a sentence. These networks are created in such a way that they approximate common sentence structures like subject/noun or verb/object grammatical relationships, based on how the training data is structured.

Teaching Tip

Teaching Tip: Generative AI - The next set of slides introduce a core innovation with language models - the Attention Mechanism. This innovation was first introduced in 2017 from a team at Google in a paper titled Attention is All You Need (what a clever name!). It introduces this Attention Mechanism, wrapped up in a special kind of computing architecture called a Transformer (essentially the combined Storage and Processing components in our model). Before presenting these slides, you should consider exploring the interactive website **"Generative AI exists because of the transformer" from the Financial Times**. This interactive website offers a similar explanation as the one to follow, with very engaging graphics and visuals. You can also consider showing it to students to reinforce the concepts on the next few slides, although it does go into more depth than is required in this unit.

Attention is All You Need - Activity

Video

Focus Question: What is attention in a language model?

GENERATIVE AI

Say: Embeddings and Neural Networks have existed for a long time, even before language models exploded in popularity. Let's learn more about how language models combine these two elements to reach the next level in language models that we use today.

Video: Watch the **Generative AI: Attention**

Discuss: What is attention in a language model?

Discussion Goal: Students should recall what they heard in the video, but it's okay if their answers aren't 100% accurate since this will be reinforced in the next slide. Once a few students have shared, transition to the next slide to present the definition.

Attention is All You Need - Activity

Attention

A process that updates the meaning of each word by using the context from other words in the input.

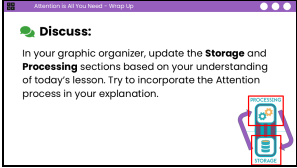
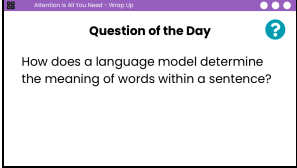
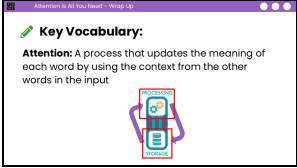
RECOGNITION

The bat lived in a cave in the park

bat [0.1] [0.2] [0.3] [0.4] [0.5]

Say: Attention is a process that updates the meaning of each word by using the context from other words in the input. Neural networks are used to detect relationships between each word in the sentence, and then the embedding of each word is updated to reflect the meaning of the word in the sentence.

Wrap Up (5 minutes)

Slide	Lesson Guide
 Discuss: In your graphic organizer, update the Storage and Processing sections based on your understanding of today's lesson. Try to incorporate the Attention process in your explanation.	Discuss: In your graphic organizer, update the Storage and Processing sections based on your understanding of today's lesson. Try to incorporate the Attention process in your explanation.
 Question of the Day How does a language model determine the meaning of words within a sentence?	Question of the Day: How does a language model determine the meaning of words within a sentence?
 Key Vocabulary: Attention: A process that updates the meaning of each word by using the context from the other words in the input.	Vocabulary: The key vocabulary from this lesson is provided on this slide and in the lesson plan.



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Lesson 9: Outputs & Probabilities

45 minutes

Overview

In this lesson, students investigate how language models generate output based on probabilities. The lesson kicks off with the class collaboratively crafting a story, with a twist to alternate between realistic and fantastical elements. This leads into a group activity where students fill in a mad lib with predetermined words, aiming for either coherent or absurd outcomes. They are then introduced to an app that aggregates responses from numerous users to the same mad lib, showcasing a variety of popular choices for each blank. Through selecting the most popular choices based on this data, students experience firsthand how language models select words with the highest probability from training data.

Standards

Full Course Alignment

AI4K12 National Guidelines 2023

- **BI-2** - Agents maintain representations of the world and use them for reasoning

CSTA K-12 Computer Science Standards (2017)

- **CS** - Computing Systems

Agenda

Before the Lesson

Student Levels

Warm Up (5 minutes)

Activity (35 minutes)

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Explain how a language model uses probabilities to generate words as output

Preparation

- Print copies of the mad lib story and cut them into quarters
- Print copies of the mad lib cards - one set per group
- Run through the slide deck and videos ahead of time
- (Optional) Print out the slide deck with speaker notes as a resource during class. [Click here for steps on how to do this.](#)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

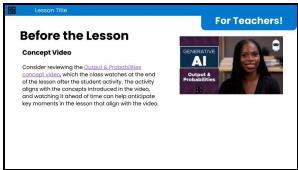
- **Mad Lib Cards** - Resource
▼ Make a Copy
- **Mad Lib Story** - Resource
▼ Make a Copy
- **Outputs & Probabilities** - Slides
▼ Make a Copy

For the students

- **Generative AI: Output & Probabilities**
- Video ([Download](#))

Teaching Guide

Before the Lesson

Slide	Lesson Guide
	Before the Lesson: This activity uses a pedagogy rooted in discovery learning, where students first complete an activity and reflect on their process, which illuminates the underlying AI concepts. A video is provided at the end of the lesson that explains and summarizes the key learning from the lesson. Concept Video: Consider reviewing the <u>Output & Probabilities concept video</u>, which the class watches at the end of the lesson after the student activity. The activity aligns with the concepts introduced in the video, and watching it ahead of time can help anticipate key moments in the lesson that align with the video.

Student Levels

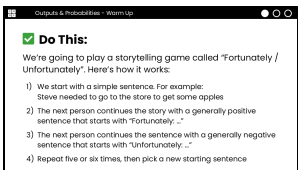
These levels contain the same questions as the **Warm Up** and **Wrap Up** from the lesson slide deck, allowing students to answer the questions within the Code.org platform. These can be used as a digital alternative to physical journals, and lets you take advantage of our [Student Summaries features](#) for checking and displaying student work.

By answering these questions in these levels, you can take advantage of discussion and differentiation strategies that elevate student voice to help you guide the discussion. See our [Teaching Guide for Incorporating Student Responses in Lessons](#) for more information on these strategies.

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Wrap Up

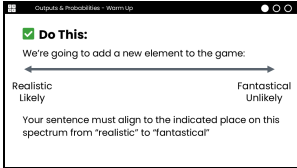
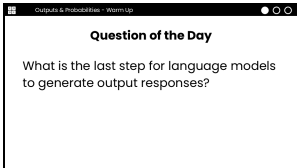
Warm Up (5 minutes)

Slide	Lesson Guide
	☒ Do This: Give students an overview of the Fortunately / Unfortunately game: - We start with a simple sentence. For example: Steve needed to go to the store to get some apples - The next person continues the story with a generally positive sentence that starts with "Fortunately: ..." - The next person continues the sentence with a generally negative sentence that starts with "Unfortunately: ..." - Repeat five or six times, then pick a new starting sentence Play 1-2 rounds of this as a class before continuing to the next slide that adds a new twist.

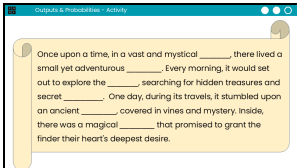
💡 Teaching Tip

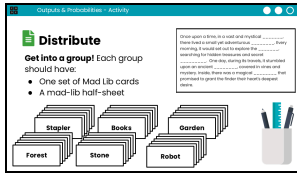


Teaching Tip: Video Example - For a video explanation of how this game works, **consider reviewing this video** before presenting the game to the class

	🗣 Say: We're going to add a new element to this: your responses must now align to a scale from "realistic" and "likely", to "fantastical" and "unlikely". For example, if our starting sentence was still "Steve needed to go to the store to get some apples", then: - A realistic "fortunately" answer might be "Fortunately, he had a full tank of gas in his car" - And a fantastical "fortunately" answer might be "Fortunately, Steve secretly a superhero and could fly wherever he wanted, so he could easily get to the store" ✅ Do This: Play and additional 1-2 rounds. While playing, point on the screen to where responses should fall on this scale, or draw the scale on a whiteboard or poster paper to point out where responses should lie. Lean into the silliness of fantastical answers, especially in contrast to more realistic ones. 🗣 Say: This storytelling game demonstrates a range of outcomes from realistic to fantastical. It turns out this same idea applies to how language models generate responses as well.
	🗣 Say: This is the final step for a language model, where it selects from possible responses weighing the probable against the improbable. Today, we'll delve into this process for a language model and how it uses probabilities to make these final choices. Question of the Day: What is the last step for language models to generate output responses?

Activity (35 minutes)

Slide	Lesson Guide
	🗣 Say: We're going to explore this last facet of language models by returning to the idea of a mad-lib. Let's read this mad-lib together: Once upon a time, in a vast and mystical _____, there lived a small yet adventurous _____. Every morning, it would set out to explore the _____, searching for hidden treasures and secret _____. One day, during its travels, it stumbled upon an ancient _____, covered in vines and mystery. Inside, there was a magical _____ that promised to grant the finder their heart's deepest desire.



Say: Unlike a traditional mad-lib where you can fill in whatever you want, we're going to play with a set amount of words that we can use.

Group: Have students get into groups, ideally 6 groups in total.

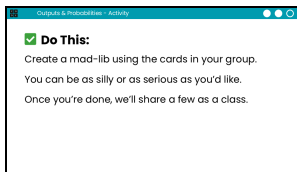
Distribute: Pass out the **Mad-Lib cards** to each group. These should have been cut into individual cards before class. There are 48 cards total - with 6 groups, each group will get 12 cards. You can keep cards in the same group on the handout, or completely randomize the cards - either will work for this activity.

Also pass out the **Mad-Lib Story** handout for easy access for groups. This also should be cut up before distributing.

Teaching Tip

Teaching Tip: Preparing Materials - There are several sets of cards and half-sheets to manage in this activity. Here are some strategies recommended by teachers who piloted this unit:

- I put all words on one sheet in 2" x 1/2" cards. I put the story on a full 8x11 sheet of paper. This allows the cards to be placed directly into the story for easy reading (and full ridiculousness).
- Get some ziplock sandwich baggies to put the cut up cards in so that you can reuse the cards. Get the cards laminated so that you can reuse them.
- I did not cut their choice of words into cards. It just made for more preparation and more clean up afterward. Keeping the words on the original sheet was much easier.



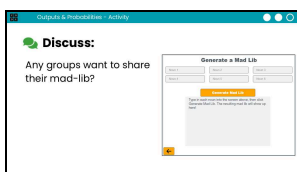
Do This: Create a mad-lib using the cards in your group. You can be as silly or as serious as you'd like. Once you're done, we'll share a few as a class.

Circulate: Check in with groups as they use the cards to complete their mad-libs. Identify a few unique stories to share out as a class.

As groups are finishing up, bring up the Mad Lib Generator app (Level 2 on studio.code.org) and go into the "Generate Mad Lib" screen. This app does not need to be used by students - instead, you can drive this app while projecting to the class.

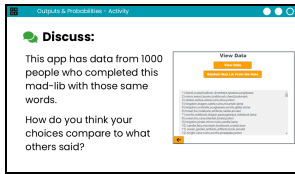
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[For Teachers] Mad Lib Widget



Discuss: Any groups want to share their mad-lib?

Model: As groups share, type in their responses to the Generate Mad Lib screen on the app, then generate the resulting story. Depending on time, see if 2-3 groups would like to share their story.

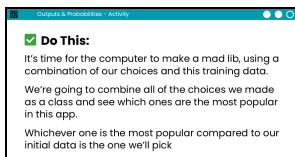


Discuss: This app has data from 1000 people who completed this mad-lib with those same words. How do you think your choices compare to what others said?

Model: Navigate to the “View Data” section of the app. Click the “View Data” button to show the data behind-the-scenes of this app. You can also press the “Random Mad Lib from Data” button to generate one of these mad libs, which is pulled from one of these 1000 examples.

Teaching Tip

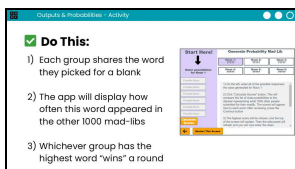
1,000 People Did This Mad-Lib? Really? No - not really; this data is generated from a program designed to create certain probabilities that will be important later in the lesson. However, the result is identical to what would happen if we did survey people and gather data. If students ask about this, feel free to tell them that the data is as real as it can be for the lesson, but there aren’t actually 1000 people who’ve done this mad-lib.



Say: It’s time for the computer to make a mad lib, using a combination of our choices and this training data. We’re going to combine all of the choices we made as a class and see which ones are the most popular in this app. Whichever one is the most popular compared to our initial data is the one we’ll pick

Teaching Tip

Teaching Tip: Family Fued - There’s an aspect to this part of the activity that is very similar to the television show Family Fued. In this game, a survey is given to 100 people and the contestants have to try and guess the “most popular” answers - the more popular the answer, the more points they get. This is similar to what’s happened here: we have data from 1000 versions of this mad-lib, and we’re trying to guess which choices are the most popular. You could decide to frame this section of the activity around this game, but be sure to explain it thoroughly (or even show a clip from the show) to make sure all students in the room understand how the show works.



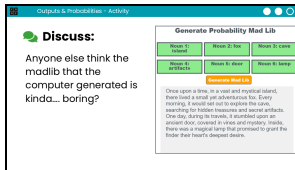
Do This: Navigate to the Probability section of the App. Follow these steps:

- 1) Ask each group to share their choice for the first noun in the mad lib. Type each noun into the inputs on the left
- 2) Press the “Calculate Scores” button. A list of numbers will appear - this represents how often that choice appears in the initial data. Point out to the class which word has the highest score - the higher the number: the more popular that choice.
- 3) Press the “Continue” button. The highest rated word will get chosen, and the top-section of the app will update. Then the side-panel will reset.
- 4) Repeat this process for each of the blanks in the mad-lib - each group shares their word, the words are compared with the input data, and the highest score is chosen.
- 5) Once all the words have been chosen, we can generate the mad lib.

Teaching Tip

Teaching Tip: Probability, not Preference - At the end of the day, this screen involves choosing the “best” word for the computer to use from the choices that each group in the class made. It can be very easy for this to slip into the realm of “favorites” or “preferences”, where groups might argue one word is better than others for a variety of reasons. Ideally, the structure of this lesson steers away from these conversations by leaving that choice up to mathematics and probability - we aren’t making any kind of “value judgement” on the words, but instead comparing them to the data and seeing how frequently those words appeared in the data we collected.

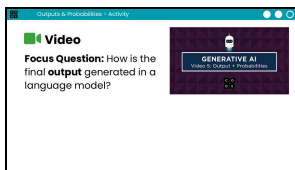
Hopefully this helps ground this decision away from ideas of “favorite” or “best”, and more towards “most probable from the data”. However, the underlying point - that there is some kind of preference or favoritism happening - is still fair, but the real place to have that conversation is when collecting the initial data used in the app. If students want the app to make different choices, they should collect more data and see how it influences the app.



Discuss: Anyone else think the madlib that the computer generated is kinda... Boring?

Have students discuss in small groups before sharing as a class.

Discussion Goal: This prompt is purposefully leading, helping students notice that the most probable responses are also the least surprising. This can lead to pretty generic responses that may seem uncreative or bland. This is also what tends to happen with real language models - when asked to be creative without special prompting, the results from a language model can be bland and lukewarm since it's pulling from the most commonly used phrases associated with the input. In the next chapter of the unit, students will learn techniques to make models more creative.

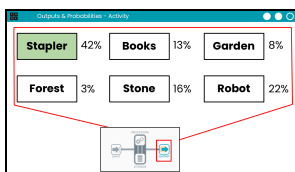


Say: By taking the highest-rated word in each group, we've created a “realistic” version of the mad-lib - even if a little unsurprising - where each response is the most probable choice based on the data already in the app. It turns out this whole process is pretty similar to what happens in actual language models

Video: Watch the [Generative AI: Output + Probabilities](#) Video

Discuss: How is the final output generated in a language model?

Discussion Goal: Students should recall what they heard in the video, but it's okay if their answers aren't 100% accurate since this will be reinforced in the next slide. Once a few students have shared, transition to the next slide to present an explanation



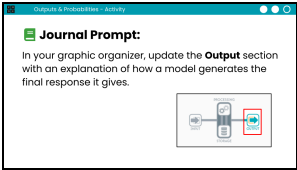
Say: The output layer uses a special neural network to take all the calculations it's done so far and generate a list of words that could potentially go in these blanks. This is similar to what each group did in this activity - as a class, we generated a list of potential words to fit in the mad-lib.

Click the animation

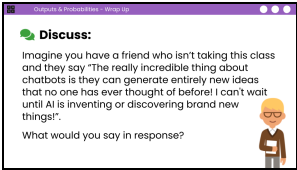
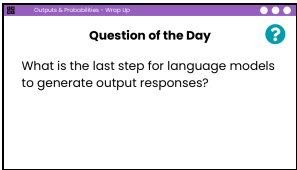
Say: This network also creates a probability for each word, representing how likely it thinks this is the best choice to go in the blank. This is what the app helped with - by comparing our choices with the training data already in the app, we can see the probabilities for how often each word appears in that blank.

Click the animation

Say: And then using this list of probabilities, it just picks the top one. That's all there is to the output layer!

	 Discuss: In your graphic organizer, update the Output section with an explanation of how a model generates the final response it gives.
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Wrap Up (5 minutes)

Slide	Lesson Guide
	 Discuss: Imagine you have a friend who isn't taking this class and they say "The really incredible thing about chatbots is they can generate entirely new ideas that no one has ever thought of before! I can't wait until AI is inventing or discovering brand new things!". What would you say in response?  Discussion Goal: Students will likely quibble with the idea that chatbots are generating "entirely new ideas" or can "invent or discover brand new things". Students should ground their thoughts in what they've seen in the lesson, particularly with how a chatbot's response is based on the training data it was given and the probabilities of certain words. This implies it may not be possible for current chatbots to "invent" or "discover" brand new things, but it may be possible certain words are strung together in unique or novel ways that haven't been seen before.
	Question of the Day: What is the last step for language models to generate output responses?



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Lesson 10: Hallucinations and Fabrications

45 minutes

Overview

In this lesson, students investigate why language models may produce inaccurate outputs, known as hallucinations. They start by understanding the probabilistic nature of generative AI and how bias in training data can lead to such errors. Following discussions on examples of AI systems recalled for bias and hallucinations, students assume the role of AI Auditors to explore proactive measures for mitigating harm caused by deployed language models. After their exploration, they watch a video of a professional AI Auditor, reflecting on the parallels between the professional's work and their class activity.

Standards

Full Course Alignment

AI4K12 National Guidelines 2023

- **BI-2** - Agents maintain representations of the world and use them for reasoning

CSTA K-12 Computer Science Standards (2017)

- **IC** - Impacts of Computing

Agenda

Student Levels

Warm Up (5 minutes)

Activity (35 minutes)

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Make recommendations regarding the input and output of a language model to avoid bias and harm

Preparation

- Print copies of the activity guide for each student
- Run through the slide deck and videos ahead of time
- (Optional) Prepare vocabulary posters for class word wall
- (Optional) Print out the slide deck with speaker notes as a resource during class. [Click here for steps on how to do this.](#)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- [Hallucinations and Fabrications](#) - Slides [▼ Make a Copy](#)

For the students

- [AI Auditing with Dr Deborah Raji](#) - Video ([Download](#))
- [AI Auditors](#) - Activity Guide [▼ Make a Copy](#)

Vocabulary

- **Hallucinations** - Inaccuracies in a language model's response. These

Teaching Guide

Student Levels

These levels contain the same questions as the **Warm Up** and **Wrap Up** from the lesson slide deck, allowing students to answer the questions within the Code.org platform. These can be used as a digital alternative to physical journals, and lets you take advantage of our **Student Summaries features** for checking and displaying student work.

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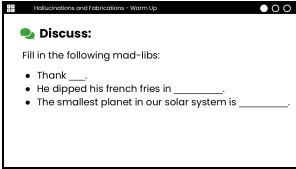


Warm Up



Wrap Up

Warm Up (5 minutes)

Slide	Lesson Guide
	 Discuss: Fill in the following mad-libs: - Thank ____. - He dipped his french fries in _____. - The smallest planet in our solar system is _____. Have students share with a partner before discussing as a full group.  Discussion Goal: Students will likely use common phrases or personal knowledge to answer these questions. Thank You will likely be a common answer to the first mad-lib. The second answer may vary depending on the region you live in or the cultural background of your students - ketchup is a common US condiment to serve with french fries, but some countries use mayonnaise or vinegar, while others may use hot sauce. If this comes up, emphasize how prior experiences inform our answers here. For the last question, students may say Pluto is the smallest planet, while others may say Pluto is no longer a planet and instead Mercury is the smallest planet. This debate is key to the learning outcome on the next slide.

	Discuss: A language model answered these mad-libs with: - Thank you - He dipped his french fries in ketchup - The smallest planet in our solar system is Pluto. Why do you think it gave these answers? Discussion Goal: The key part of this discussion is whether or not Pluto is the "correct" way to complete the last sentence, since Pluto is no longer a planet which would make Mercury the smallest planet. To help verify this, considering searching the internet for this question - as of the time this lesson was written, most online articles cite Mercury as the smallest planet instead of Pluto. This leads to a discussion of why the language model would choose Pluto as its answer, when this is no longer true. From last class, students should recall that language models use probability to choose the words they generate. And since Pluto was the smallest planet for hundreds of years, most articles used to train a language model probably included this fact. This means the language model might see "smallest planet" and "Pluto" as so highly correlated that the language model always picks this word to fill in this blank.
	Say: Language models are capable of incredibly complex behavior, but they can also be wrong. In fact, there are some very simple things that it can be wrong about. In this example, you can see the language model says there are 14 words in the sentence... when there are actually 17 words.

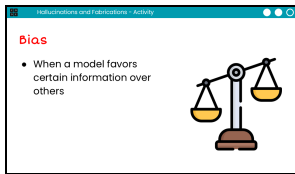
Teaching Tip

Teaching Tip: "Strawberry" and other AI mistakes - At the time that this lesson is being written, another example of a "simple" mistake by a language model is asking how many 'r's are in the word "strawberry" - there are 3 'r's, but **some chatbots will say there are only 2**. Language models are good at receiving updates that help fix these errors when they're encountered, so some of these tricks may not work if you try them yourself. Instead, consider searching for any other recent examples of "simple" mistakes that you can highlight for the class.

	Say: Today we'll examine other ways a language model can be inaccurate and explore how we, as designers and users of these models, can respond. Question of the Day: What happens when a language model gives inaccurate information?
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Activity (35 minutes)

Slide	Lesson Guide
	Say: When a language model gives inaccurate information, those are called hallucinations. These occur when the most probable response isn't accurate or factual. The example we saw in the warm-up is a hallucination - Roosevelt isn't the actual president, even if he happens to be the most probable answer. image: Flaticon.com



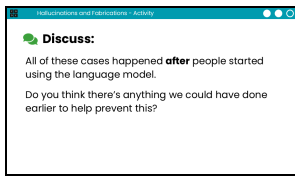
Say: Hallucinations can be an example of how bias manifests in a language model. The reason the model generates these inaccurate responses is because that information may have been overrepresented in the training data, influencing the responses that the model generates.

image: Flaticon.com



Say: Issues with bias and hallucinations in chatbots are significant issues that impact real people. For example,

- Amazon had to shut down a project to help screen applicants because it showed a bias against women.
- An eating disorder helpline needed to stop its chatbot from giving advice when it started giving counterproductive advice
- Research into popular language models shows it can provide debunked medical theories that disproportionately impact Black patients.

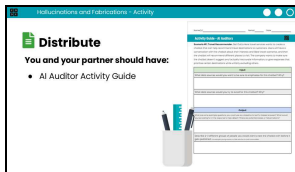


Discuss: All of these cases happened after people started using the language model. Do you think there's anything we could have done earlier to help prevent this?

Have students discuss in small groups before taking 2-3 volunteers as a class.

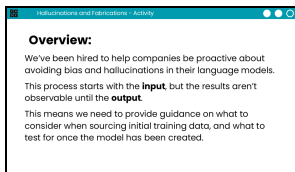
Discussion Goal: This is a purposefully open-ended question meant to generate solution-oriented hypotheses and thoughts. There's no one right answer, but students might suggest:

- Adding some kind of filter or guardrail before anything gets sent back to the user to make sure the response is okay
- Making sure the data used to train the model didn't include the kinds of sources that lead to this type of behavior
- Adding more data when training the model to make sure enough perspectives are represented so this kind of bias response doesn't exist



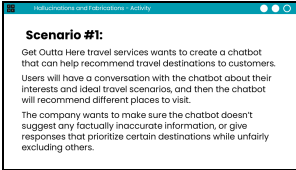
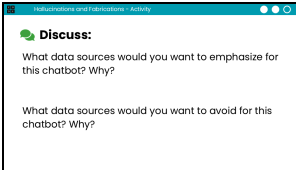
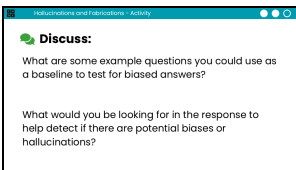
Say: Today we're going to put ourselves in the shoes of the people who find these kinds of issues with AI systems - an AI auditor. Our job will be to come up with different recommendations we might make to a company trying to make sure their chatbot doesn't create headlines like these.

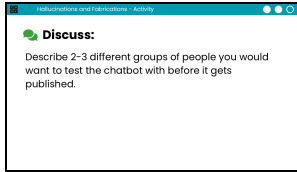
Distribute: Pass out the **AI Auditor Activity Guide**



Do This: Read the overview as a class.

Say: We've been hired to help companies be proactive about avoiding bias and hallucinations in their language models. This process starts with the input, but the results aren't observable until the output. This means we need to provide guidance on what to consider when sourcing initial training data, and what to test for once the model has been created.

	☑ Do This: Read the first scenario as a class. This is also on the activity guide. 🗣 Say: Get Outta Here travel services wants to create a chatbot that can help recommend travel destinations to customers. Users will have a conversation with the chatbot about their interests and ideal travel scenarios, and then the chatbot will recommend different places to visit. The company wants to make sure the chatbot doesn't suggest any factually inaccurate information, or give responses that prioritize certain destinations while unfairly excluding others.
	🗣 Say: Remember that most chatbots use huge chunks of the internet as their training data. Imagine you could emphasize certain sources and avoid others - which ones might you try to consider? For example, you might want to emphasize the official travel websites from countries all over the world since those are likely to be trusted factual sources, and try to exclude forums or message boards from people who travel just to antagonize the people who live there since those might influence the model to consider travel recommendations that are unacceptable. 🗣 Discuss: Have students discuss the prompts on the board in small groups. 🔄 Circulate: Check in on groups as they discuss, listening in on the types of sources students might want to emphasize or avoid. Listen for the rationale they provide for why to emphasize or avoid certain sources, especially as it relates to potential bias and hallucinations. This can be a very open-ended discussion, but ideally the rationale focuses on centering sources that provide a wide representation of experiences and opportunities, and decenters sources that may provide inaccurate or intentionally biased content. Share Out: Ask 1-2 people to share out with the fill group, then have students fill in this section of their activity guide
	🗣 Say: We can only really test for bias or hallucinations once the chatbot has been trained. What are some example questions you might ask as a baseline? For example, you might ask "Where are good places to travel in Europe?" and look for responses that include a wide variety of countries and not just larger western European countries. Or you could ask "where are good places to travel in the USSR" and check that the chatbot realizes that the USSR doesn't exist anymore but recommends other places to travel within the region. 🗣 Discuss: Have students discuss the prompts on the board in small groups. 🔄 Circulate: Check in on groups as they discuss, listening in on the types of questions they may want to ask. Listen for the rationale they provide for how the responses can help detect bias or hallucinations, being strategic about what baseline responses they would expect to help convince them the model has not been compromised. This can be a very open-ended discussion. Share Out: Ask 1-2 people to share out with the fill group, then have students fill in this section of their activity guide



Say: As we saw in some of the headlines, technology doesn't always work the same for everybody. It is also important to make sure the chatbot will work for a wide variety of users and will take into account their unique needs when making travel recommendations. Who are some folks you would want to make sure are able to test the chatbot before it gets published and verify it works for them? Why are they an important group to ensure the chatbot works correctly for them?

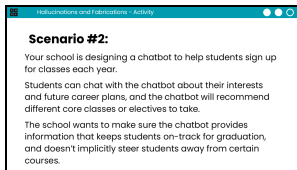
Discuss: Have students discuss the prompts on the board in small groups.

Circulate: Check in on groups as they discuss, listening in on the users they would want to make sure are included in this vetting process. This can be a very open-ended discussion, but ideally students latch onto how folks are affected by technology in different ways, and inviting those groups to be a part of testing and feedback before release can be a part of the auditing process.

Share Out: Ask 1-2 people to share out with the fill group, then have students fill in this section of their activity guide

Ethics Opportunity

Ethics Opportunity: Airbags - You may decide to share the history of the airbag in cars with students. When airbags were first taken to market and installed in cars, data began to show that women were being severely injured or dying at much higher rates than men. It turns out this is because all of the testing for airbags was done by men or assuming weight and height averages associated with men - so, when folks outside this norm would drive, the airbags would cause damage that had been undetected in the tests. This scenario underscores the need to have a wide variety of users involved in the design and testing of products.



Do This: Read the first scenario as a class. This is also on the activity guide.

Say: Your school is designing a chatbot to help students sign up for classes each year. Students can chat with the chatbot about their interests and future career plans, and the chatbot will recommend different core classes or electives to take. The school wants to make sure the chatbot provides information that keeps students on-track for graduation, and doesn't implicitly steer students away from certain courses.

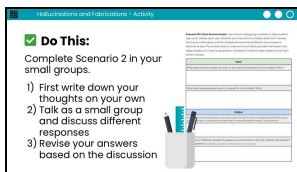
Do This: Have students work through Scenario 2 in small groups:

- 1) First write down your thoughts on your own
- 2) Talk as a small group and discuss different responses
- 3) Revise your answers based on the discussion

Circulate: Listen in on students as they complete this task. Help facilitate discussion amongst groups as needed so students can discuss this scenario openly with their peers.

Formative Assessment

While circulating, you can listen in to groups to get a pulse on how the class is thinking about this scenario. If most groups seem stuck in the discussion, consider regrouping as a full class and repeating the same structure as Scenario #1. If groups seem able to discuss in small groups, be sure to collect the activity guide after class to review how well students were able to articulate these AI auditing practices.



	Discuss: (Optional) Invite groups to share out from their conversations if they wish.
	Say: As we saw in the headlines, the impact of biased models can be profound and deeply affect users and certain communities. Luckily, there are folks working in the field to help discover these flaws in models and advocate for change to improve them. This is a career known as AI Auditing - let's look at someone actively working in this field to improve AI technologies. This is Dr Deborah Raji describing some of her work auditing facial recognition systems. Video: Watch the AI Auditing with Deborah Raji video

Wrap Up (5 minutes)

Slide	Lesson Guide
	Discuss: A friend said "I love using LLMs to help me with my homework. I asked it to help me write my history paper - it came up with all these interesting sources and information I had never heard of before". What would you say to this person? Discussion Goal: Students should be skeptical that all of these "interesting sources" are true and accurate, citing the work from today's lesson as evidence that language models can sometimes hallucinate. Additionally, consider coaching students on how to broach this subject with a friend from an empathic perspective and what they might suggest so their friend uncovers these hallucinations themselves. For example, instead of bluntly telling their friend "those sources are probably wrong", they could suggest their friend check their sources via the internet and verify they're correct before continuing with their assignment.
	Question of the Day: What happens when a language model gives inaccurate information?
	Vocabulary: The key vocabulary from this lesson is provided on this slide and in the lesson plan.



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Lesson 11: Project: Demystifying Generative AI

90 minutes

Overview

In this lesson, students create their project for the unit. In the project, students read several generative AI opinion articles that contain misrepresentations or exaggerations of what generative AI is capable of and how it works. Using their newfound knowledge from this unit, students draft a rebuttal and create an artifact to help visualize or conceptualize an aspect of their argument. This project acts as a summative assessment for the content of this unit.

Standards

Full Course Alignment

AI4K12 National Guidelines 2023

- ▶ **BI-2** - Agents maintain representations of the world and use them for reasoning

CSTA K-12 Computer Science Standards (2017)

- ▶ **CS** - Computing Systems
- ▶ **IC** - Impacts of Computing

Agenda

Warm Up (5 minutes)

Activity (80 minutes)

Wrap Up (5 minutes)

End of the Unit

Student End-Of-Unit Survey

Teacher End-Of-Unit Survey

Objectives

Students will be able to:

- Create artifacts that explain generative AI concepts in a clear and engaging manner
- Identify and address misconceptions about generative AI

Preparation

- Print copies of the project guide for each student
- Print multiple copies of the four Generative AI Opinion Articles for students to review
- Decide how students will submit their work for the project
- Decide which options for student artifacts you would like to emphasize for students
- Decide how to break up the multi-day lesson with your class schedule
- Run through the slide deck and videos ahead of time

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **Project - Demystifying AI** - Slides

▼ Make a Copy

For the students

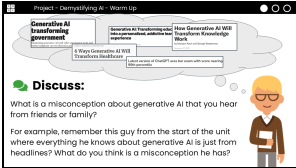
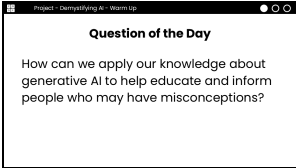
- **Generative AI Opinion 1** - Resource

▼ Make a Copy

- [Generative AI Opinion 2](#) - Resource
▼ Make a Copy
- [Generative AI Opinion 3](#) - Resource
▼ Make a Copy
- [Generative AI Opinion 4](#) - Resource
▼ Make a Copy
- [Project - Demystifying AI](#) - Rubric
▼ Make a Copy
- [Project Guide - Demystifying AI](#) - Activity Guide
▼ Make a Copy

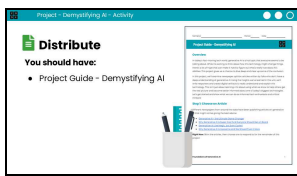
Teaching Guide

Warm Up (5 minutes)

Slide	Lesson Guide
	Discuss: What is a misconception about generative AI that you hear from friends or family? For example, remember this guy from the start of the unit where everything he knows about generative AI is just from headlines? What do you think is a misconception he has? Have students discuss in small groups before regrouping. Discussion Goal: Invite students to share from personal experience any misconceptions or exaggerations about generative AI that they’ve heard from friends or family. Generating a “correct” answer isn’t important in this discussion - instead, the goal is to generate many possible ideas and validate students’ lived experiences thinking and talking about generative AI. This discussion will segue into the main project, where students will be looking at written works with misconceptions about generative AI and try to address them. For now,
	Say: With all of the headlines and discussion about generative AI, it’s common to find people with misconceptions or misunderstandings about generative AI. This creates a risk where these folks could get taken advantage of or not know how to use this new technology effectively. Today, we’ll address these misunderstandings and explore how our newfound knowledge can help educate them. Question of the Day: How can we apply our knowledge about generative AI to help educate and inform people who may have misconceptions?

Activity (80 minutes)

Slide	Lesson Guide
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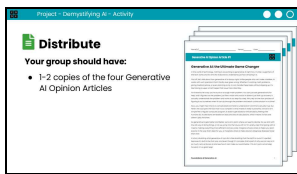


Distribute: Pass out the **Project Guide - Demystifying AI**



Do This: Read the overview of the project. Students can read individually, or you can read the overview as a class.

Students will be examining newspaper opinion articles written by folks who don't have a deep understanding of generative AI. Using the insights they've learned in this unit, they will write responses and create digital artifacts to help rebut and educate the authors of these articles.

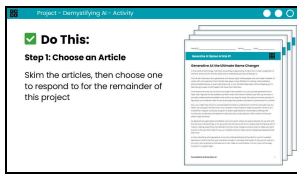


Distribute: Pass out 1-2 copies of each opinion article to each group. Students can share articles when deciding which one to respond to.

- **Generative AI - the Ultimate Game Changer**
- **Why Generative AI is Super Cool And Everyone Should Get on Board**
- **Generative AI: Like Magic, but Even Cooler!**
- **Why Generative AI is Awesome and We Should Trust It More**

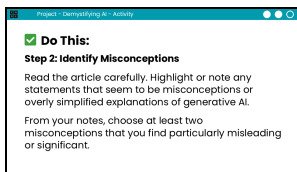
Teaching Tip

Teaching Tip: Where Did These Articles Come From? - For more details about how these articles were created and the specific misconceptions baked into the article, refer to the **Behind the Activity Teacher Resource**



Do This: Read through Step 1: Choose an Article in the project guide.

Students will skim the articles, then choose one to respond to for the remainder of the project

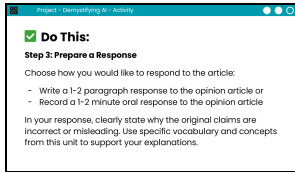


Do This: Read through Step 2: Identify Misconceptions.

Students will read the article and highlight or note any statements that seem to be misconceptions or overly simplified explanations of generative AI. Then students will choose at least two misconceptions to address.

Students should complete these step on Page 2 of the activity guide

Circulate: Give ample time for students to read these articles and plan accommodations ahead of time for students who may need reading supports (such as offering a read-aloud option via the online Google Doc). These articles are designed so there are at least 3 misconceptions or exaggerations in each article for students to identify - if students struggle to identify these misconceptions, consider asking leading questions about each individual sentence in the article and whether or not that feels like a grounded, accurate statement.



☑ Do This: Read through Step 3: Prepare a Response

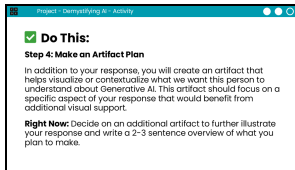
Students will choose how they want to respond to the article - either as a written response or an oral response. Students should focus on using specific vocabulary or concepts from the unit to support their explanations.

Students can complete their written responses on Page 3 of the activity guide. For the oral responses, plan for how you would like students to generate and submit their responses.

🔄 **Circulate:** Give students ample time to prepare their response. Depending on your class lengths, this may take the remainder of the first day of this project. If so, before breaking for the day, consider completing Step 4 (on the next slide) as the wrap-up.

💡 Teaching Tip

Teaching Tip: Check with the Rubric - Make sure students have access to the **rubric** for this project so they can compare their responses with the rubric. This is an opportunity for students to self-reflect and course-correct their own work.



☑ Do This: Read through Step 4: Make an Artifact Plan

Students will choose an additional artifact they would like to create to support their argument. The artifact should help visualize or contextualize a concept or process involved in generative AI. At this stage, students are just deciding the type of artifact they would like to make and writing a brief outline.

Students can complete this on Page 4 of the project guide.

💡 Teaching Tip

Teaching Tip: Break for the Day and Check In - This can be a good point to break for the day and continue with the rest of the activity in the next class period. If you decide to do so, consider collecting student activity guides and reviewing them before the next class. This is an opportunity to check in on how students are developing their ideas in the first part of the activity, and you can make notes of students to check in with individually during the next class period. Or if you notice a consistent misunderstanding of the assignment, consider using the warm-up in the next day to address these misunderstandings with the entire class. When students return, they can pick up where they left off with the project.



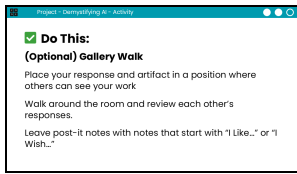
☑ Do This: Have students complete Step 5: Make the Artifact.

Decide ahead of time how you'd like to support students in this project - for example, providing paper supplies for the cartoon strip or suggesting a specific program for 3D models. Also decide how you would like students to submit their artifacts.

🔄 **Circulate:** Give students ample time to create their artifacts. Encourage students to focus on a specific concept in their explanation that the artifact can help illustrate. In some ways, this artifact is a reframing of content from the unit graphic organizer, but in a student's own words and in a different modality.

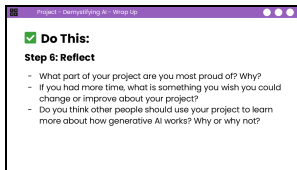
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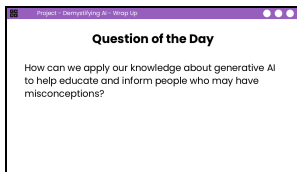
☒ **Do This:** (Optional) Gallery Walk. Have students place their responses and artifacts in a position where others can interact - for students with oral responses, this may mean computers with headphones. Have students interact with their peers work and leave post-it notes with positive feedback starting with "I Like..." or "I Wish..."

Wrap Up (5 minutes)



☒ **Do This:** Have students complete Step 6: Reflect.

Students will answer three reflection questions on the project guide page 5



Question of the Day: How can we apply our knowledge about generative AI to help educate and inform people who may have misconceptions?

End of the Unit

Student End-Of-Unit Survey

Please have students take the post-survey. Although their answers are anonymous, the aggregated data will be available to you once at least five students have completed the survey.



Student Post-Survey

Teacher End-Of-Unit Survey

We also have a teacher end-of-unit survey to learn more about how the unit went for you and your students. While students take their survey, **please complete this end of unit survey for teachers** as well. Your feedback is valued and appreciated!



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